



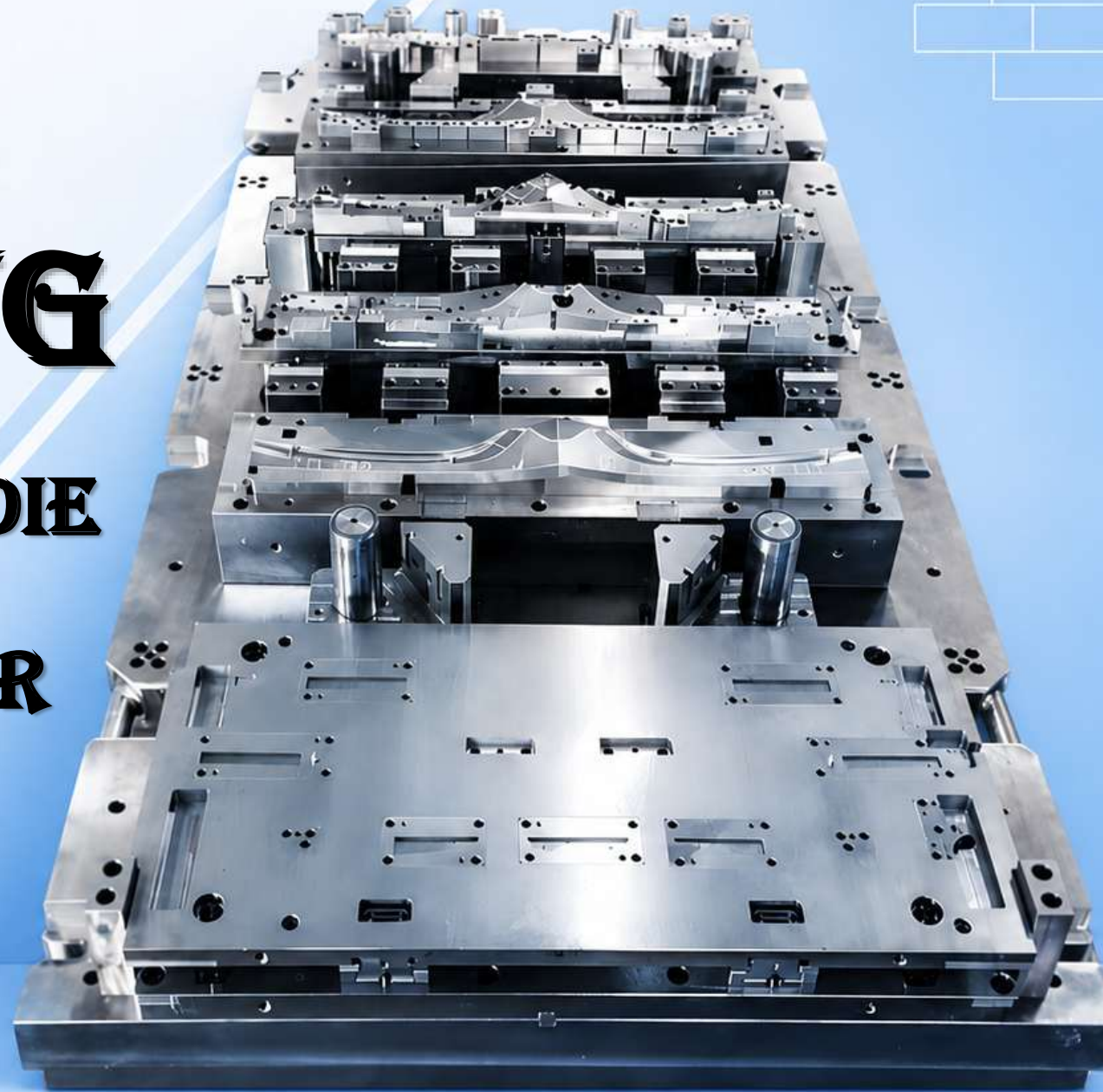
MLD TOOLING

PROFESSIONAL STAMPING DIE

ONE-STOP MANUFACTURER

2026

**PRESENTED BY
MLD BD TEAM**



CONTENTS



MLD TOOLING CO.
PRECISION • QUALITY • PERFORMANCE

- 01** | About Us
- 02** | Capability
- 03** | Design Showcases
- 04** | Product Catalog
- 05** | Contact

01

ABOUT

US



Company Overview

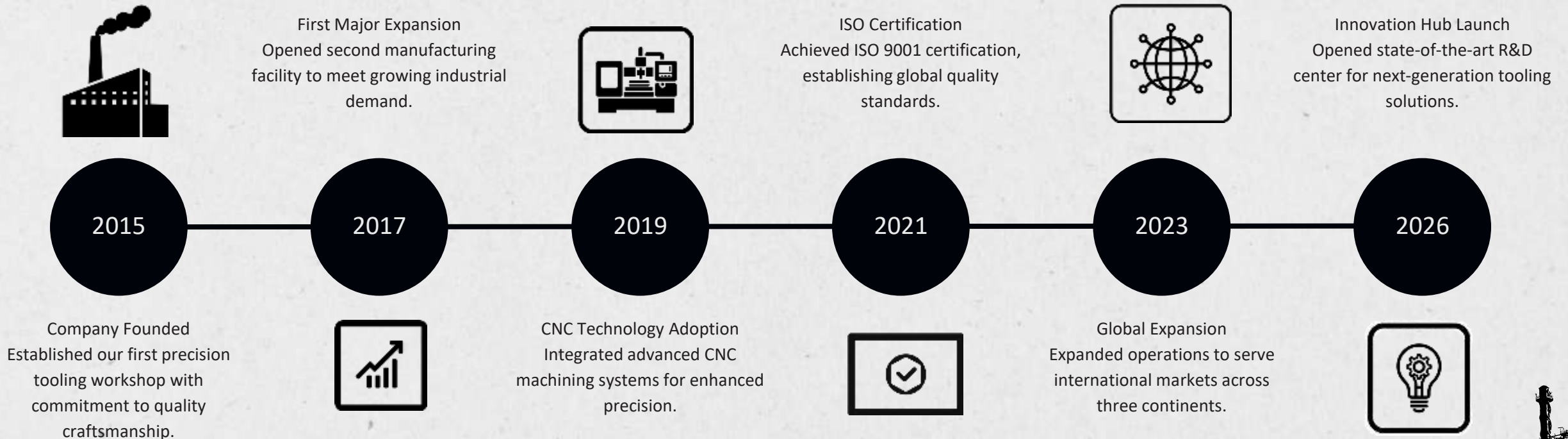
WITH OVER 10 YEARS OF PRECISION TOOLING EXPERTISE, WE SERVE AUTOMOTIVE AND INDUSTRIAL CLIENTS ACROSS NORTH AMERICA, EUROPE, AND ASIA. OUR MISSION: DELIVERING ENGINEERING EXCELLENCE THROUGH INNOVATION AND PARTNERSHIP.



Founded in 2015, Huizhou MLD Tooling Co., Ltd. specializes in metal stamping die design and manufacturing. With more than 20,000 m² of production space, 120 employees, and a 24-member engineering team skilled in 2D and 3D tooling design, we support customers with reliable technical capability, modern equipment, and stable quality performance. Committed to integrity, innovation, and continuous improvement, we focus on delivering dependable tooling solutions and building long-term customer trust.

Our Journey

Key Milestones in Our Journey of Precision and Innovation





MLD

Revenue Growth Overview

Consistent year-on-year growth driven by stable OEM/Tier 1 programs and expanding global business.



01

Sales in 2023

US\$ 10.2M



02

Sales in 2024

US\$ 12.0M



03

Sales in 2025

US\$ 14.1M

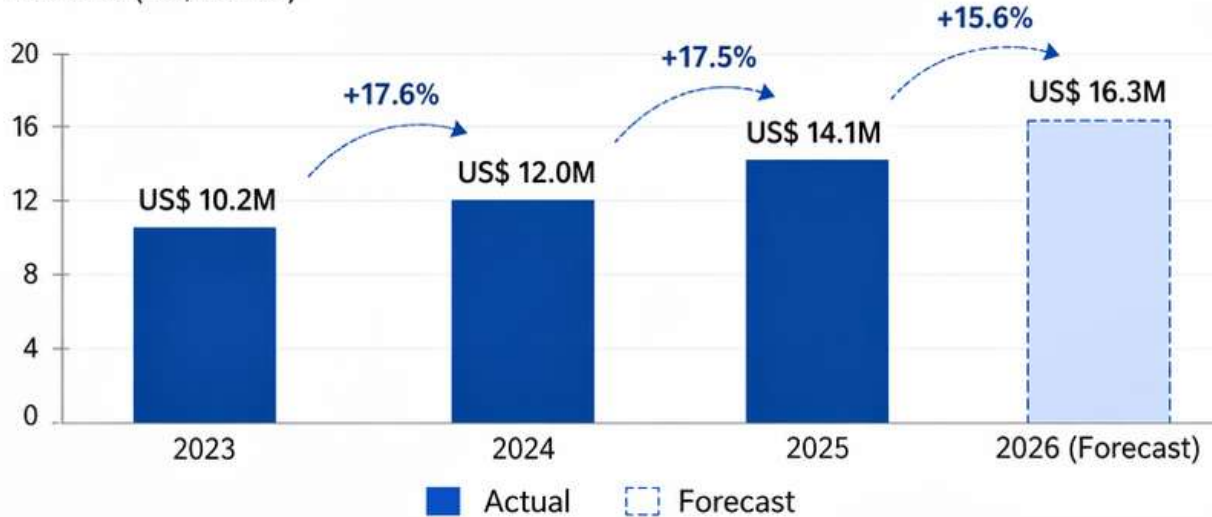


04

Projected Sales in 2026

US\$ 16.3M

Revenue (US\$ Million)



Note: 2026 represents projected sales based on current program visibility and market outlook.



Steady expansion
across export markets



Sustained revenue growth
with strong customer retention



2026 forecast supported by
ongoing tooling and
program launches



MLD

Why Choose Us

— • PRECISION • QUALITY • RELIABILITY



20,000+
m² Factory Area



120+
Skilled Employees



50+
Advanced Equipments



Quality
You Can Trust



Precision Engineering Excellence
in every tooling solution



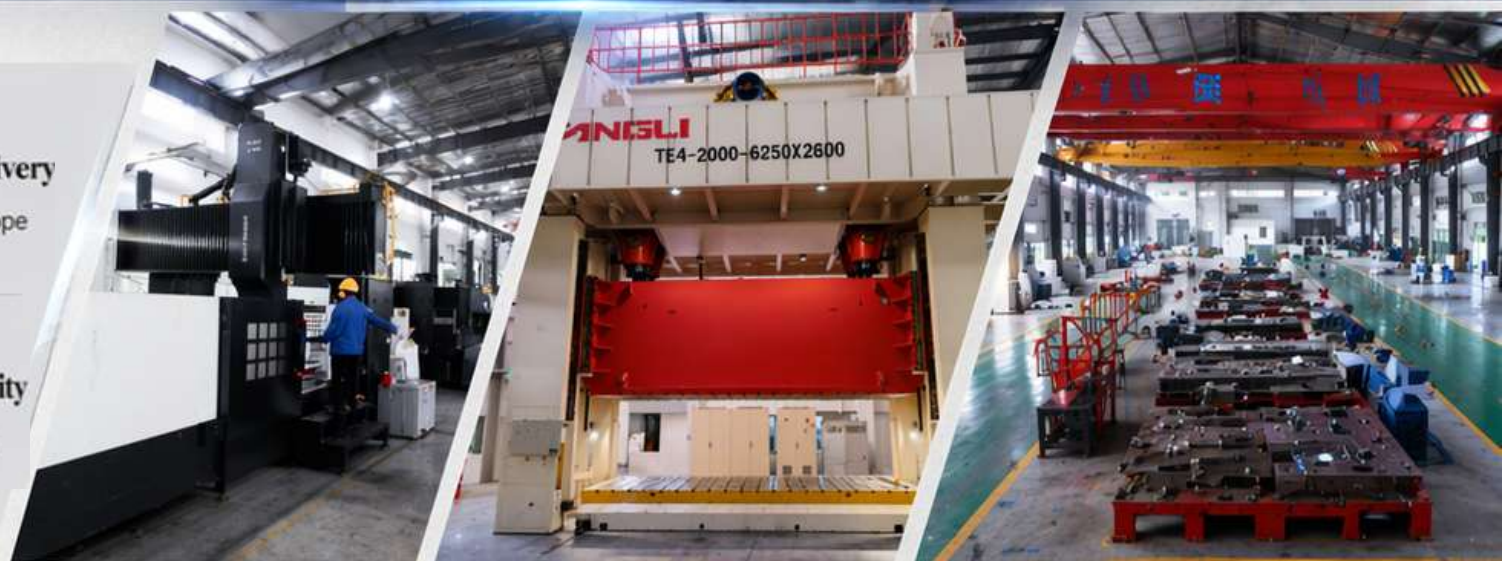
On-time Global Delivery
to North America & Europe



Dedicated Customer Partnership
from concept to completion



Industry-leading Quality
certifications & standards



— CLEAN ENVIRONMENT • ADVANCED EQUIPMENT • STRICT MANAGEMENT • RELIABLE DELIVERY —



MLD

Shopfloor Environment

Clean, organized and efficient production workshop with advanced equipment and strict management system to ensure stable quality and on-time customer delivery.



ORGANIZED
WORKSHOP



5S MANAGEMENT

Clean, tidy and well-organized work environment



SAFETY FIRST

Strict safety control for people, tools and process



SKILLED TEAM

Well-trained staff with rich production experience



CONTINUOUS IMPROVEMENT

Lean production and continuous optimization



ADVANCED
EQUIPMENT



PRECISION
INSPECTION



QUALITY &
RELIABILITY



MLD

Company Culture New Year Party





MLD

Company Culture Team building





MLD

02

CAPABILITY



MLD

MLD TOOLING CO., LTD

Machining & Tryout Facilities

Our state-of-the-art machining centers deliver precision tolerances within **$\pm 0.01\text{mm}$** .

Dedicated tryout stations and advanced inspection equipment ensure every tool meets exact specifications before delivery.



OUR EQUIPMENT CAPABILITIES



MACHINING (CNC MACHINES)

CNC Machining Centers	7 Sets
Surface Grinding Machine	1 Set
Precision Grinders	2 Sets
Drilling / Tapping Machines	4 Sets
Perforating Machine	1 Set



High-precision machining for complex tools and parts



WIRE CUT MACHINES

Medium Wire Cut	4 Sets
Small Wire Cut	1 Set
Perforating Wire Cut	1 Set



Accurate contour cutting with high surface quality



TRYOUT & SPOTTING

Press Machines (with Coil Feeder)	5 Sets
Hydraulic Presses	4 Sets
3D Coordinate Systems	2 Sets
Blue Scan Machine	1 Set
Plug Gauge	1 Set



Complete tryout support and dimensional verification



INSPECTION EQUIPMENT

3D Coordinate Systems	2 Sets
Hardness Tester	1 Set
Crane Scale	—
Height Gauge	—
Plug Gauge	1 Set
Fork Lifter	2 Sets
Electric Fork Lifter	1 Set



Precision inspection ensures consistent quality



HANDLING / CRANE

Cranes (30 Tons)	2 Sets
Cranes (20 Tons)	2 Sets
Cranes (15 Tons)	1 Set
Cranes (10 Tons)	2 Sets
Cranes (2.8 Tons)	1 Set
Fork Lifters	2 Sets
Electric Fork Lifter	1 Set
Manual Pallet Trucks	2 Sets



Safe and efficient material handling for all operations



ENGINEERED FOR PRECISION. COMMITTED TO QUALITY.



Precision Tolerance: **$\pm 0.01\text{mm}$**



MLD

PRECISION. QUALITY. RELIABILITY.

Press Shop Facilities – Factory 1

Large-tonnage stamping capability for
tryout, production support, and precision forming.

MAX

6250MM X 2600MM X 1600MM



**YANGLI
2000T**

Max Press Capacity



**YADON
WTS-1250T**

- High efficiency
- Large tonnage capacity
- Stable performance



JM 36-400

- Strong and durable
- Reliable operation
- Widely applied



SEYI SN2-250

- High precision
- Stable performance
- Suitable for repeatable press operations



2000T

Max Press Capacity



Multiple
Press Tonnage Options



Stable
Quality Control



Tryout & Production
Support



MLD

PRECISION. QUALITY. RELIABILITY.

Press Shop Facilities – Factory 2

Dedicated large-tonnage press capacity for tryout, forming support, and stable production operations. **FEEDER THICKNESS RANGE: 0.6 MM-6 MM**
MAXIMUM FEEDER WIDTH: 2000 MM



Dual Press Configuration

Two 2000T-class presses operating side by side for maximum flexibility.



2000T Class Capability

Engineered for heavy-duty stamping and large part forming.

MAX

5000MM X 2500MM X 2000MM



Factory 2

Dual Large-Tonnage Presses



YANGLI Press

- Large working area
- Stable press performance
- Suitable for heavy-duty stamping support



TIANDUAN TT27-2000L

- High tonnage capability
- Robust frame structure
- Reliable operation for large parts



Dual Press Layout

Parallel Press Configuration



Large-Tonnage Capacity

2000T Class Press Support



Stable Quality Control

Consistent & Reliable Output



Production & Tryout Support

Flexible for Various Needs



MLD

PRECISION. QUALITY. RELIABILITY.

Machining Facilities – Factory 2

Comprehensive precision machining support for large tooling components, structural parts, and stable in-house manufacturing operations.



Integrated Machining Capacity

Multiple machining processes supporting tool manufacturing.



Large-Part Processing Support

Gantry equipment and machining centers ensure precision, efficiency, and consistency.

MAX

4000MM X 2500MM X 1200MM



Factory 2

Precision Machining Workshop



Gantry Machining Center

- Large machining envelope
- Stable heavy-duty structure
- Suitable for large tooling components



CNC Machining Centers

- Multi-machine layout
- Efficient batch processing
- Consistent dimensional control



Flexible Workshop Layout
Integrated Process Support



Precision Machining
High Accuracy & Repeatability



Stable Quality Control
Consistent & Reliable Output



Production Support
Efficient Tooling Component
Manufacturing



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PRECISION. QUALITY. RELIABILITY.

Wire-Cut Facilities – Factory

Advanced wire EDM cutting support for tooling inserts, contour machining, and in-house precision component processing.



Precision Wire-Cut Capacity

High-accuracy EDM machines deliver clean, precise cuts with tight tolerance.



Insert & Profile Processing

Ideal for tooling inserts, intricate profiles, and precision part fabrication.

MAX
500MM X 400MM



Factory EDM Wire-Cut Workshop



Wire EDM Machines

- Reliable wire-cut machines
- Consistent cutting performance
- Suitable for various materials



Contour Cutting Support

- Accurate profile cutting
- Complex contour machining
- Supports fine detail requirements



Stable Machining Accuracy

- Tight tolerance consistency
- Excellent surface finish quality
- Repeatable and dependable results



Precision Cutting
Accurate Profile Processing



Wire EDM Support
Stable & Repeatable Output



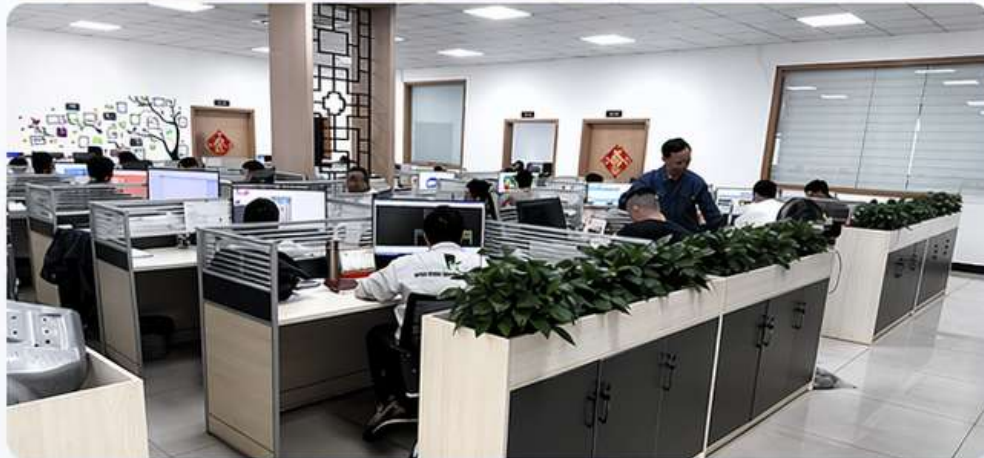
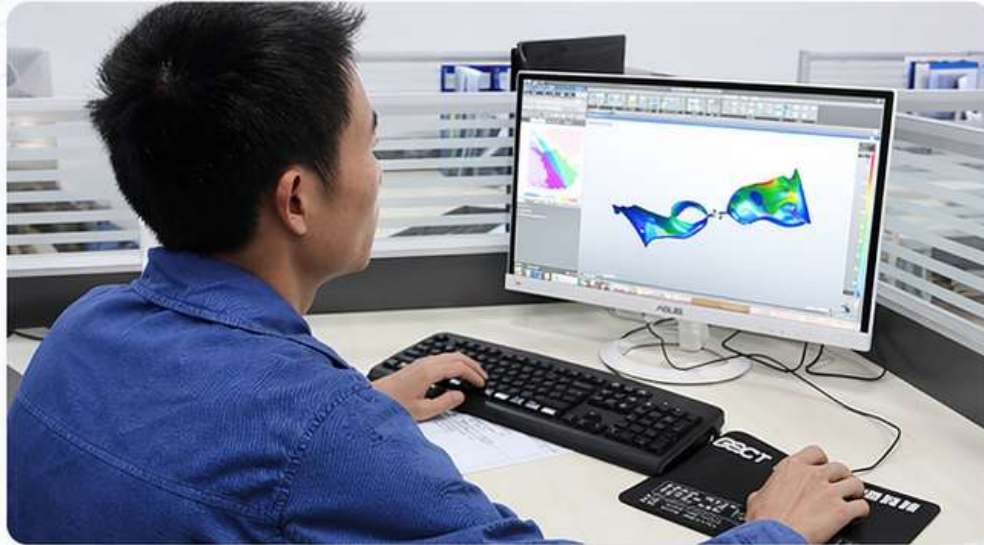
Tooling Components
Insert & Detail Manufacturing



Production Support
In-house Machining
Coordination

Die Design & Engineering

Design Capability – Professional Team



OUR TEAM CAPABILITY



**Simulation
Analysis**

4 senior engineers



3D Design

15 senior engineers

6 junior engineers



2D Design

3 senior engineers

SOFTWARE



NX
3D Design



AutoForm
Forming Simulation



AutoCAD
2D Design &
Detail Drawings

Design Capability - Coil Fed Transfer Die Process

• Precision Design • Advanced Simulation • Efficient Manufacturing • Reliable Performance



PROCESS FLOW

From Coil to Component



TOOLING HIGHLIGHTS



Coil fed progressive transfer structure
卷材连续传递结构



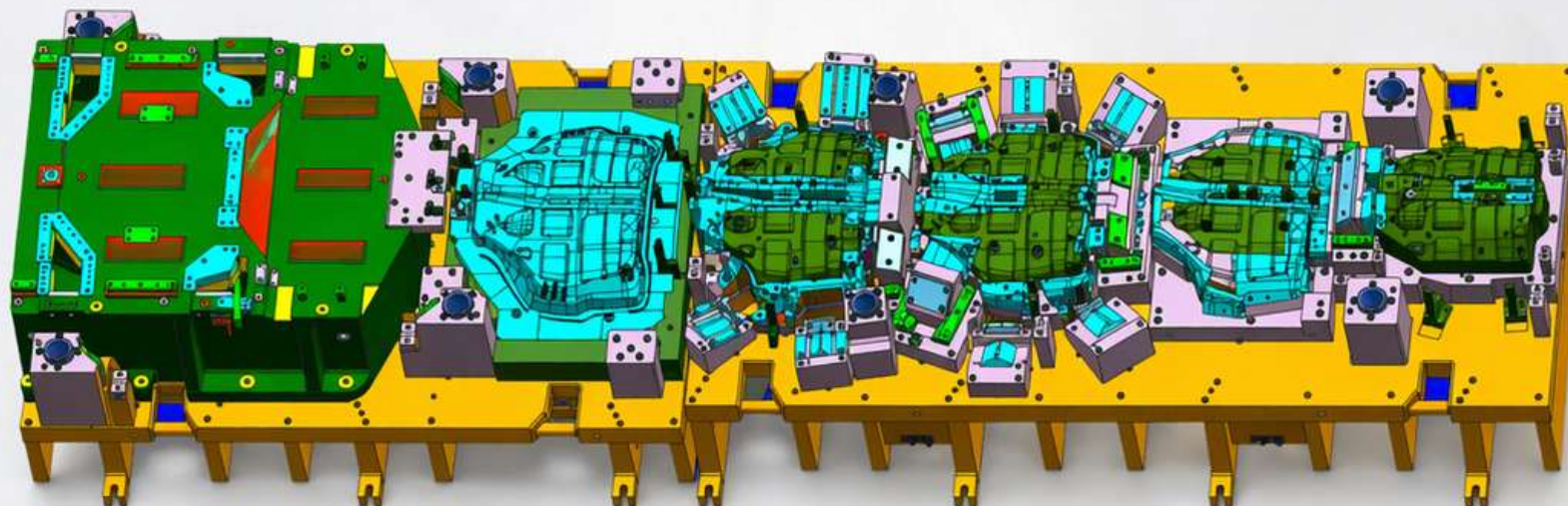
Multi-station integrated design
多工序一体化设计



High precision locating & guiding system
高精度定位与导向系统



Optimized for high efficiency & stability
高效率与稳定性优化



Optimized Material Flow Design for Smooth Progression
优化材料流动设计



Accurate Simulation & Interference Check
精准模拟与干涉检查



Modular Design for Easy Maintenance
模块化设计，便于维护



High Strength & Wear-resistant Components
高强度耐磨零件



12 WEEKS

Rapid Tooling Design & Development



80% PASS RATE BY T0

High Quality from First Tryout



High EFFICIENCY

Coil Fed Transfer Production



Reliable PERFORMANCE

Stable Process, Consistent Quality



MLD

Engineering Capability

ADVANCED DIE DESIGN. PRECISE ENGINEERING. RELIABLE SOLUTIONS.

We leverage advanced 3D design, analysis and simulation to deliver high-precision, efficient, and reliable progressive die solutions from concept to production.



Advanced Design

3D design, analysis and simulation to optimize performance.



Precise Engineering

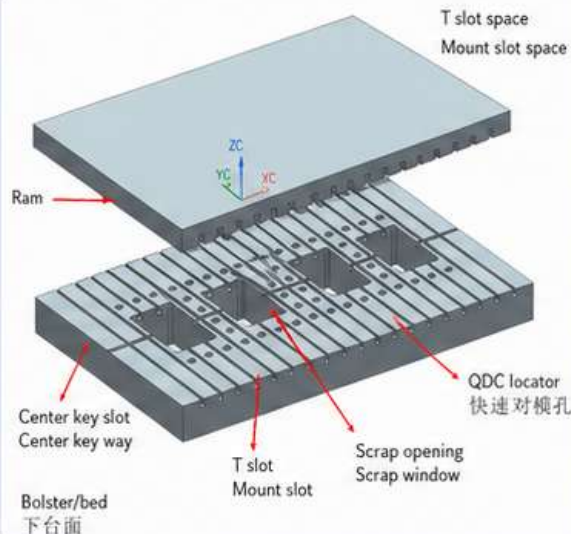
High precision tooling and process control for consistent quality.



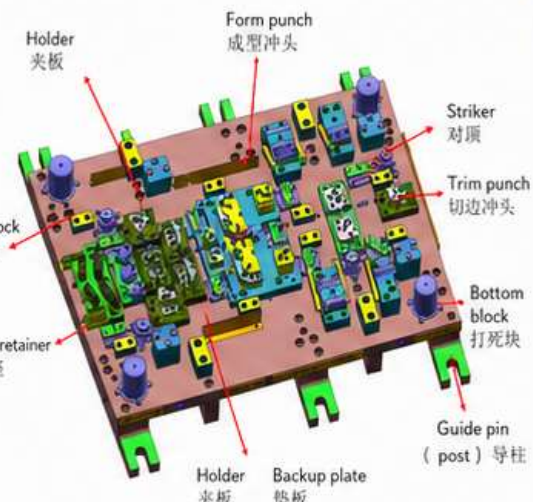
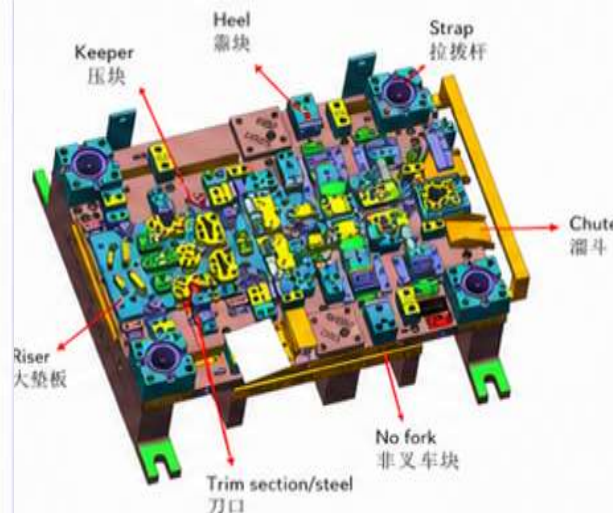
Reliable Solutions

Engineered for durability, efficiency and long-term reliability.

1 PRESS / BOLSTER STRUCTURE

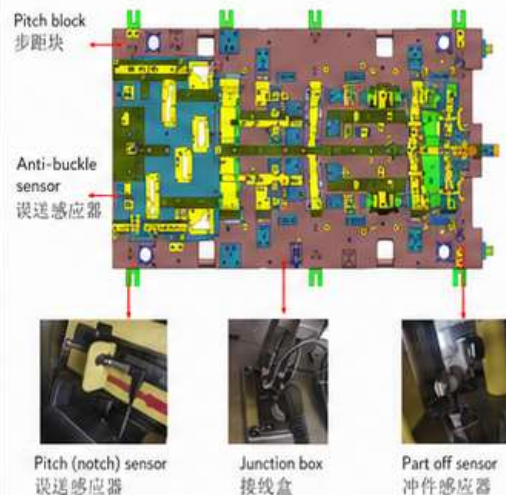


2 DIE ASSEMBLY EXAMPLES

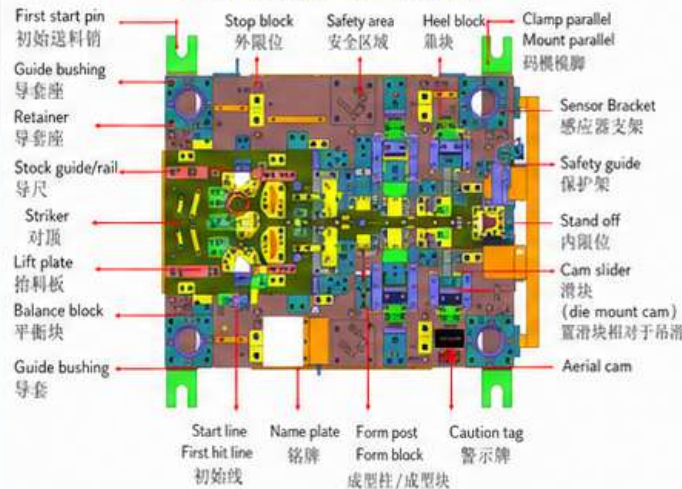


3 TYPICAL TOOL LAYOUTS

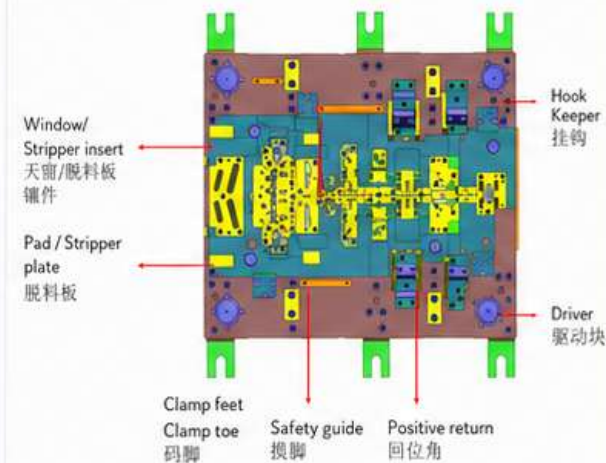
PILOT / SENSOR LAYOUT



PROGRESSIVE DIE LAYOUT (TOP VIEW)



STRIPPER & SUPPORT LAYOUT (BOTTOM VIEW)



HIGH-STRENGTH STEEL APPLICATION & PROCESS DESIGN

MATERIAL CLASSIFICATION | BIW APPLICATION | FORMING FEASIBILITY



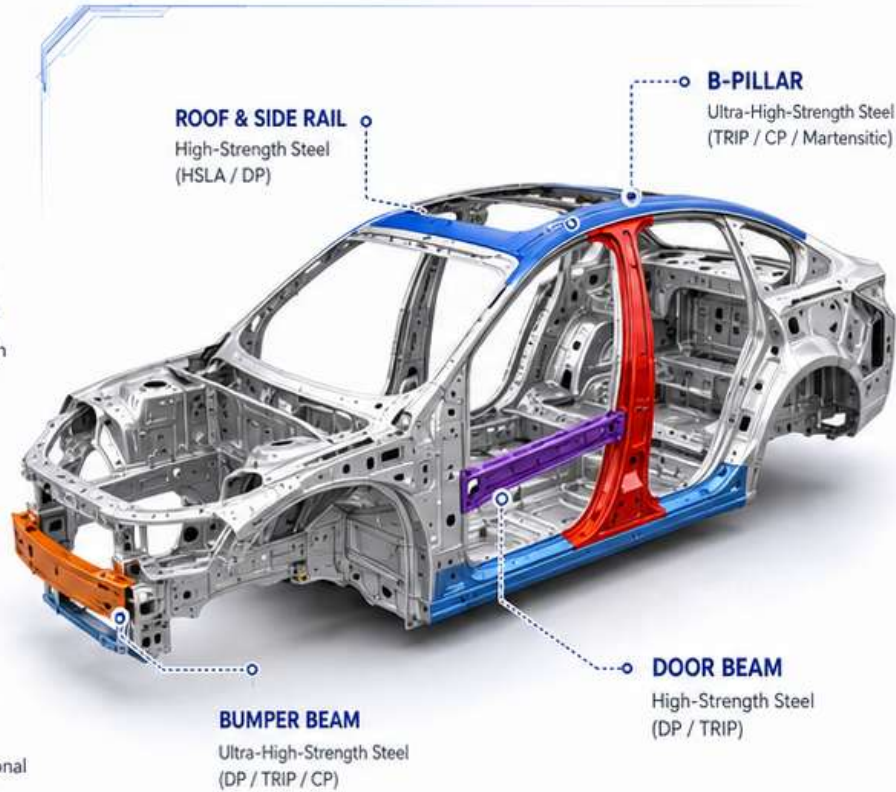
MLD

ENGINEERING EXCELLENCE.
DRIVING INNOVATION.

STRENGTH TREND

800 MPa+
Ultra-High Strength

270 MPa
Conventional Strength



1 CONVENTIONAL STEEL



Low Carbon Steel
低碳钢



IF Steel
超低碳钢

2 HIGH-STRENGTH STEEL



HSLA
高强度低合金钢



BH
烘烤硬化钢



C-Mn
碳锰钢



DP
双相钢

3 ULTRA-HIGH-STRENGTH STEEL



TRIP
相变诱导塑性钢



CP
复相钢



TWIP
孪晶诱导塑性钢



Martensitic
马氏体钢

BIW APPLICATION EXAMPLES

OUTER PANEL



Low Carbon / IF Steel

STRUCTURAL BRACKET



HSLA / DP Steel

REINFORCEMENT PANEL



DP / TRIP Steel

DOOR BEAM



DP / TRIP Steel

BUMPER BEAM



DP / TRIP / CP Steel

B-PILLAR PART



TRIP / CP / Martensitic Steel



MATERIAL SELECTION

Optimal grade selection for strength, weight & cost balance.



FORMING FEASIBILITY

CAE-driven forming analysis to ensure shape feasibility.



SPRINGBACK CONTROL

Accurate springback prediction and compensation strategy.



PROCESS DEVELOPMENT

Robust process design for consistent part quality.



TOOLING OPTIMIZATION

Tool design refinement for durability and performance.

From material behavior to tooling strategy, MLD supports robust development for high-strength automotive structural parts.



INNOVATE | DESIGN | ENGINEER | DELIVER

BUILDING STRONGER STRUCTURES. DELIVERING GREATER VALUE.

BIW B-PILLAR DESIGN & SIMULATION

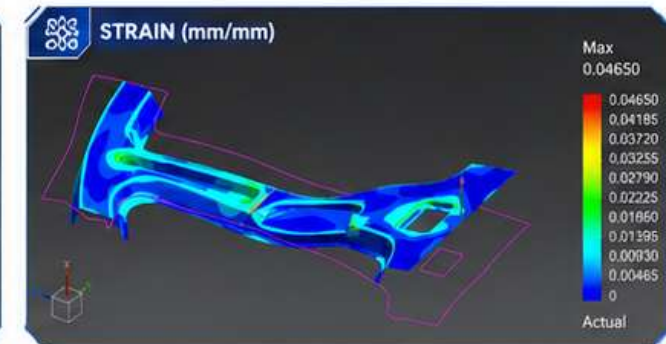
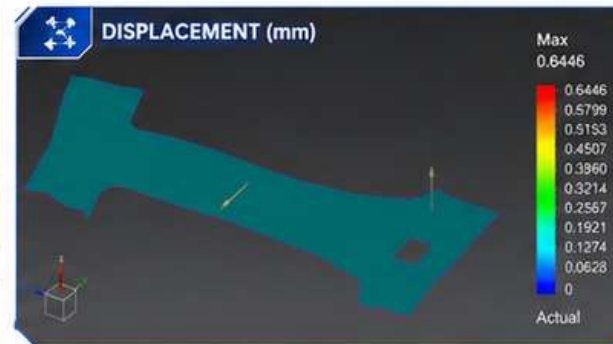
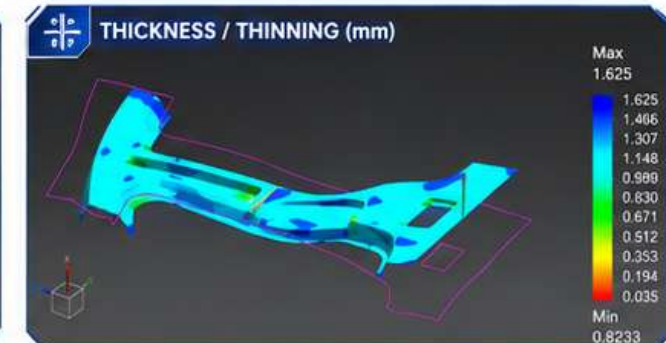
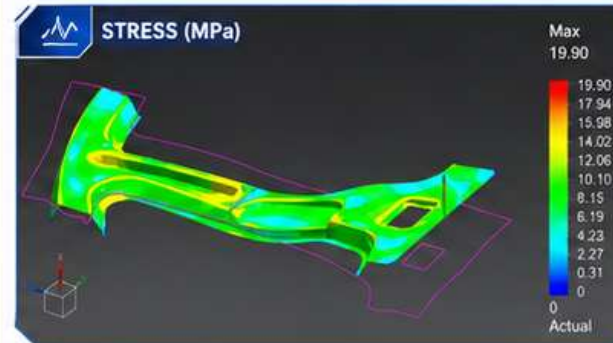
PRODUCT STRUCTURE | CAE ANALYSIS | ENGINEERING DEVELOPMENT



MLD

ENGINEERING EXCELLENCE.
DRIVING INNOVATION.

CAE SIMULATION & ANALYSIS



BIW STRUCTURAL PART | B-PILLAR

B-PILLAR PRODUCT DESIGN

BIW B-PILLAR
(REV02)



BIW B-PILLAR
(FINAL DESIGN)



CAE VALIDATION

Verified structural performance through multi-physics analysis.



SURFACE DEVELOPMENT

Class-A quality surface for manufacturability and assembly.



FORMING FEASIBILITY

Optimized geometry ensures robust forming performance.



MANUFACTURING-ORIENTED DESIGN

Designed for efficient production and process reliability.

CATIA



3DEXPERIENCE

NX

SIEMENS
NX

DIGITAL INDUSTRIES
SOFTWARE



INNOVATE | DESIGN | ENGINEER | DELIVER

BUILDING STRONGER STRUCTURES. DELIVERING GREATER VALUE.



BIW A-PILLAR DESIGN & SIMULATION

PRODUCT STRUCTURE | CAE ANALYSIS | ENGINEERING DEVELOPMENT



MLD ENGINEERING EXCELLENCE.
DRIVING INNOVATION.



BIW STRUCTURAL PART | A-PILLAR

A-PILLAR PRODUCT DESIGN

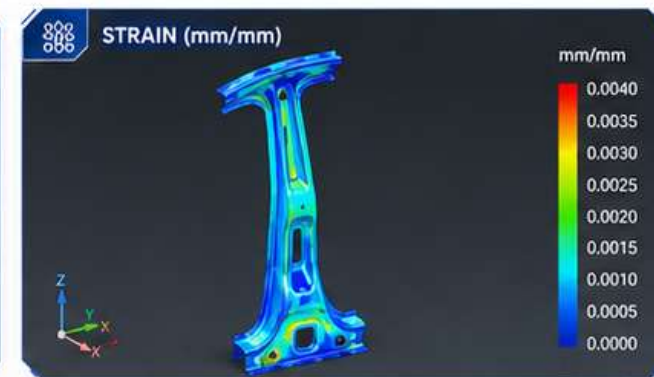
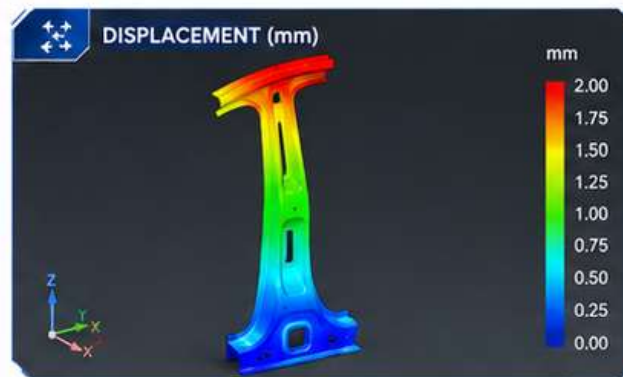
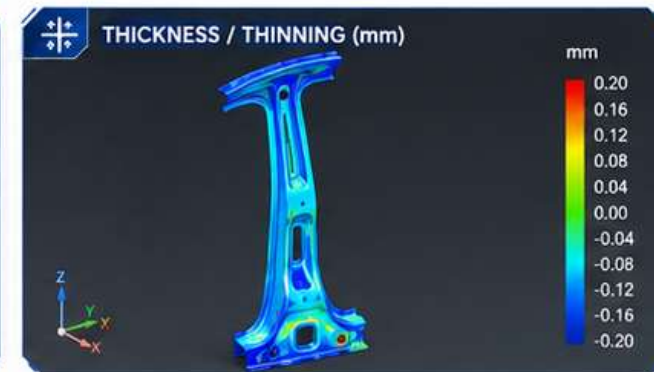
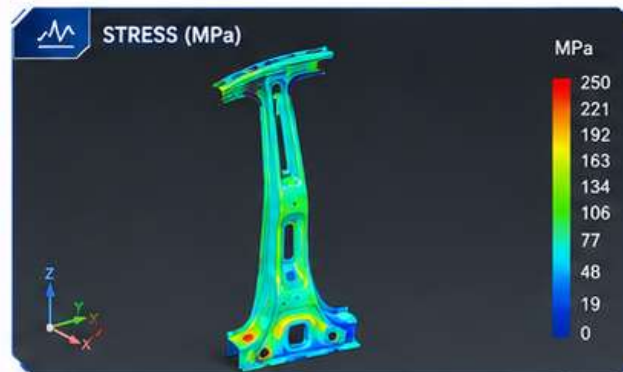
BIW A-PILLAR
(REV02)



BIW A-PILLAR
(FINAL DESIGN)



CAE SIMULATION & ANALYSIS



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INNOVATE | DESIGN | ENGINEER | DELIVER

BUILDING STRONGER STRUCTURES. DELIVERING GREATER VALUE.

TYPICAL PRODUCT PROCESS CASE STUDY

REAR LONGITUDINAL BEAM
UPPER & LOWER



MLD

PRECISE ENGINEERING
CREATING FUTURE

Product Process Case Study / Forming Analysis & Process Design

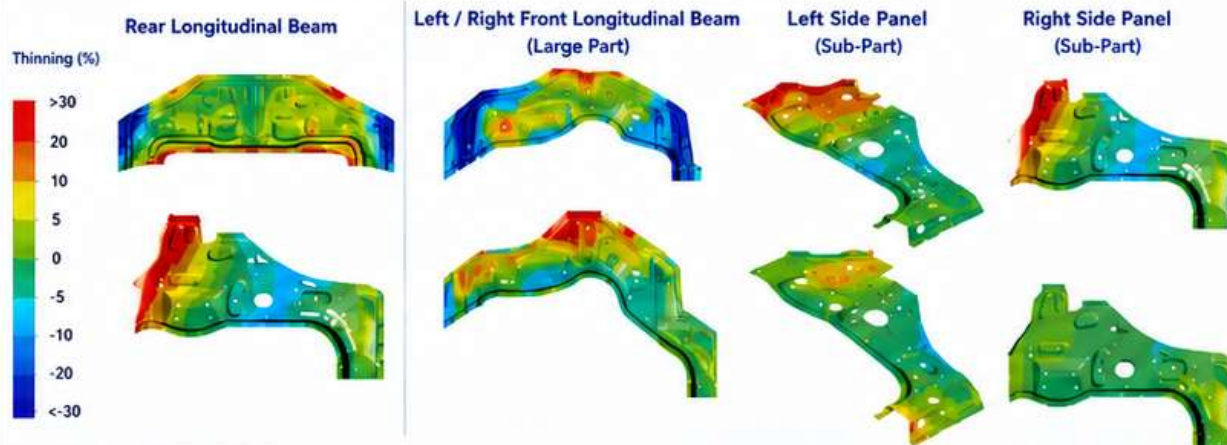
1 PROCESS DESIGN

Left / Right Front Longitudinal Beam Forming Process

① Blanking → ② Reshaping → ③ Forming → ④ Trimming / Piercing → ⑤ Final Part



3 SIMULATION RESULTS

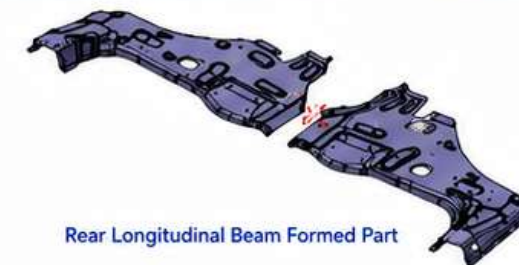


Note: Colors indicate thickness variation. Red regions represent thinning, blue regions represent thickening. Please pay attention to thinning areas.

2 MATERIAL & FORMABILITY

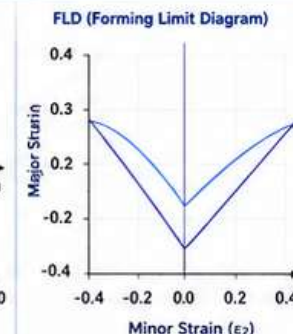
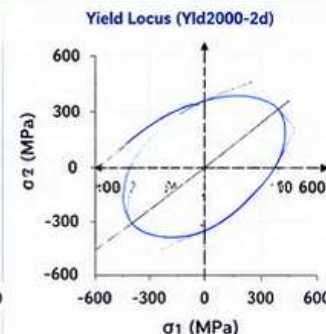
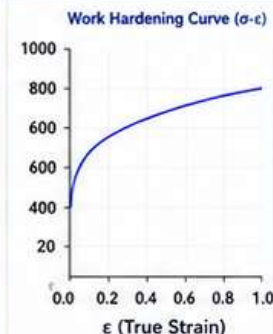
MATERIAL INFORMATION

- Material: DP590
- Thickness: 2.0 mm
- Yield Strength (σ_y): 390 MPa
- Tensile Strength (σ_b): 642.3 MPa



Rear Longitudinal Beam Formed Part

FORMABILITY ANALYSIS & MATERIAL CURVES



KEY PARAMETERS

σ_0 (MPa)	390.0
n	0.154
K (MPa)	955.0
$\bar{r}$	0.727
r_0	0.8
r_{45}	0.8
r_{45}	1.1
$r_{0,0^\circ}$	0.177
$R_{0,0^\circ}$	0.144
$R_{0,90^\circ}$	727.7
S_0	1.1
M	6
Bias	1

4 TECHNICAL HIGHLIGHTS



High Process Feasibility

Multi-stage forming with good material flow control ensures process feasibility.



Excellent Formability

DP590 high strength with optimized formability parameters, good anti-cracking capability.



Risk Controllable

FLD & thinning analysis accurately identify risk areas. Process design avoids cracking risks.



High Surface Quality

Reasonable die design and pressure control minimize springback and improve surface quality.



Efficient & Cost-Effective

Clean process, structure optimized, improved efficiency, reduced cost.



INNOVATE | DESIGN | ENGINEER | DELIVER

BUILDING STRONGER STRUCTURES. DELIVERING GREATER VALUE.

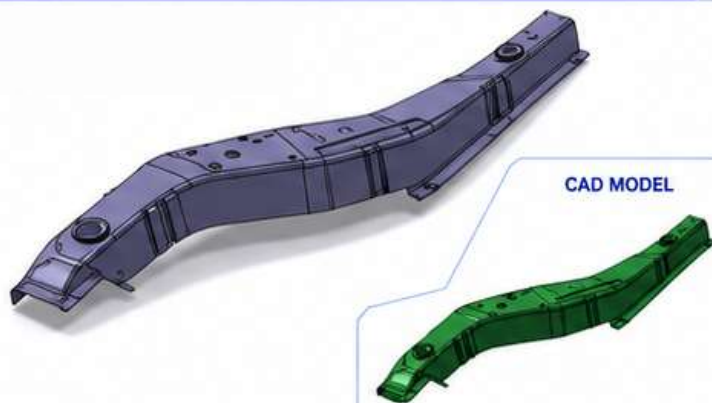
HIGH-STRENGTH STEEL CASE STUDY & PROCESS DEVELOPMENT

PART OVERVIEW | MATERIAL DATA | FORMING PROCESS | SIMULATION RESULTS



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ENGINEERING EXCELLENCE.
DRIVING INNOVATION.



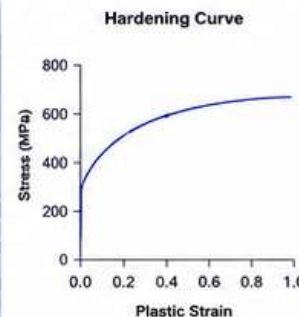
CAD MODEL

1 PROJECT OVERVIEW

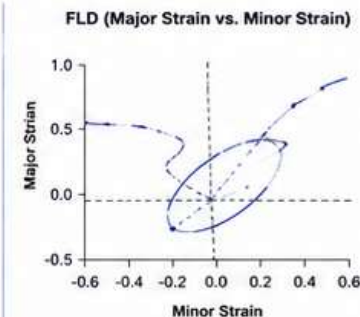
Case Part	Left / Right Front Pillar Inner Panel
Material	SAPH440
Thickness	2.0 mm
Yield Strength (σ_y)	375 MPa
Tensile Strength (σ_b)	514 MPa
Elongation	18%
Hardening	Work Hardening
Process	Hot Stamping Total

The process plan was defined based on part geometry, dimensional accuracy requirements, functional performance, and material characteristics.

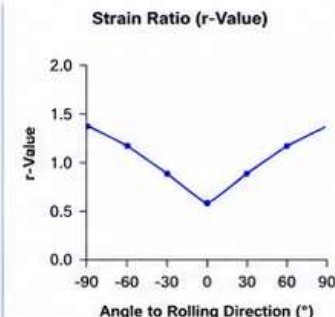
2 MATERIAL / FORMABILITY ANALYSIS



$\sigma_y = 375 \text{ MPa}$ $\sigma_b = 514 \text{ MPa}$



$r_0 = 0.165$

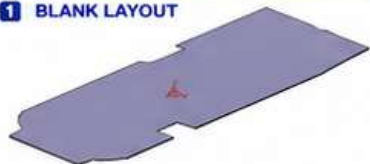


$r_{45} = 0.134$

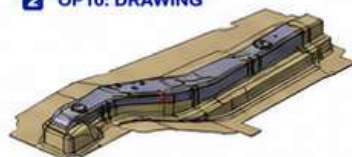
$r_{90} = 0.186$

3 FORMING PROCESS DEVELOPMENT

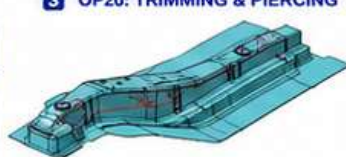
1 BLANK LAYOUT



2 OP10: DRAWING



3 OP20: TRIMMING & PIERCING



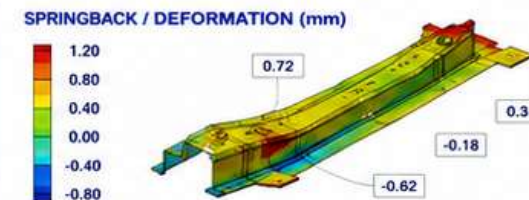
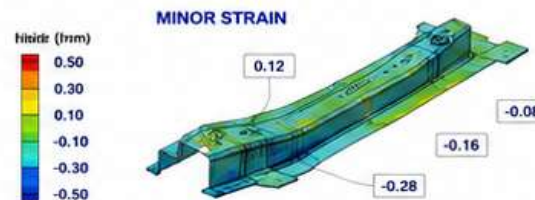
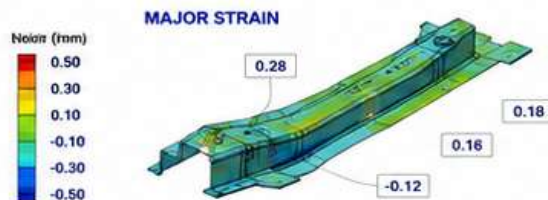
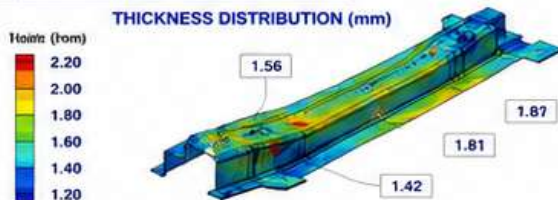
4 OP30: FLANGING & PIERCING



5 OP40: FINAL FLANGING / RESTRAINING



4 SIMULATION RESULTS (CAE)



5 KEY CAPABILITIES & VALUE



THICKNESS DESIGN

Robust forming process development for high-strength steel components.



FORMABILITY CONTROL

Optimization using FLD, r-value, and material data.



ACCURACY & QUALITY

Dimensional control through CAE and tryout validation.



SPRINGBACK / COFORMATION

CAE-driven springback prediction and die compensation.



MASS PRODUCTION READINESS

Reliable, repeatable process designed for mass production.

From material analysis to process validation, MLD develops robust forming solutions for high-strength structural components.



INNOVATE | DESIGN | ENGINEER | DELIVER

BUILDING STRONGER STRUCTURES. DELIVERING GREATER VALUE.

HIGH-STRENGTH STEEL CASE STUDY & PROCESS DEVELOPMENT

PART OVERVIEW | MATERIAL DATA | FORMING PROCESS | SIMULATION RESULTS

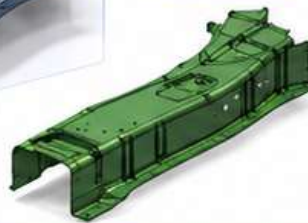


MLD

ENGINEERING EXCELLENCE.
DRIVING INNOVATION.



PART CAD MODEL

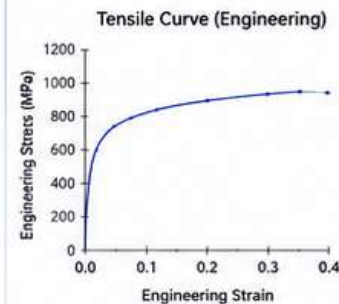


1 PROJECT OVERVIEW

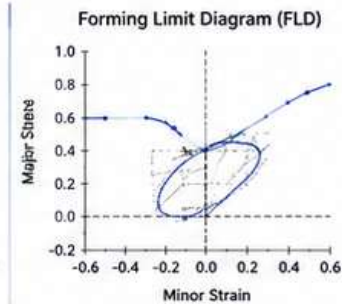
	Case Part	Left / Right Front Longitudinal Beam
	Material	DP590
	Thickness	1.8 mm
	Yield Strength (σ_s)	390 MPa
	Tensile Strength (σ_b)	642.3 MPa

The process plan was defined based on part geometry, dimensional accuracy requirements, functional performance, and material characteristics.

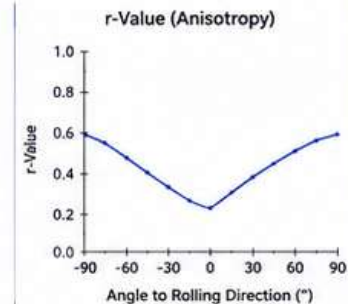
2 MATERIAL / FORMABILITY ANALYSIS



$\sigma_s = 390 \text{ MPa}$ $\sigma_b = 642.3 \text{ MPa}$



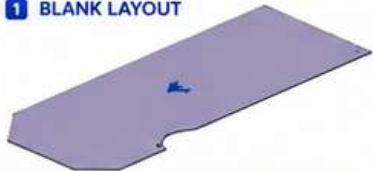
$r_0 = 1.54$



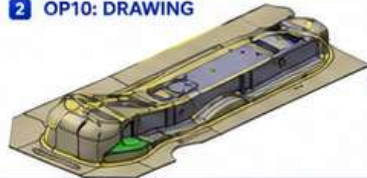
$r_{45} = 0.164$ $r_{90} = 0.184$

3 FORMING PROCESS DEVELOPMENT

1 BLANK LAYOUT



2 OP10: DRAWING



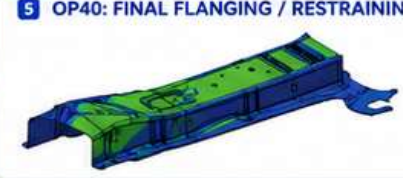
3 OP20: TRIMMING & PIERCING



4 OP30: FLANGING & PIERCING

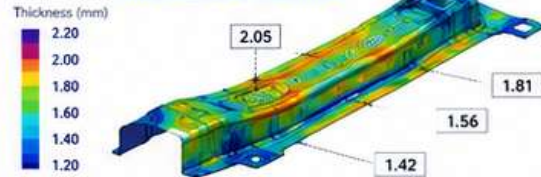


5 OP40: FINAL FLANGING / RESTRAINING

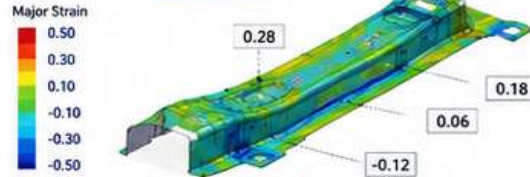


4 SIMULATION RESULTS (CAE)

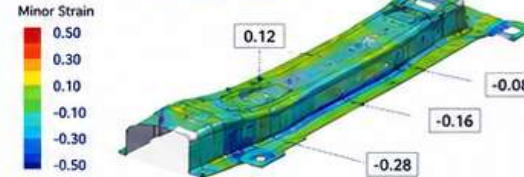
THICKNESS DISTRIBUTION



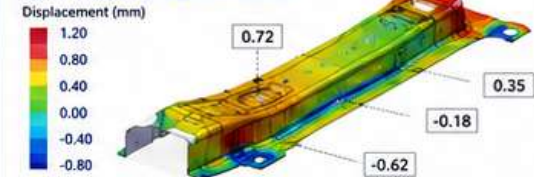
MAJOR STRAIN



MINOR STRAIN



SPRINGBACK / DEFORMATION



5 KEY CAPABILITIES & VALUE



THICKNESS FEASIBILITY

Validated through CAE and tryout to ensure manufacturability.



FORMING STABILITY

Optimized draw design and material flow to prevent defects.



DIMENSIONAL CONTROL

Precise trimming, piercing, and flanging for tight tolerance.



SPRINGBACK MANAGEMENT

Accurate compensation strategy to ensure assembly fit.



PRODUCTION READINESS

Robust and repeatable process designed for mass production.

From material analysis to process validation, MLD develops robust forming solutions for high-strength structural components.



HIGH-STRENGTH STEEL SPRINGBACK CONTROL CASE STUDY & PROCESS OPTIMIZATION



ENGINEERING EXCELLENCE.
DRIVING INNOVATION.

PART OVERVIEW | ROOT CAUSE ANALYSIS | KEY FACTORS | PROCESS OPTIMIZATION | VALIDATION RESULTS

1 PROJECT OVERVIEW

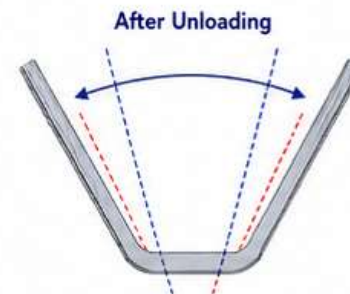
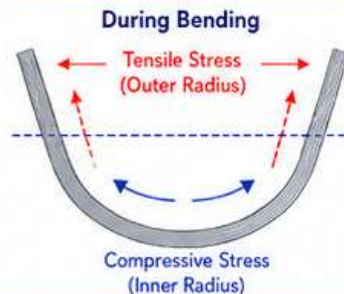


CAD MODEL



- Case Part:** Front Longitudinal Beam
- Material:** DP590 High-Strength Steel
- Thickness:** 1.8 mm
- Main Issue:** Springback / Angle Change
- Target:** Stable dimensional accuracy and assembly fit

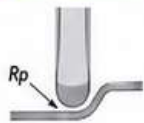
2 SPRINGBACK MECHANISM



During bending, tensile stress develops at the outer radius while compressive stress forms at the inner radius. After unloading, elastic recovery releases the residual stress, causing a change in angle and springback.

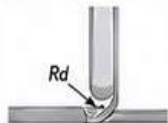
3 KEY INFLUENCING FACTORS

1 Punch Radius (R_p)



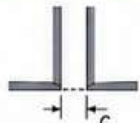
Smaller R_p increases plastic strain and helps reduce springback.

2 Die Radius (R_d)



Larger R_d increases elastic recovery and affects springback.

3 Die Clearance (C)



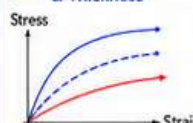
Larger clearance increases angle change and part variation.

4 Part Geometry / Section Height



Section geometry and height strongly affect springback sensitivity and dimensional stability.

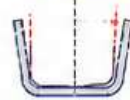
5 Material Strength & Thickness



Higher strength increases elastic recovery; thicker sheet reduces springback.

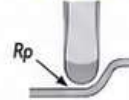
4 PROCESS OPTIMIZATION STRATEGY

1 Die Compensation (Angle Relief)



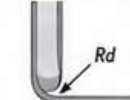
- Add 3°–6° compensation in die design.
- Pre-counteract expected springback.

2 Reduce Punch Radius (R_p)



- Use the smallest R_p allowed by the part geometry.
- Increases plastic strain at bending.

3 Reduce Die Radius (R_d)



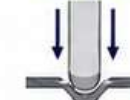
- Minimize R_d to reduce elastic recovery.
- Improves angle prediction.

4 Optimize Die Clearance (C)



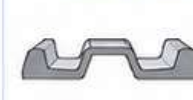
- Tighten clearance within process window.
- Reduces angle change and dimensional variation.

5 Increase Binder / Hold-Down Force



- Apply higher restraining force within limits.
- Reduces material flow and increases angle stability.

6 Secondary Bending & Stiffening Features



- Apply restrike / coining or secondary bend.
- Add beads / ribs on side walls to increase stiffness.

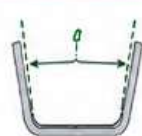
5 VALIDATION RESULTS

Before Optimization



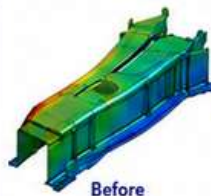
- Flange Angle Deviation: 4.8°
- Dimensional Deviation: ± 2.2 mm
- Springback Distribution: Unstable / Non-uniform

After Optimization

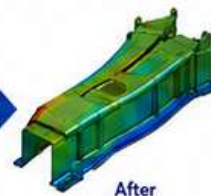


- Flange Angle Deviation: 1.1°
- Dimensional Deviation: ± 0.6 mm
- Springback Distribution: Improved / Stable

Springback Contour (CAE Result)



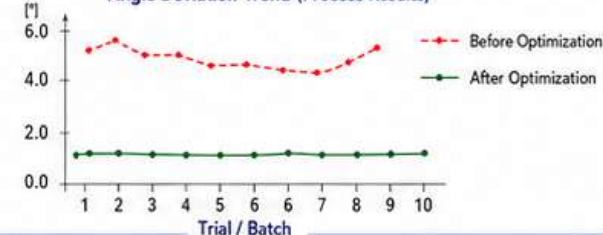
Before



After



Angle Deviation Trend (Process Results)



- ✓ Angle deviation reduced by ~77%
- ✓ Dimensional variation improved by ~73%
- ✓ Significant improvement in forming stability and repeatability

6 MLD VALUE DELIVERY



ROOT-CAUSE ANALYSIS

Deep understanding of springback mechanism and key influencing factors in high-strength steels.



DIE COMPENSATION EXPERTISE

Advanced compensation engineering and optimization to ensure ideal geometry and assembly fit.



CAE-ASSISTED OPTIMIZATION

Validated by advanced CAE to predict springback and shorten development cycles.



DIMENSIONAL CONTROL

Tight tolerance and angle control ensures consistent fit and functional performance.



PRODUCTION READINESS

Robust tooling process with stable springback for mass production and consistent quality.

Through engineering insight and process innovation, MLD delivers robust solutions that control springback and ensure perfect fit.

BUILDING STRONGER STRUCTURES. DELIVERING GREATER VALUE.



SIMULTANEOUS ENGINEERING FOR DESIGN & MANUFACTURING EXCELLENCE

Early feasibility validation, structural optimization and manufacturing readiness for automotive stamping tools

**MLD**

ENGINEERING EXCELLENCE.
DRIVING INNOVATION.

1 WHY SIMULTANEOUS ENGINEERING MATTERS

TRADITIONAL SERIAL DEVELOPMENT

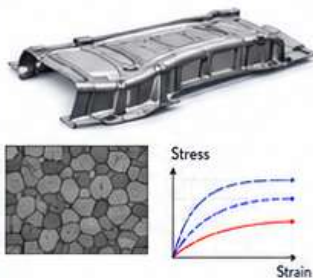


SIMULTANEOUS ENGINEERING



2 CORE SE VALUE IN BIW DEVELOPMENT

MATERIAL DEFINITION & FEASIBILITY



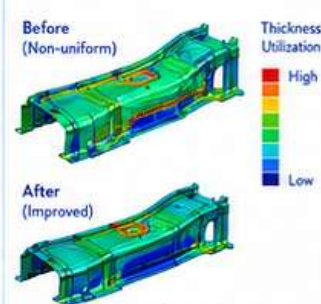
Select optimal steel grade and thickness based on product requirements and formability. Validate material feasibility with CAE and simulation.

STRUCTURAL RATIONALIZATION



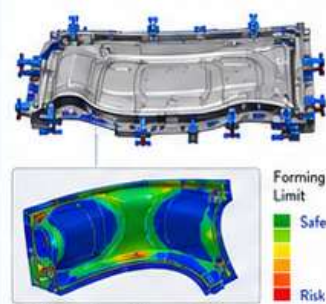
Rationalize part geometry, features and section strategy to ensure structural performance while improving manufacturability.

MATERIAL UTILIZATION OPTIMIZATION



Improve material utilization rate through thickness control, trim line optimization and feature refinement to reduce weight and material cost.

PROCESS VALIDATION FOR LARGE PANELS



Validate formability and process feasibility with CAE (Stamping Simulation). Define process windows and risk mitigation strategies early.

3 ENGINEERING WORKFLOW



4 MLD ENGINEERING VALUE DELIVERY



DELIVERING MEASURABLE RESULTS



Through simultaneous engineering, MLD delivers optimal solutions that balance performance, manufacturability and cost—enabling our customers to build better vehicles, faster.

BUILDING STRONGER STRUCTURES. DELIVERING GREATER VALUE.





Quality Control Capabilities

Precision inspection, certified systems, and full traceability for reliable tooling quality.



Advanced CMM Inspection

Micron-level dimensional verification.



Certified Quality System

ISO 9001:2015 and IATF 16949 aligned processes.



Material & Hardness Verification

Comprehensive inspection before release.



Full Traceability

Complete records and documentation for every component.



MEISTER 8126



MEISTER 153010



20,000+ m²
Factory Area



50+ Sets
Advanced
Inspection Equipment



Certified Quality
ISO 9001:2015
IATF 16949



Reliable Results
Consistent Accuracy.
On-Time Delivery.



Project Management & Customer Support

Comprehensive support from project kickoff through production and beyond.



Dedicated project managers
12/6 project managers



24/7 technical support
AI-assisted service network



Training support
On-site & remote training programs



 **Aligned. Responsive. Results.**



 **From design to delivery. We're with you.**

AI-ENABLED SUPPORT



Claude

Advanced reasoning and analysis



ChatGPT

Smart answers and insights



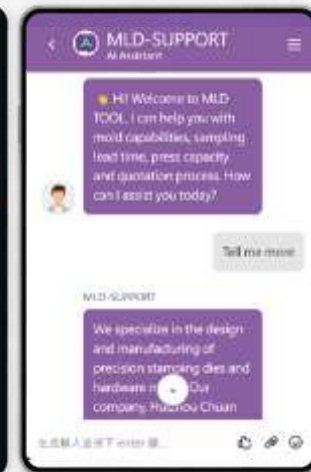
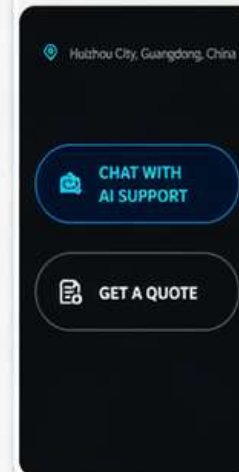
DeepL

Accurate translation and localization

Powering faster, smarter support
with generative AI.



Digital Support Experience
Fast. Convenient. Always available.



Global Customer Coverage



Serving customers across key international markets





MLD

SERVICE & REFERENCE

Trusted by OEMs and Tier 1 / Tier 2 customers worldwide.

OEM



CHRYSLER



CITROËN



NISSAN

LUCID



TIER₁ / TIER₂



Multimatic





MLD

Export & Logistics Management

Reliable global delivery supported by experienced logistics coordination and proactive risk control.



Working with multiple leading freight forwarders
for flexible and efficient global shipment arrangements



On-time delivery supported by
coordinated scheduling, documentation control, and transit monitoring



Industrial-grade export packaging
for heavy tooling and equipment protection



Strong logistics risk response capability
for customs clearance, shipment changes, and urgent delivery support



Global Shipping Coordination

Seamless operations across 40+ countries and regions.



On-Time Delivery Commitment

Reliable schedules and real-time monitoring for dependable delivery.



Fast Response to Logistics Risks

Proactive risk management and rapid support when it matters.



MLD

03

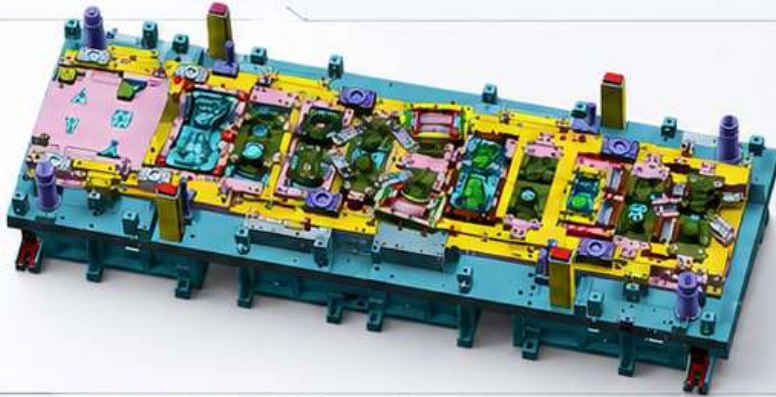
DESIGN SHOWCASES

3D Die Design Capabilities

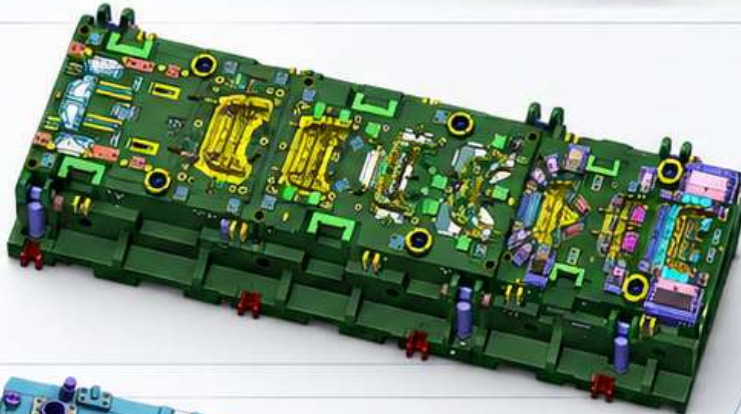
Progressive & Transfer Die Engineering | Multi-station tooling layout



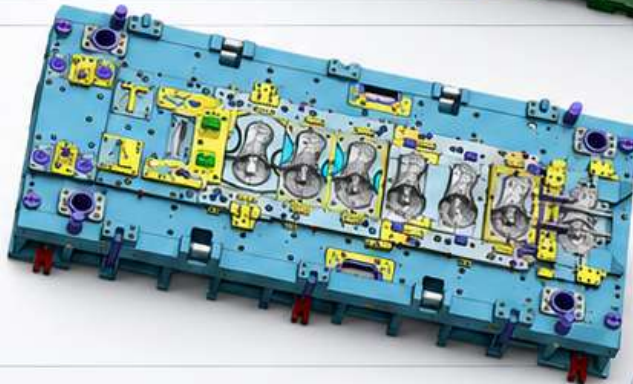
Progressive Die



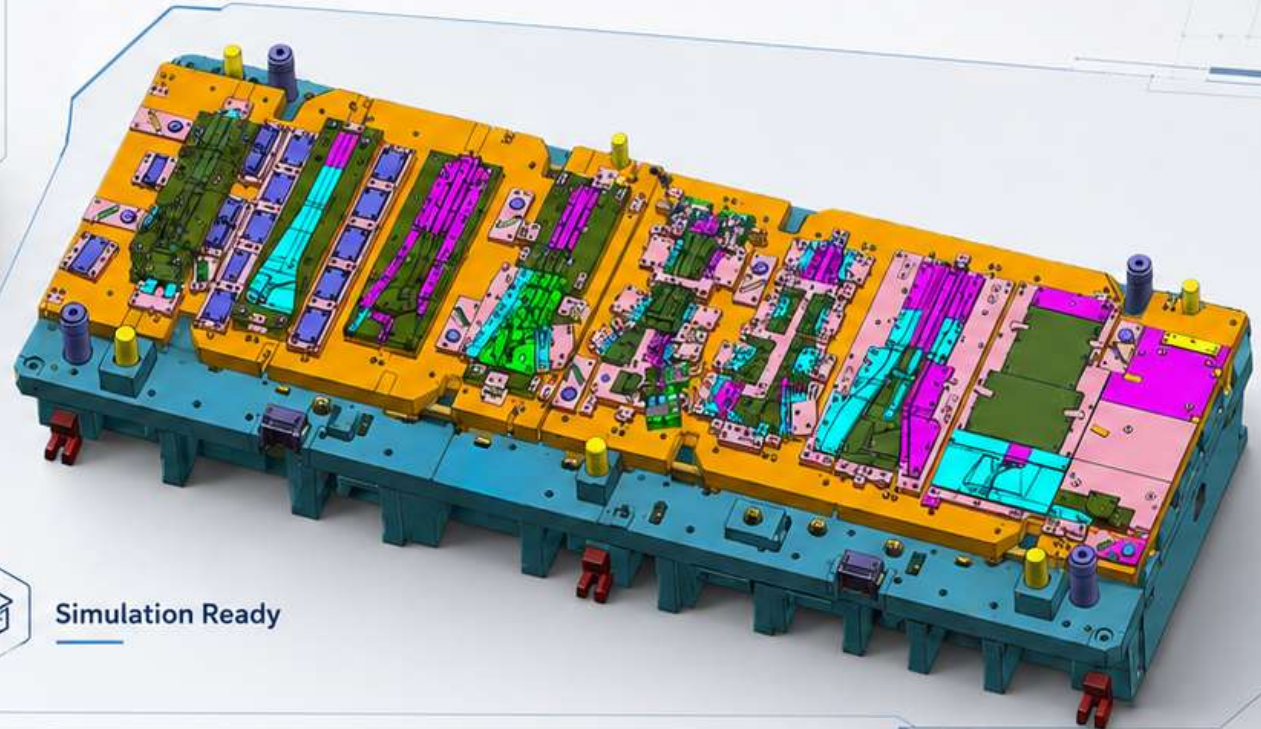
Transfer Die



Multi-station Layout



Simulation Ready



Optimized for
Performance



Precision
& Quality



Efficiency
& Productivity



Engineering Expertise
You Can Trust



ADVANTAGE TOOLING
Driving Precision. Delivering Value.



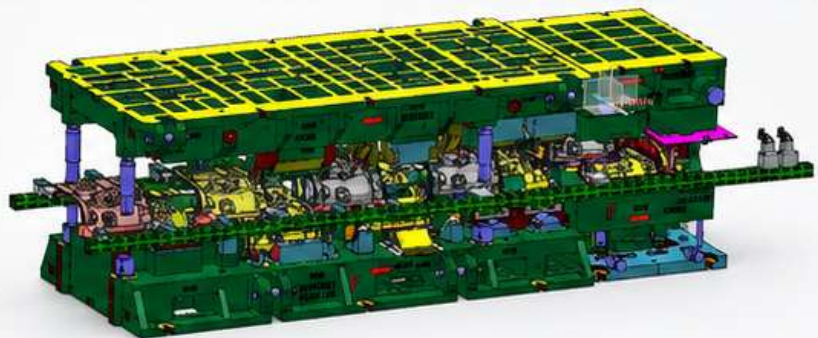
MLD

3D Transfer Simulation System

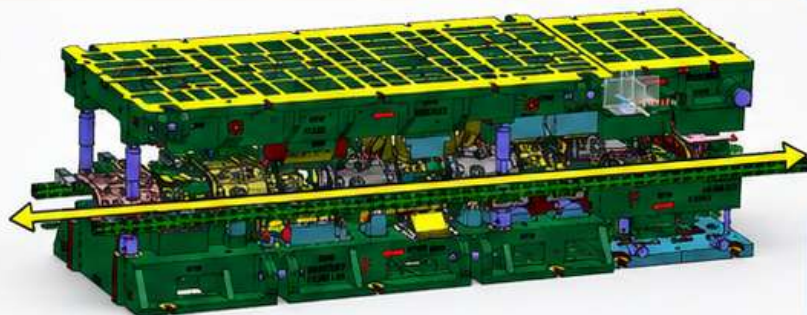
Upper & Lower Die Assembly | Transfer Path Verification



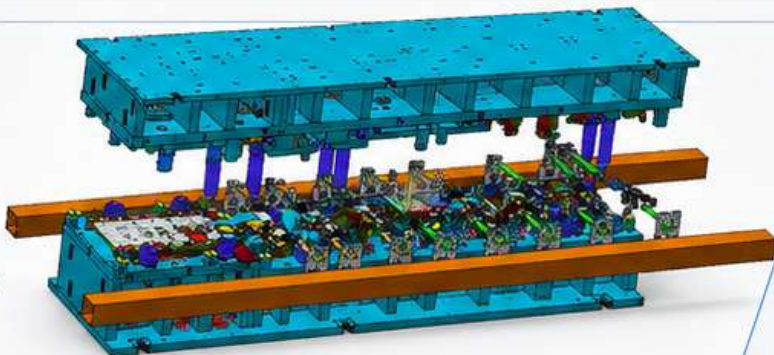
Closed Die Position



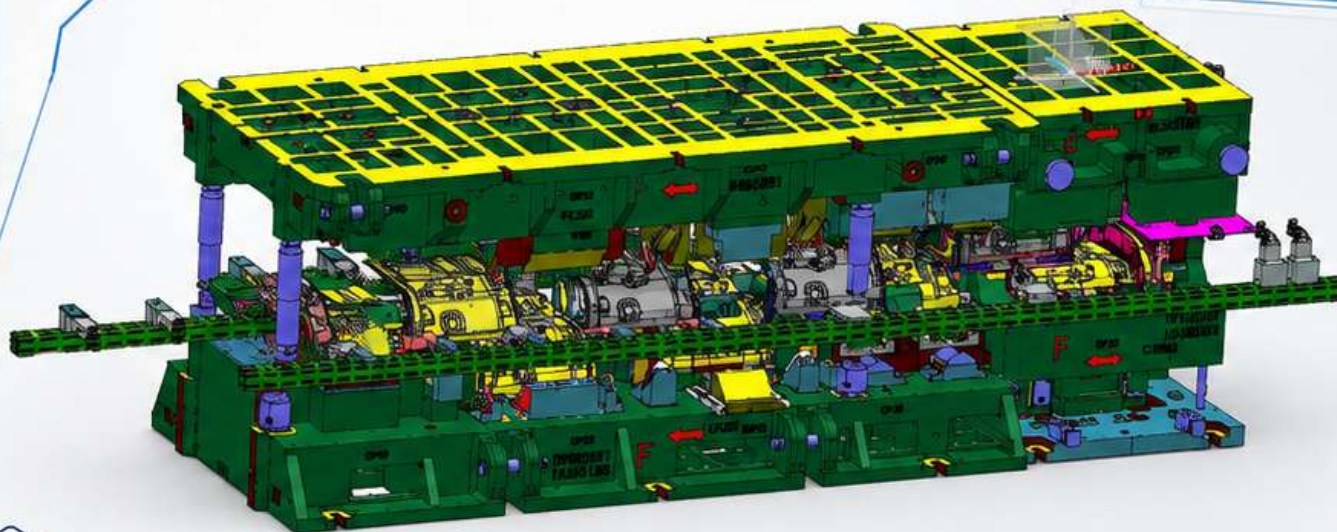
Transfer Rail Path



Open Die Clearance



Simulation Ready



Collision Check



Transfer Timing



Stroke Clearance

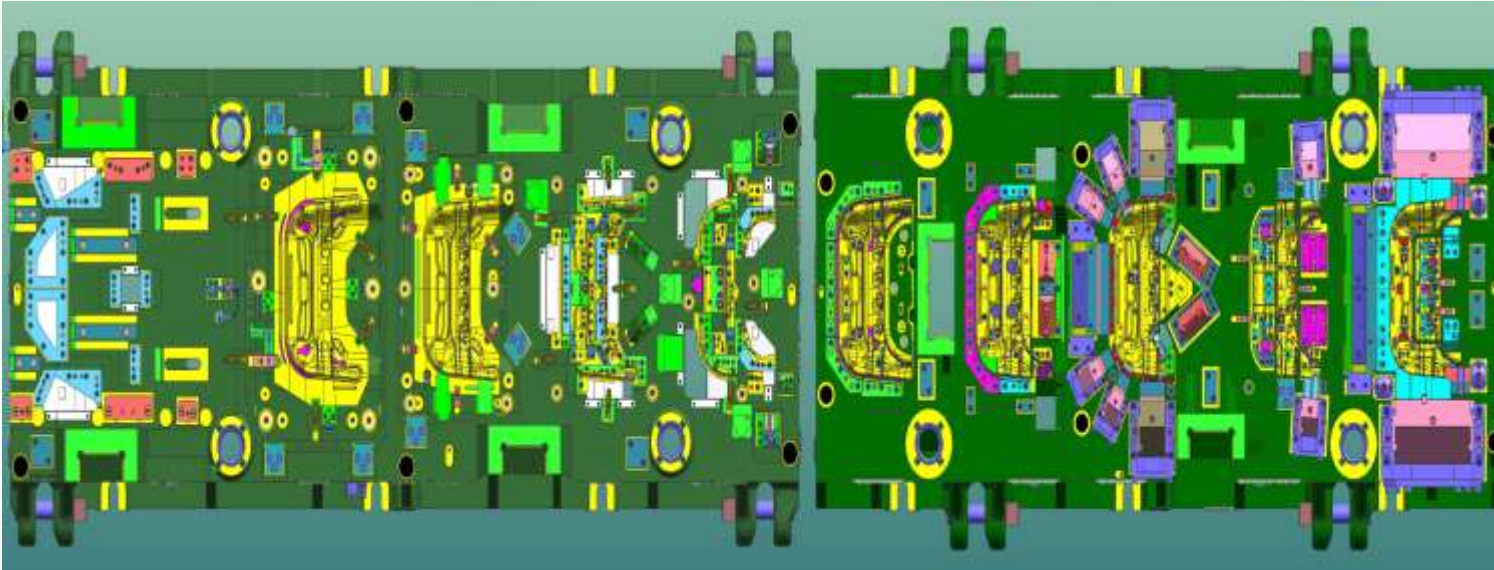
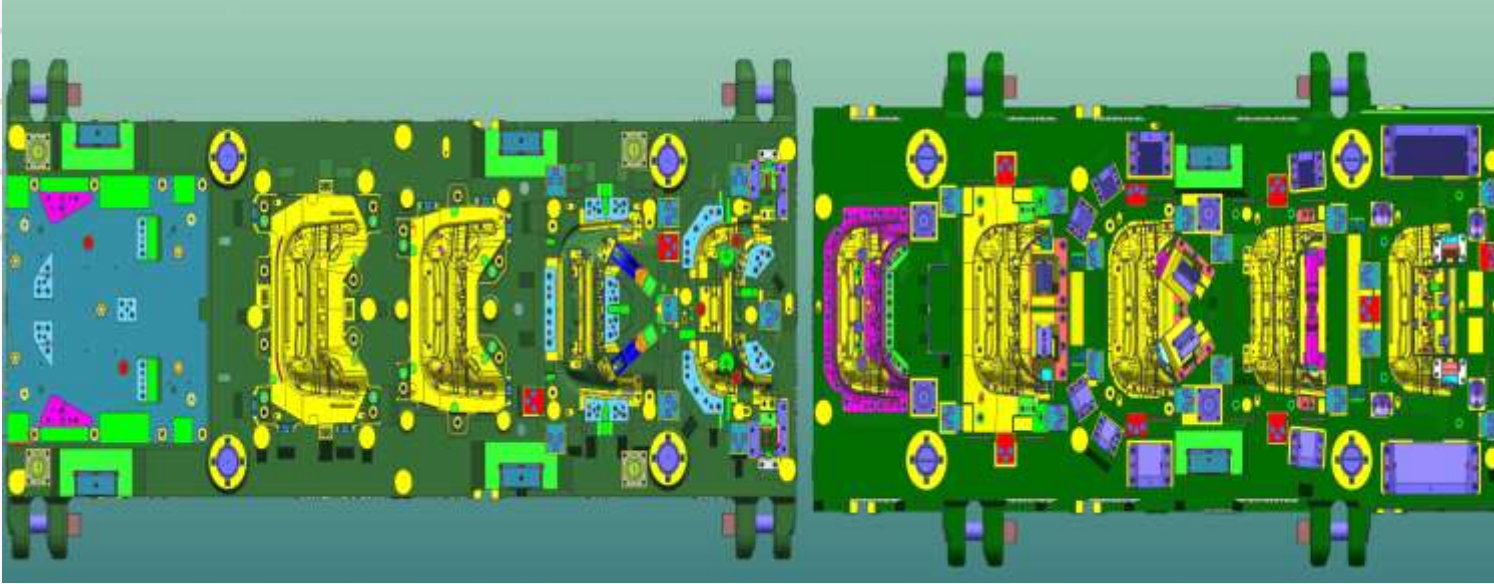


Process Validation

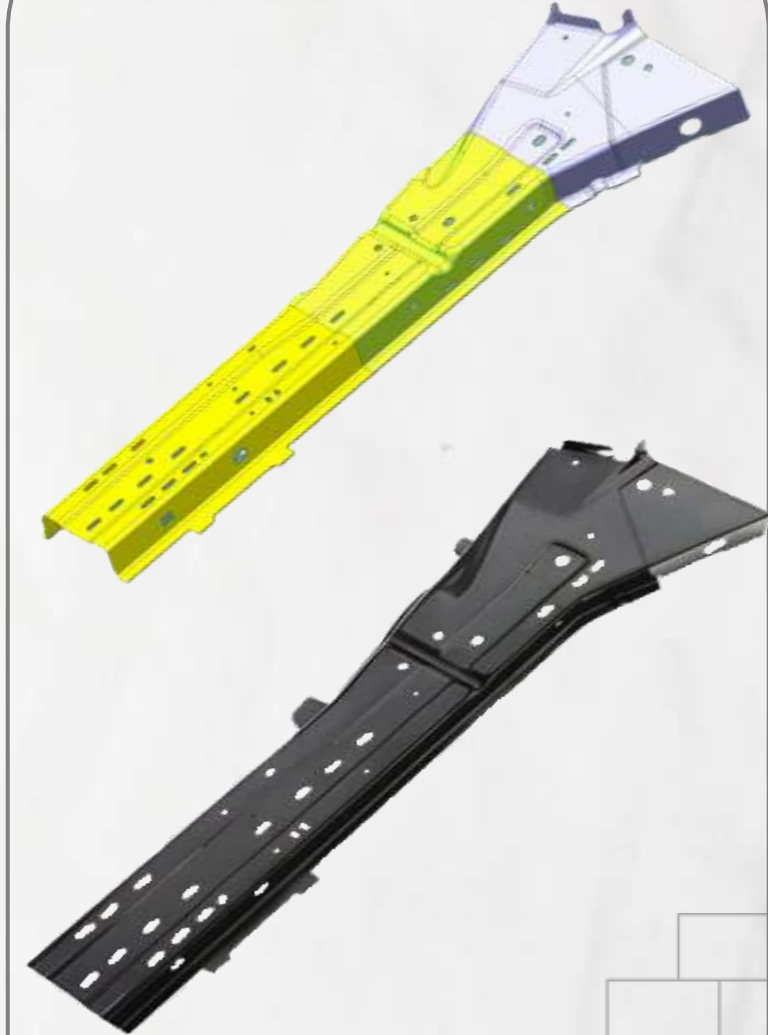
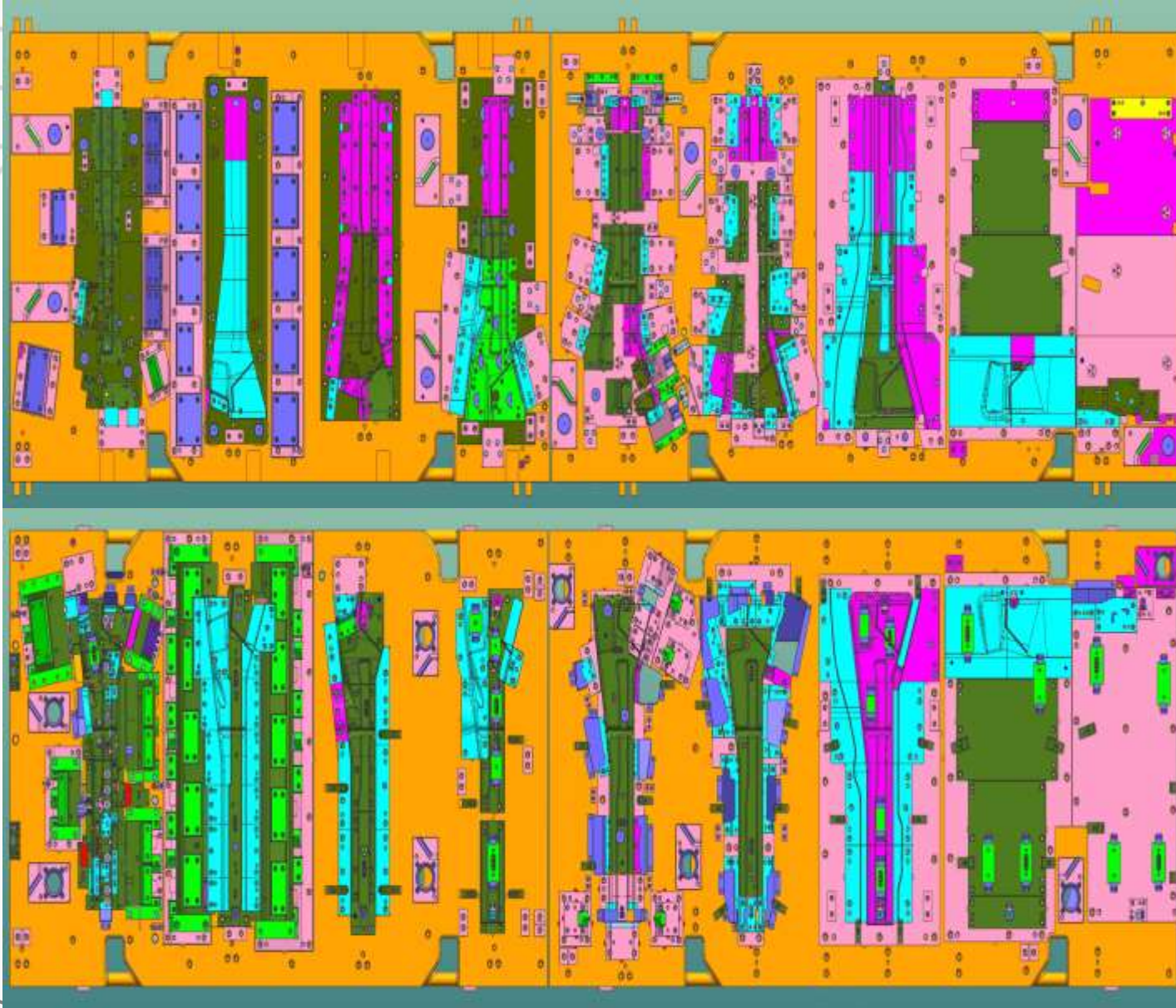


ADVANTAGE TOOLING
Driving Precision. Delivering Value.

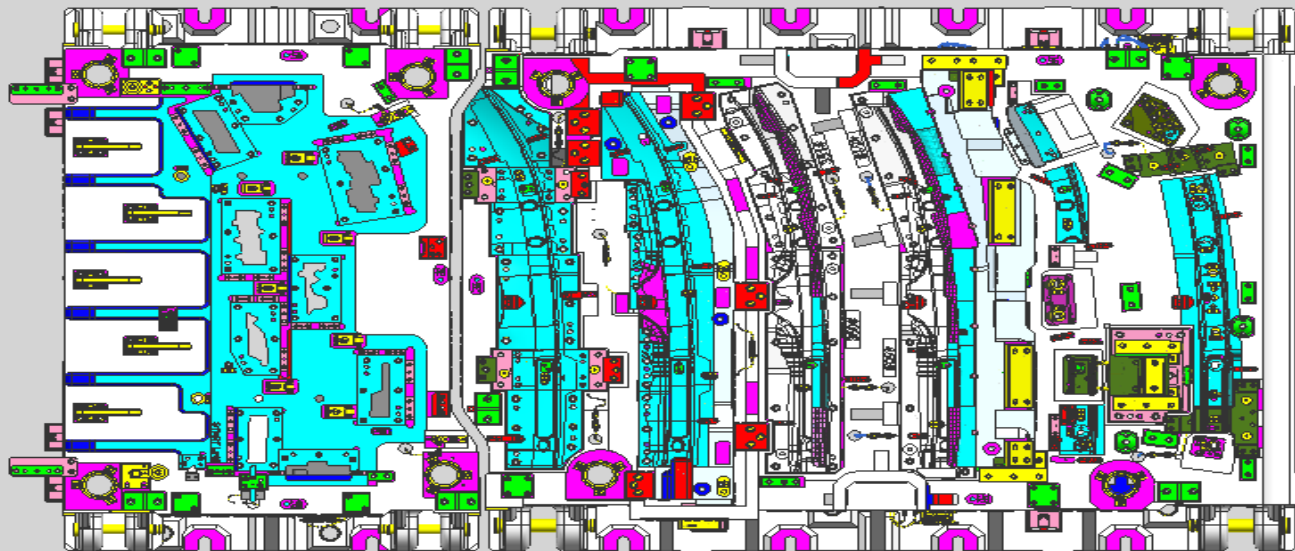
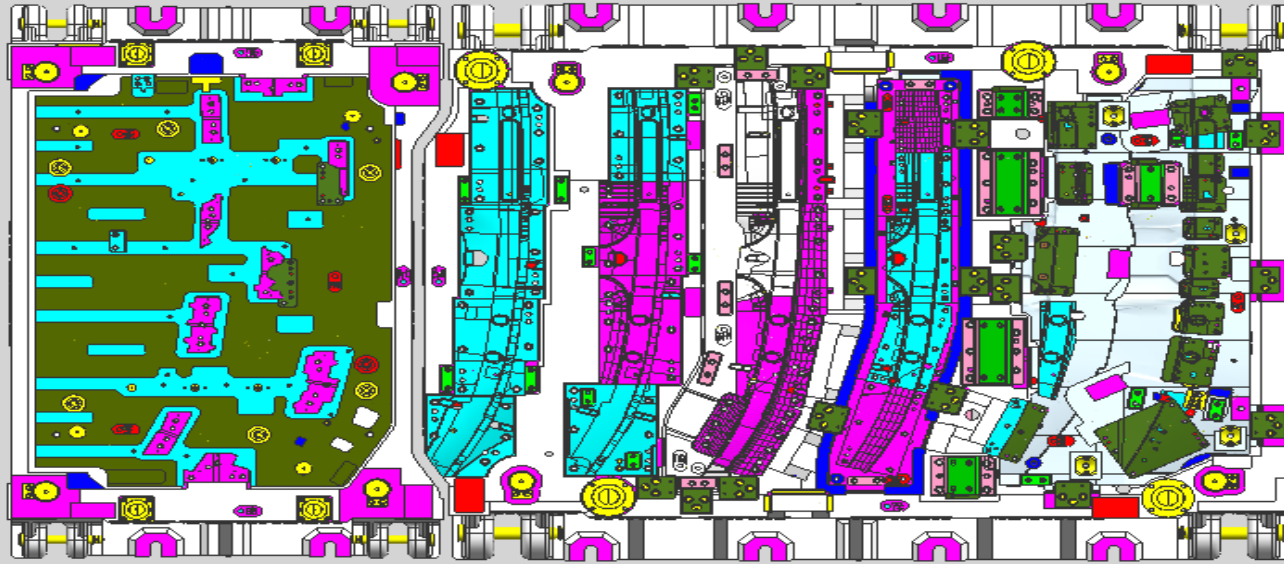
Design Capability - Showcases



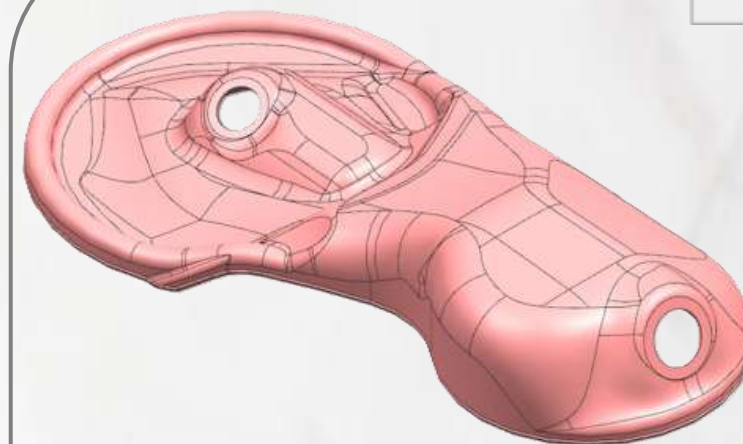
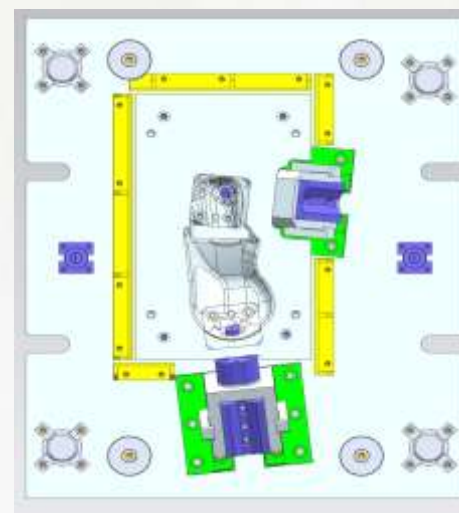
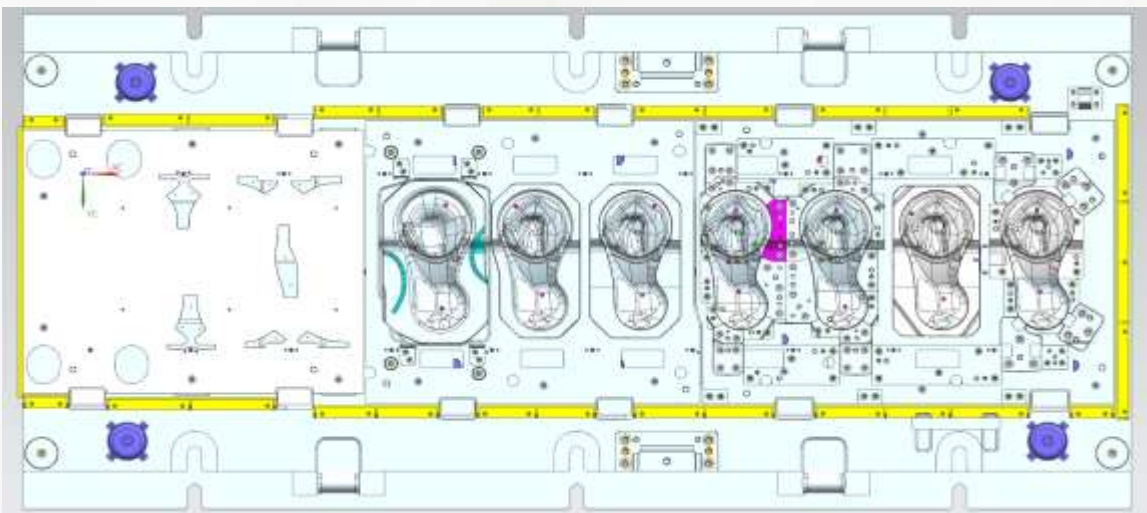
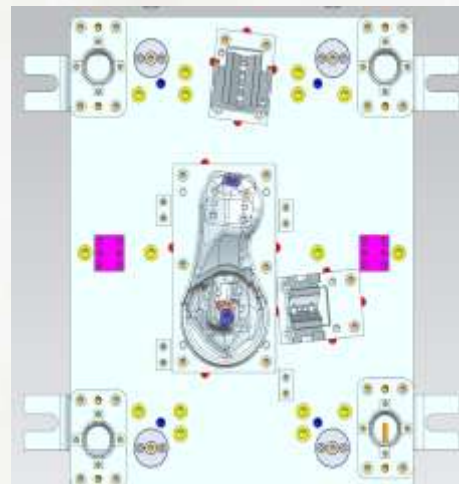
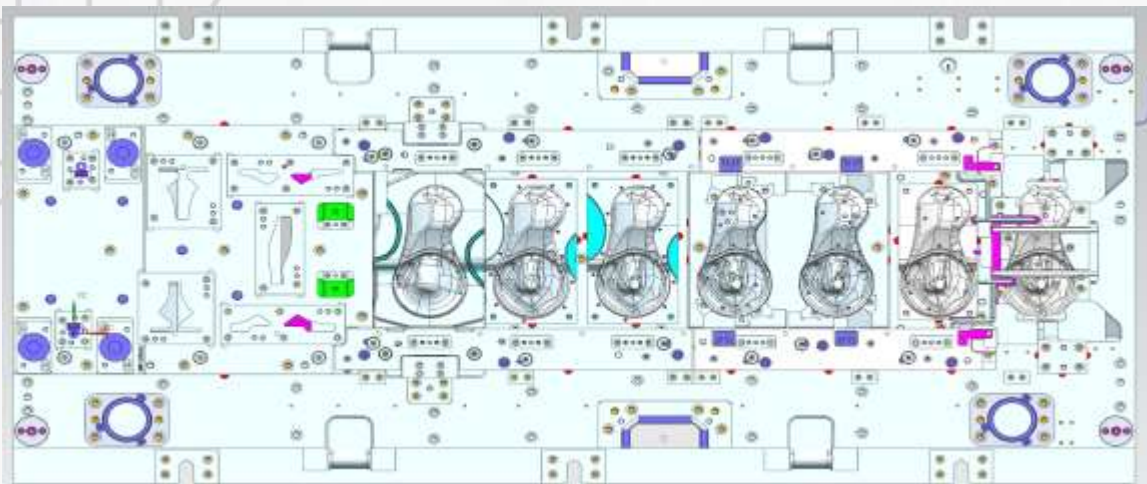
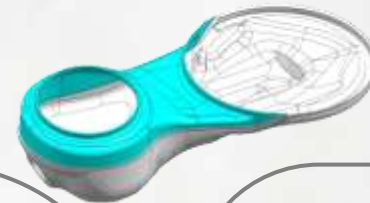
Design Capability - Showcases



Design Capability - Showcases

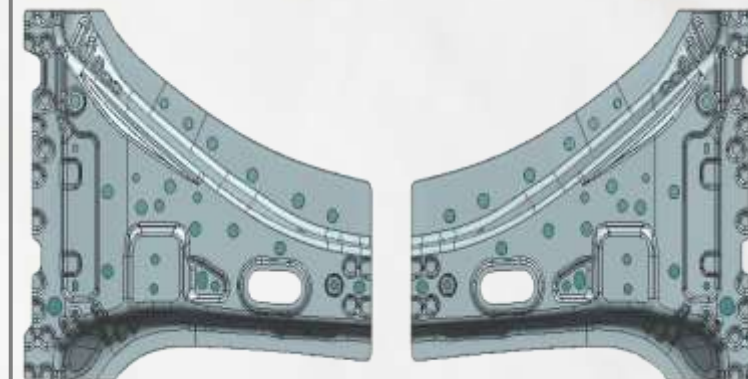
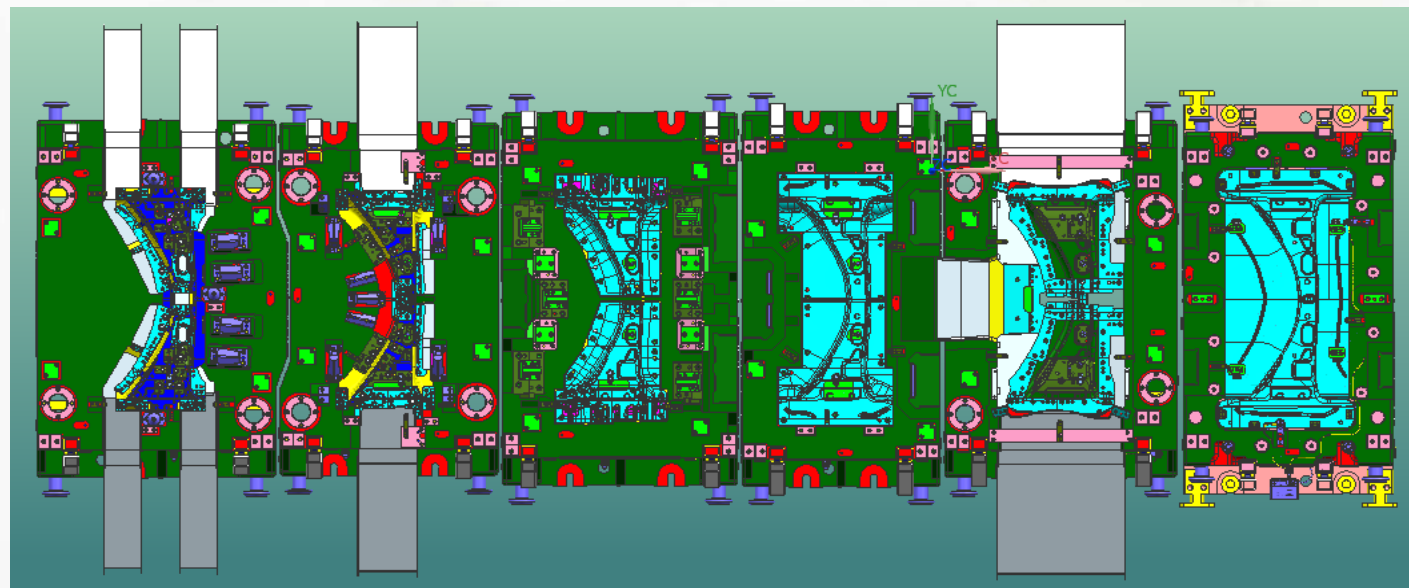
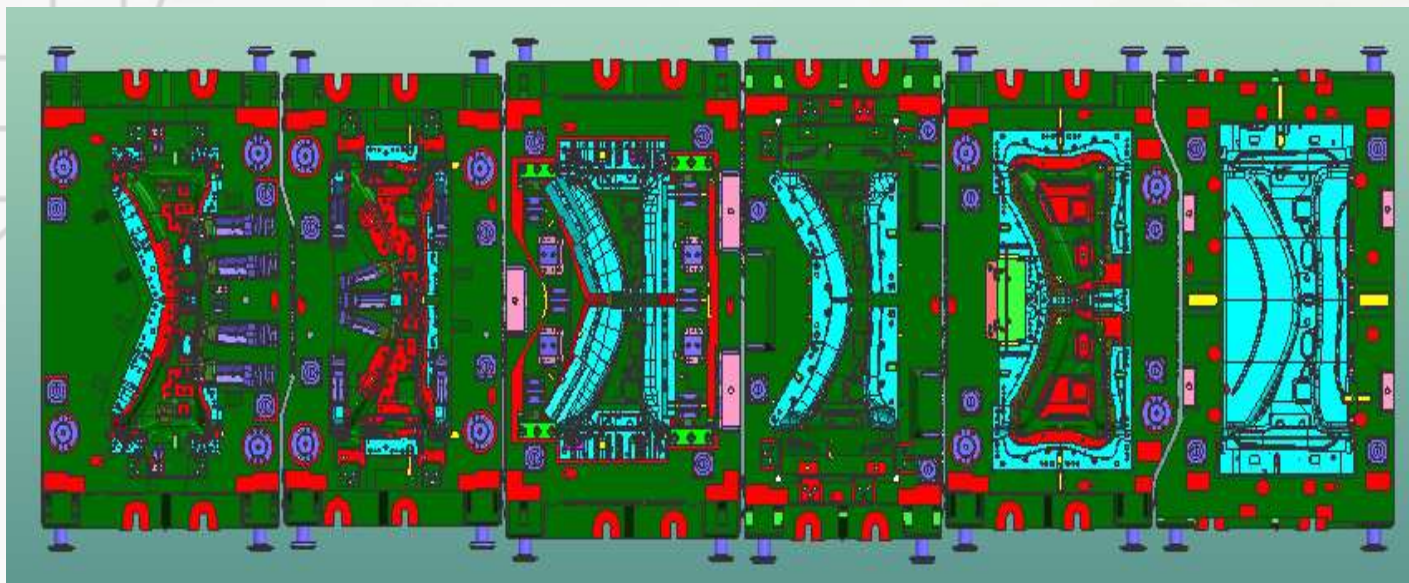


Design Capability - Showcases

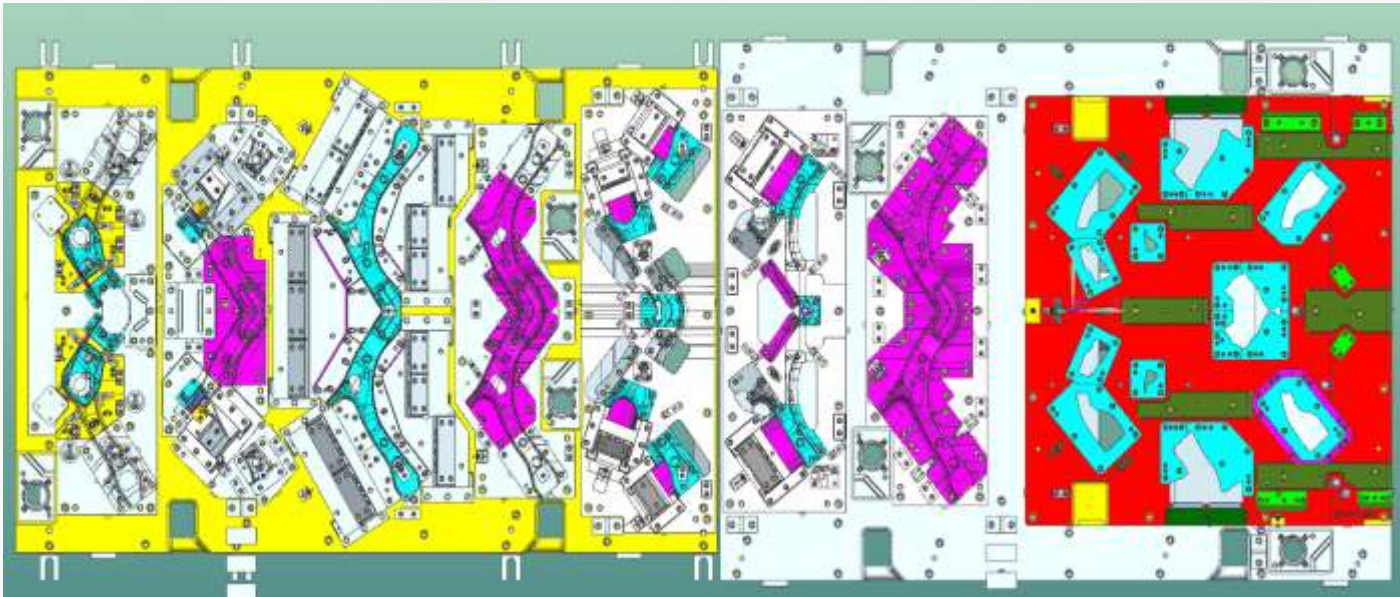
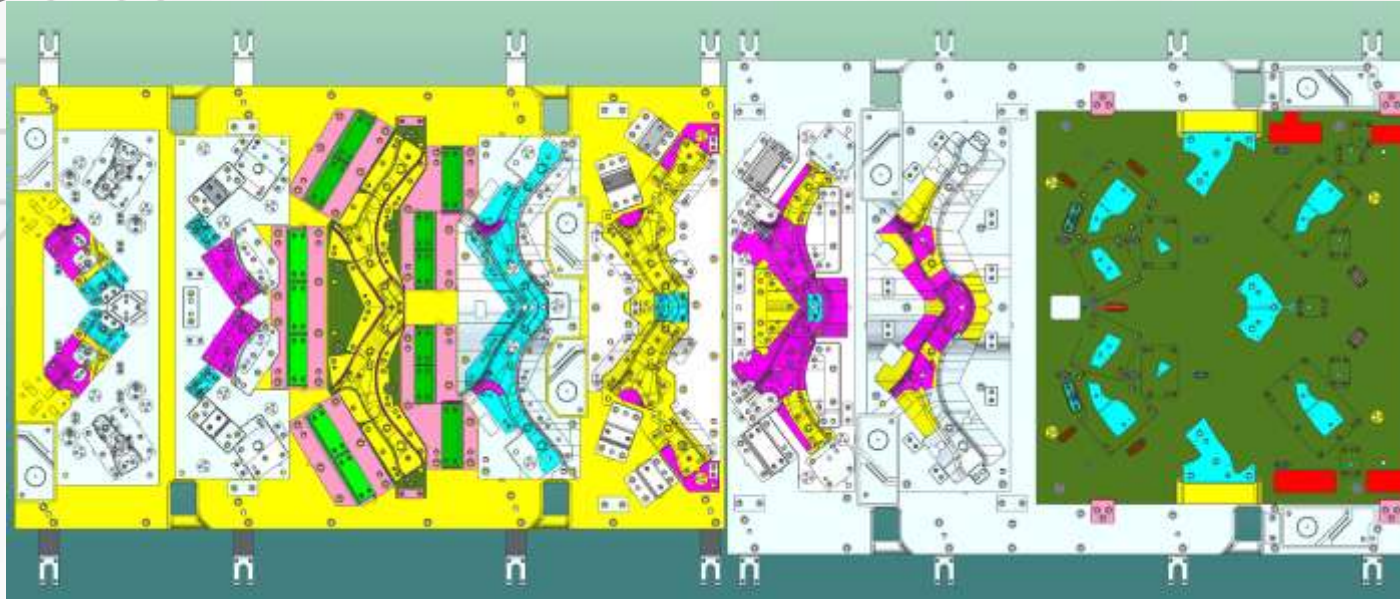


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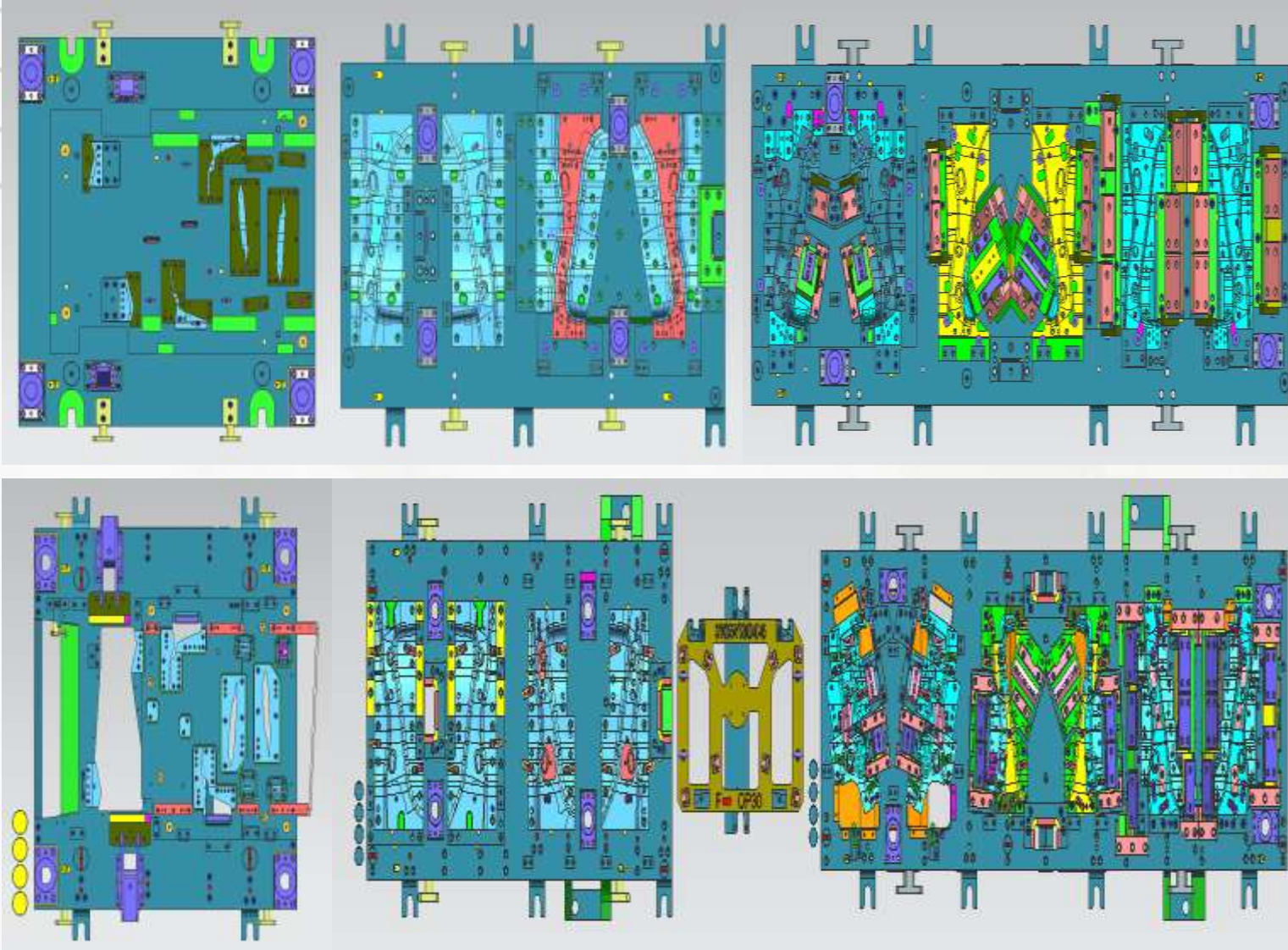
Design Capability - Showcases



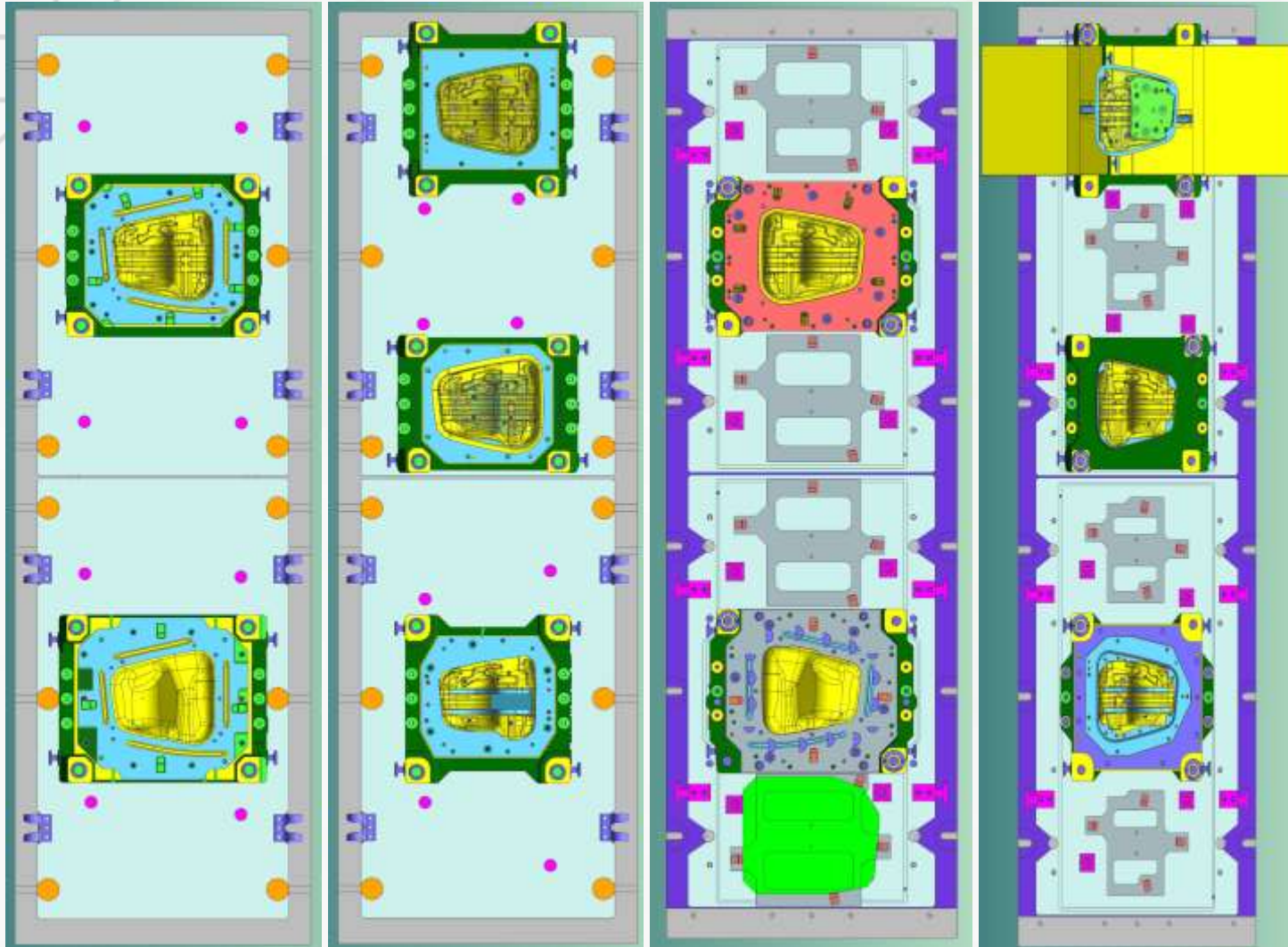
Design Capability - Showcases



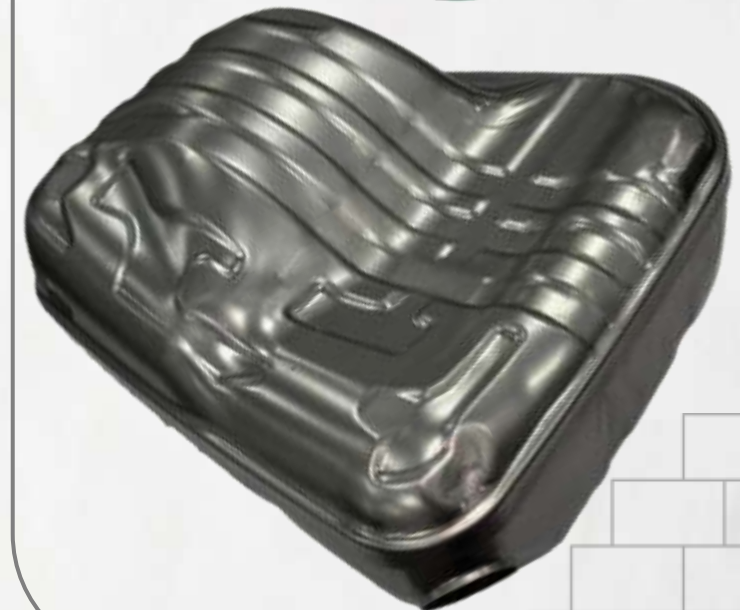
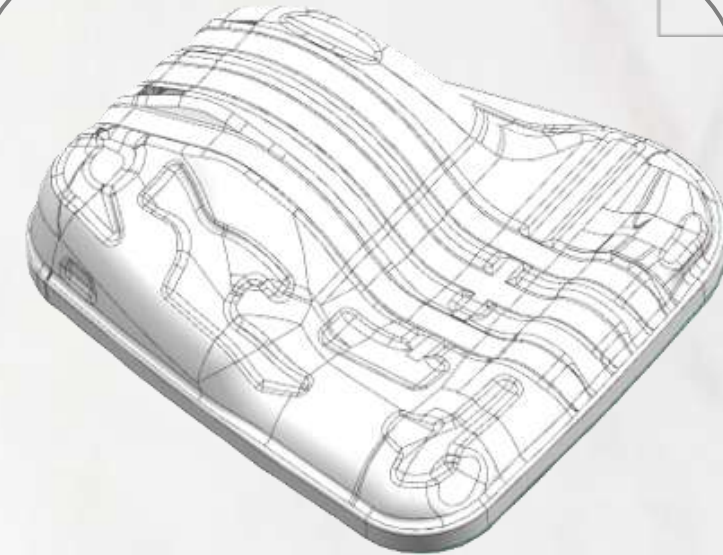
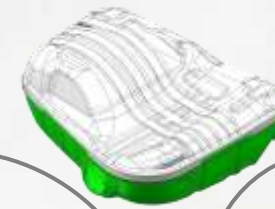
Design Capability - Showcases



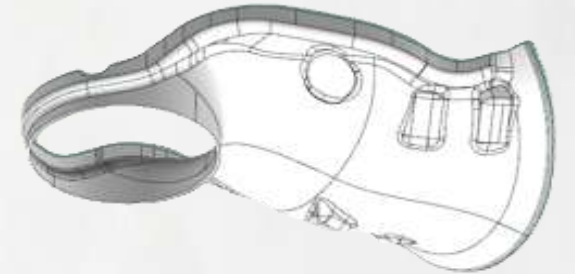
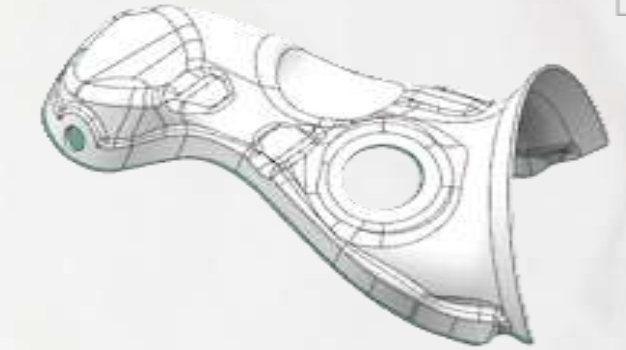
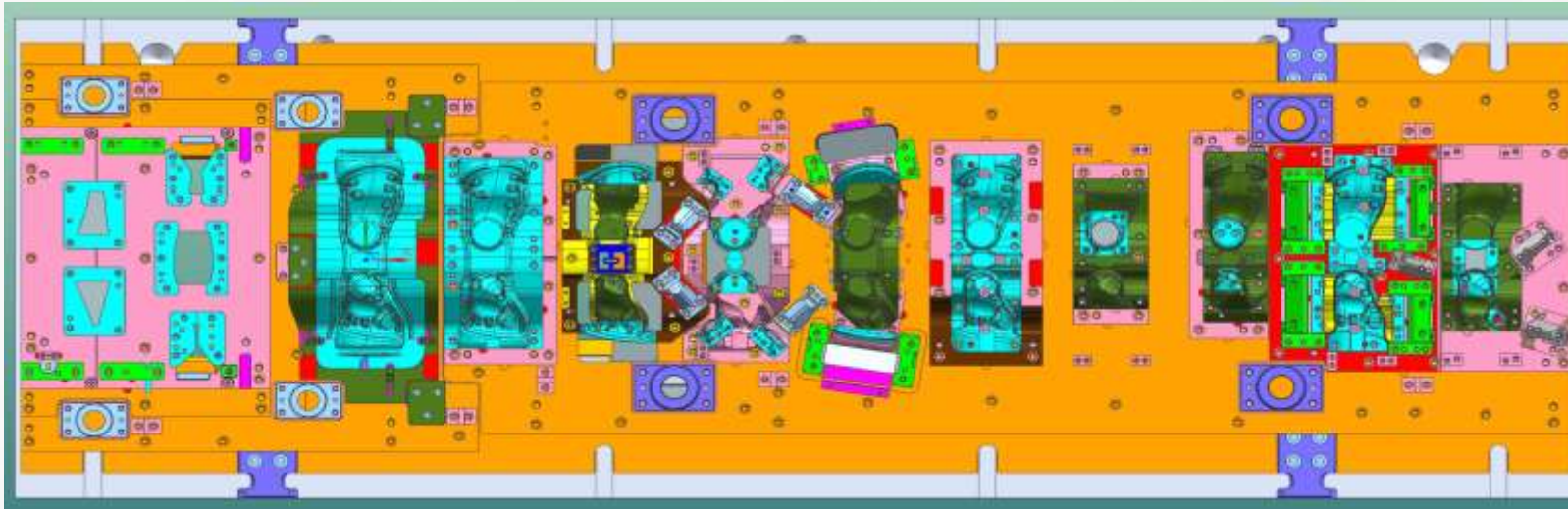
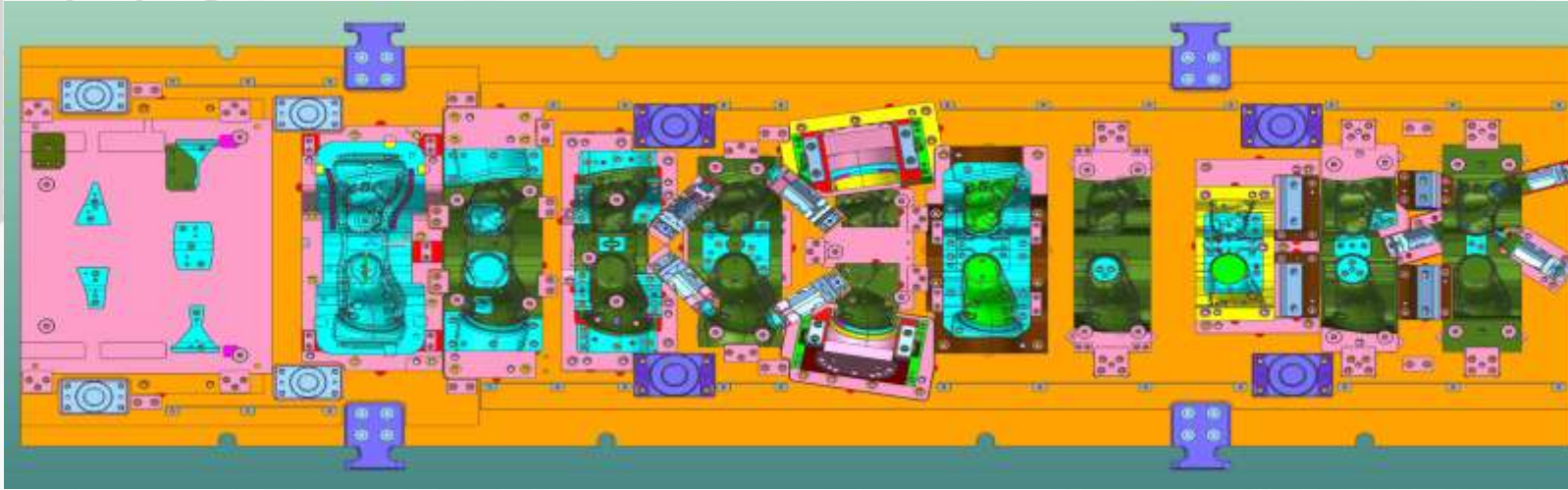
Design Capability - Showcases



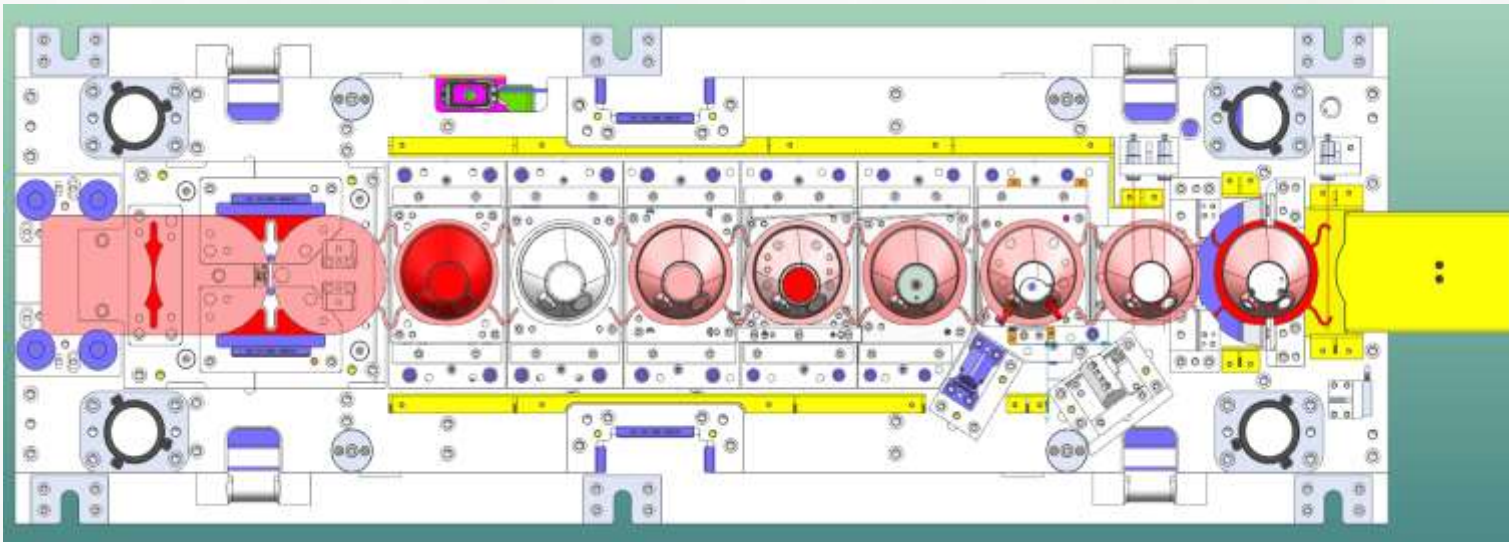
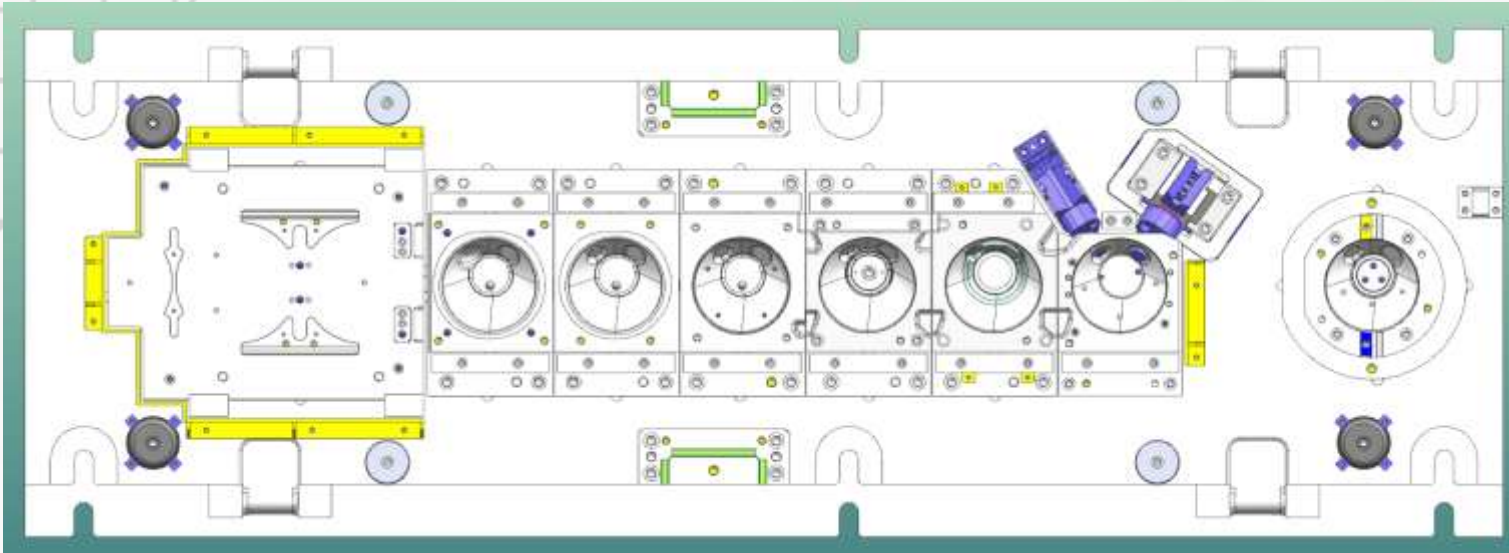
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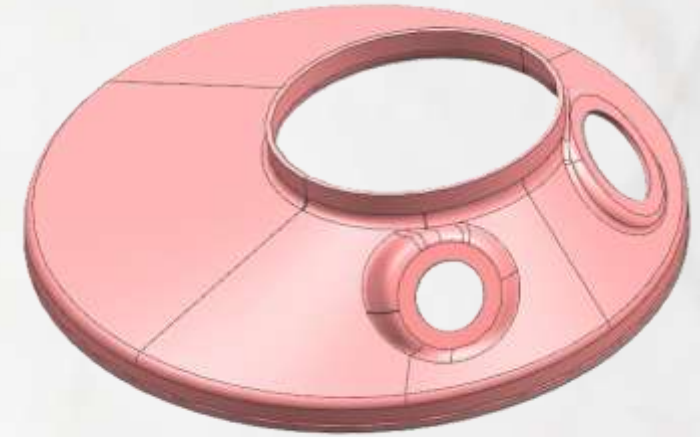
Design Capability - Showcases



Design Capability - Showcases



FAU19-3534258





MLD

04

PRODUCT CATALOG

AUTOMOTIVE SOLUTIONS.

Reliable Tooling for Critical Components



PROFESSIONAL AUTOMOTIVE TOOLING MANUFACTURER



Bracket Parts

Bracket / Housing / Support



Body Structural Parts

Rail / Cross Member /
Body Panel



Reinforcement Parts

Reinforcement / Side Member /
Support



Heat Shields / Thermal Parts

Heat Shield / Thermal Cover /
Insulator



Exhaust System

Muffler / Exhaust Cover /
Pipe Shield



DRIVING EXCELLENCE
THROUGH PRECISION TOOLING

Your Trusted Partner in Automotive
Tooling Solutions.

AUTOMOTIVE SOLUTIONS.

PRECISION TOOLING • RELIABLE QUALITY • DRIVING INNOVATION



BODY STRUCTURE

High precision stamping & assembly for stronger, safer, lighter vehicle structure

CHASSIS COMPONENTS



SUBFRAMES

Robust design for maximum structural performance



CONTROL ARMS

Precision engineered for ride comfort and handling stability



LINKS

High strength, high accuracy, longer service life



EXHAUST SYSTEM COMPONENTS



OUR ADVANTAGES



Precision Engineering

Advanced 3D design & simulation



High Quality

Strict process control & inspection



Integrated Solution

From design to tooling and production



On-time Delivery

Efficient project management & fast response



Reliable Performance

Consistent quality for automotive excellence

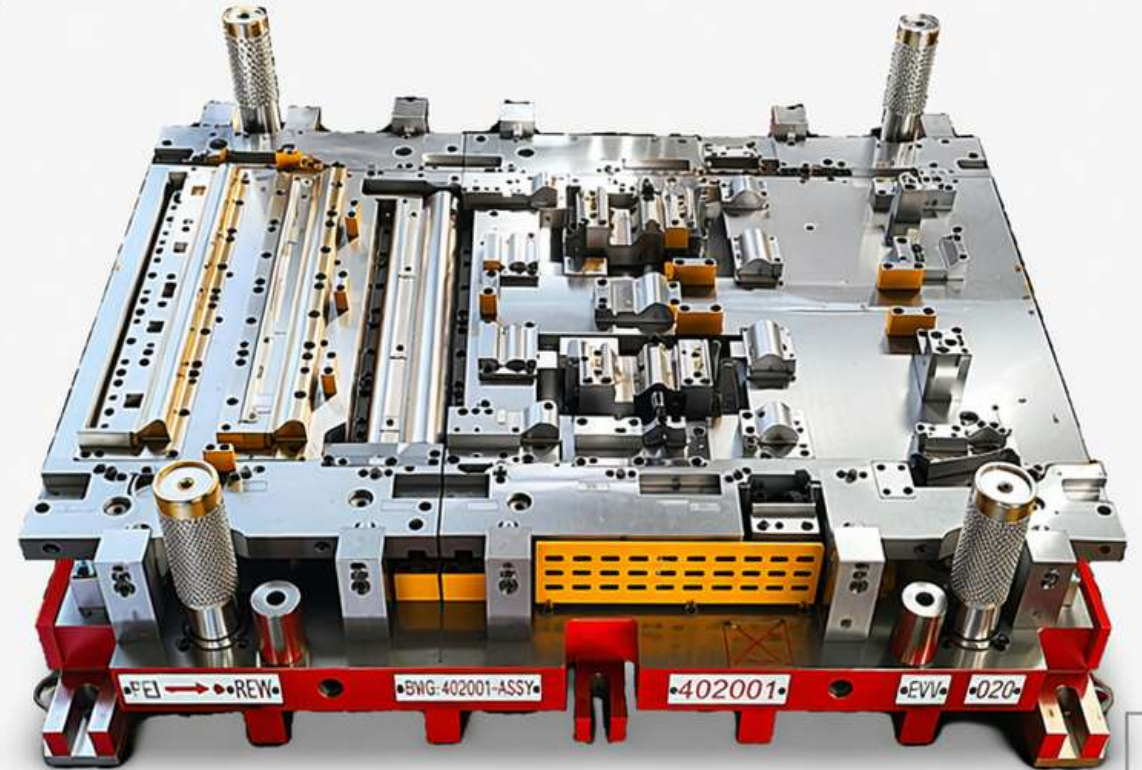
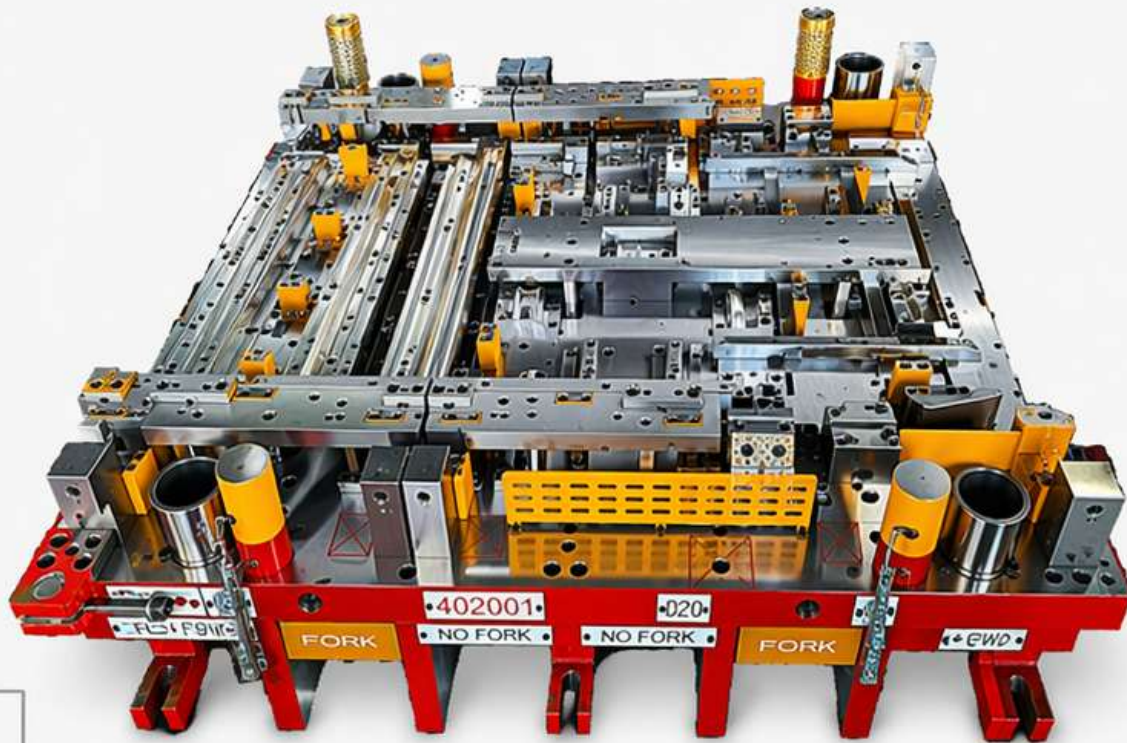

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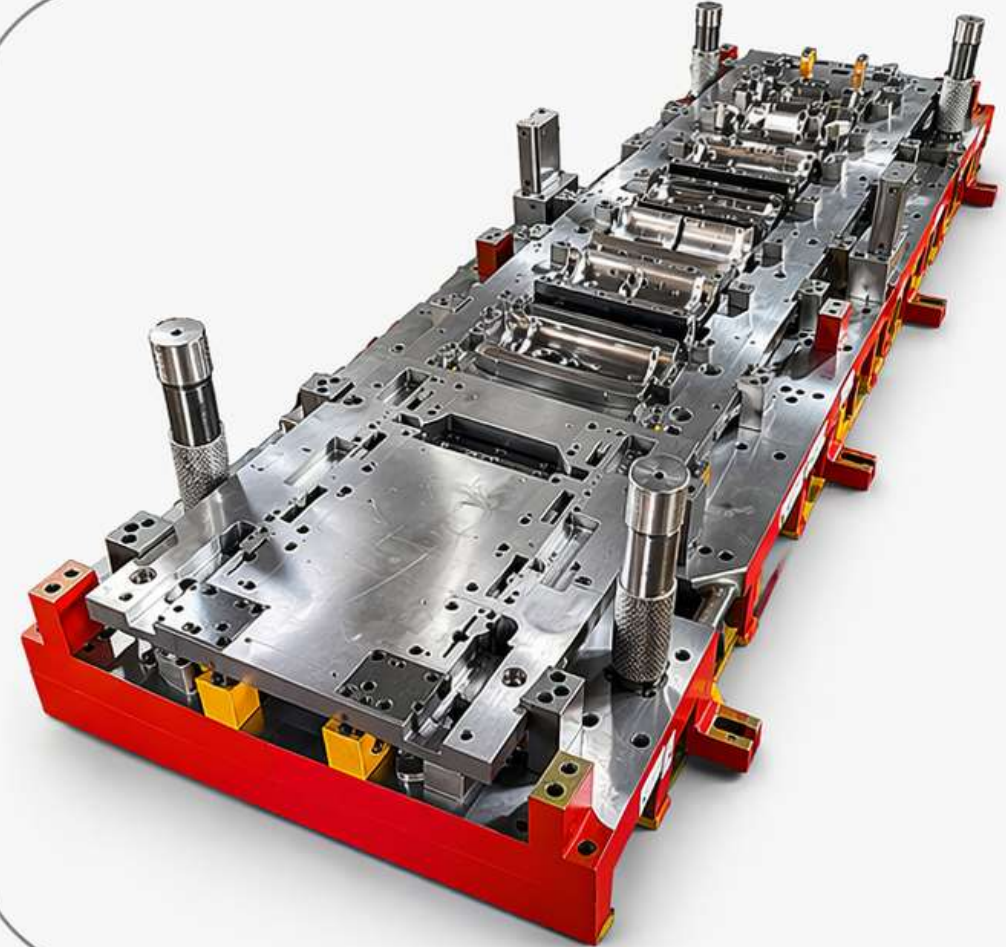
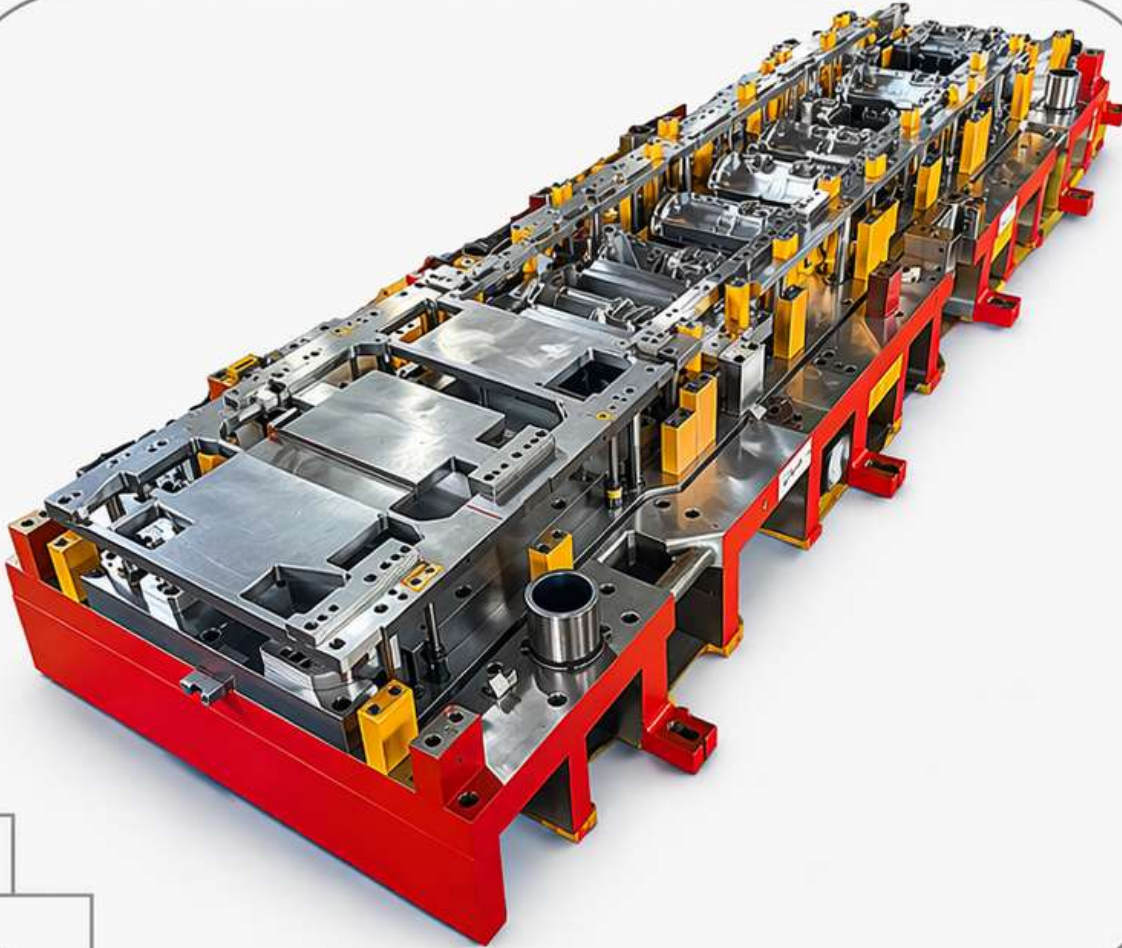
MLD

PROGRESSIVE DIE DISPLAY





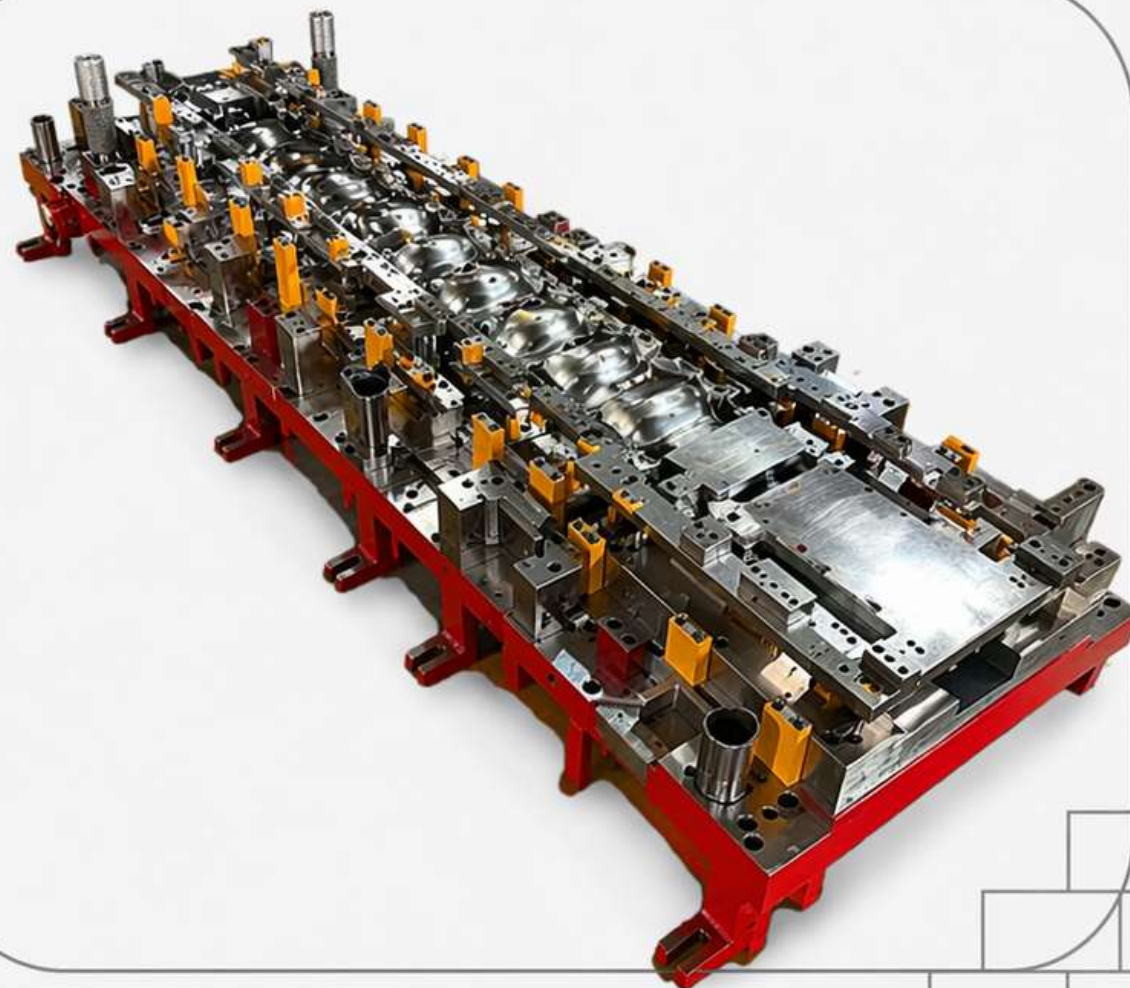
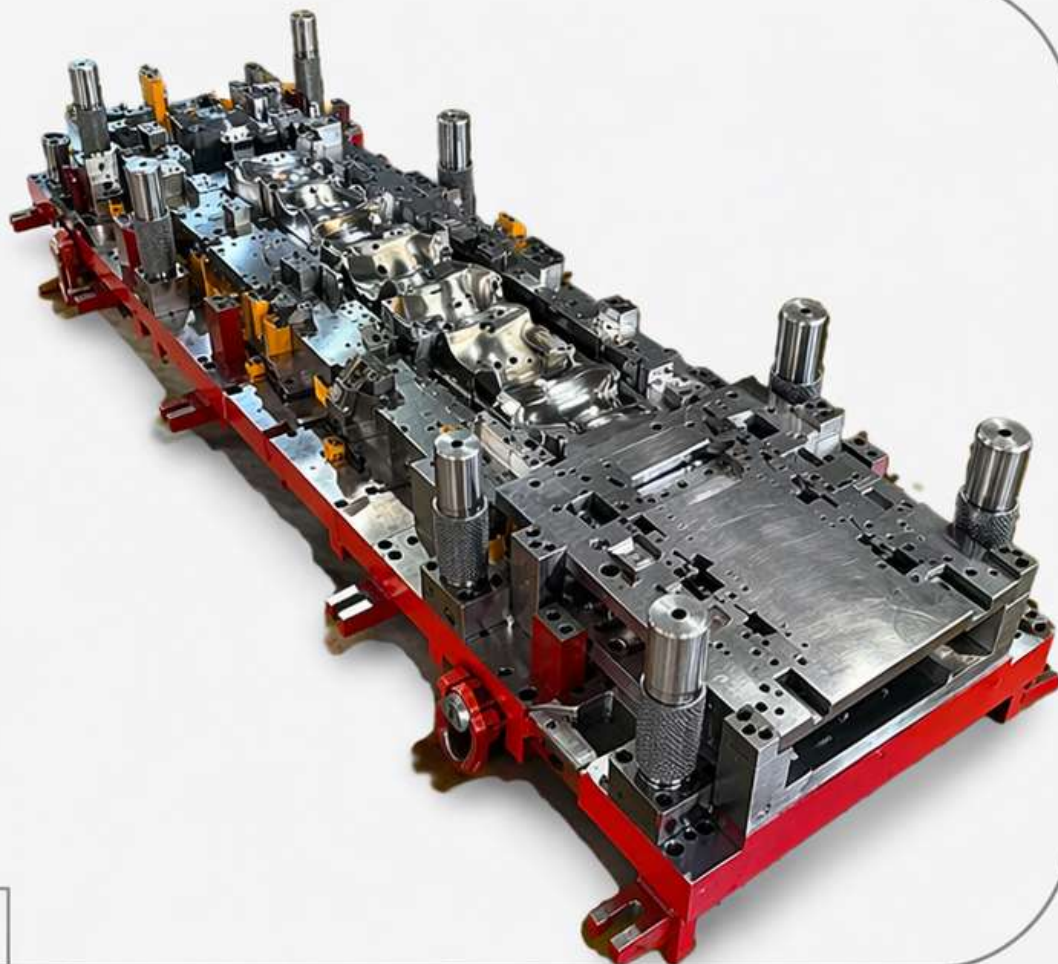
PROGRESSIVE DIE DISPLAY





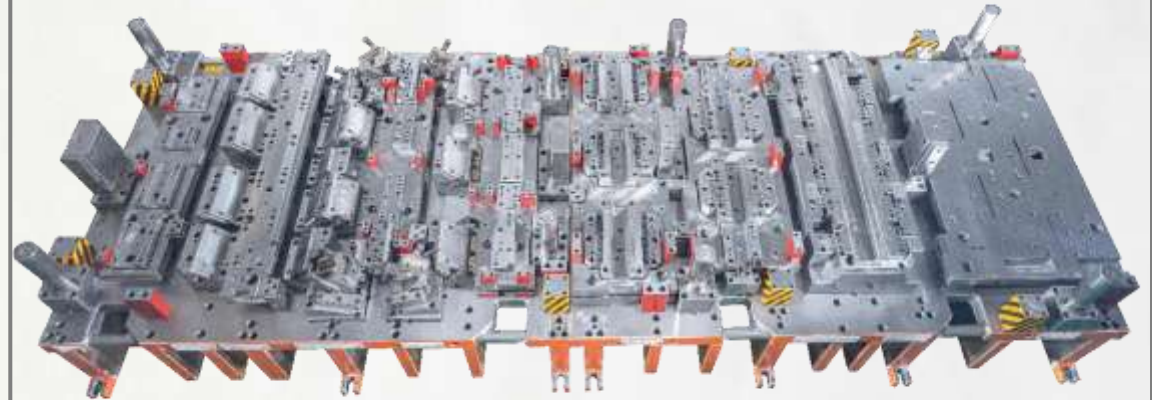
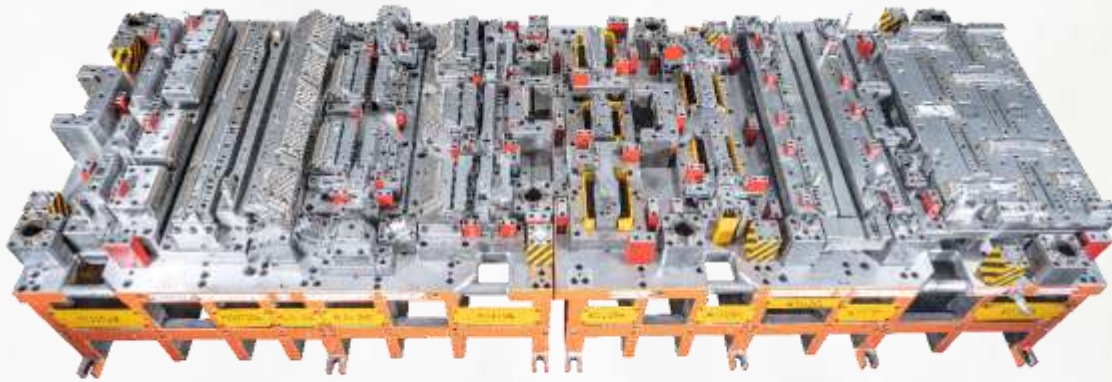
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PROGRESSIVE DIE DISPLAY





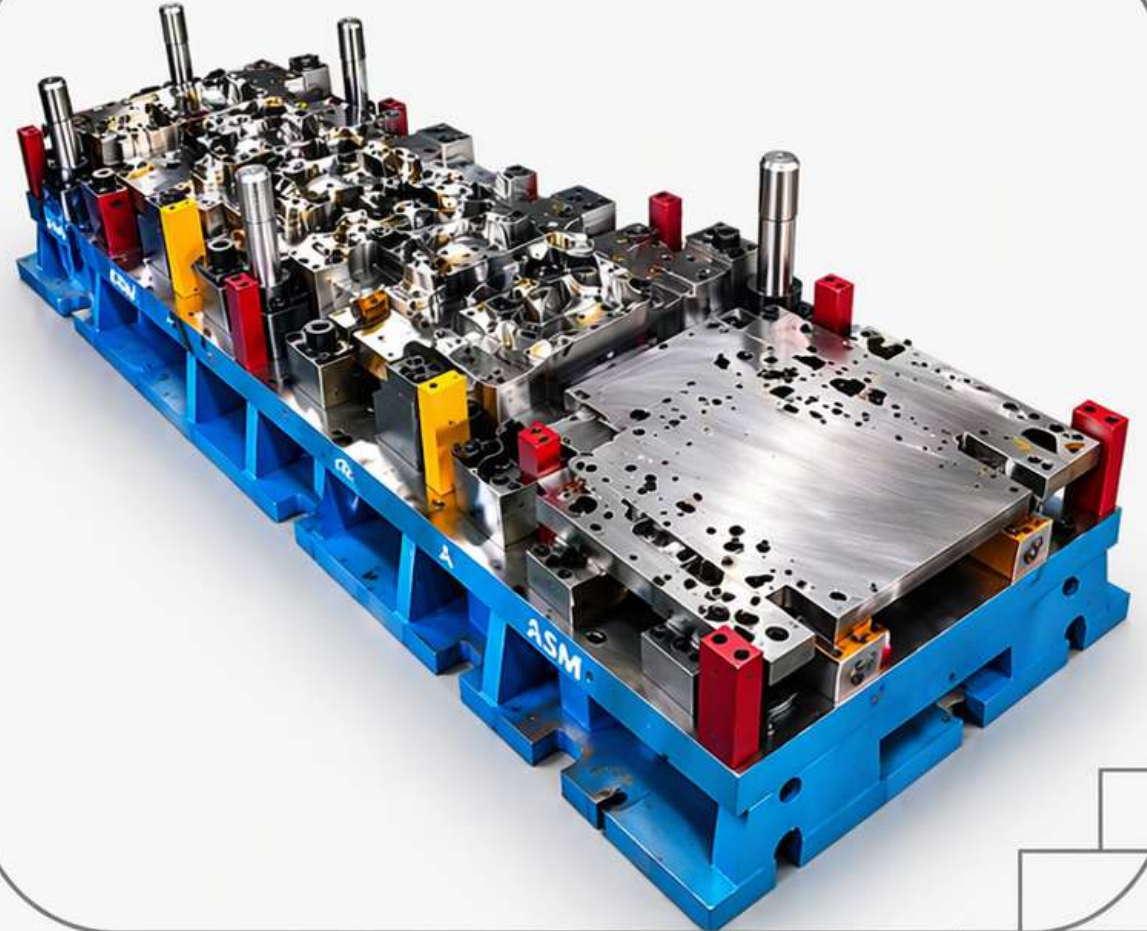
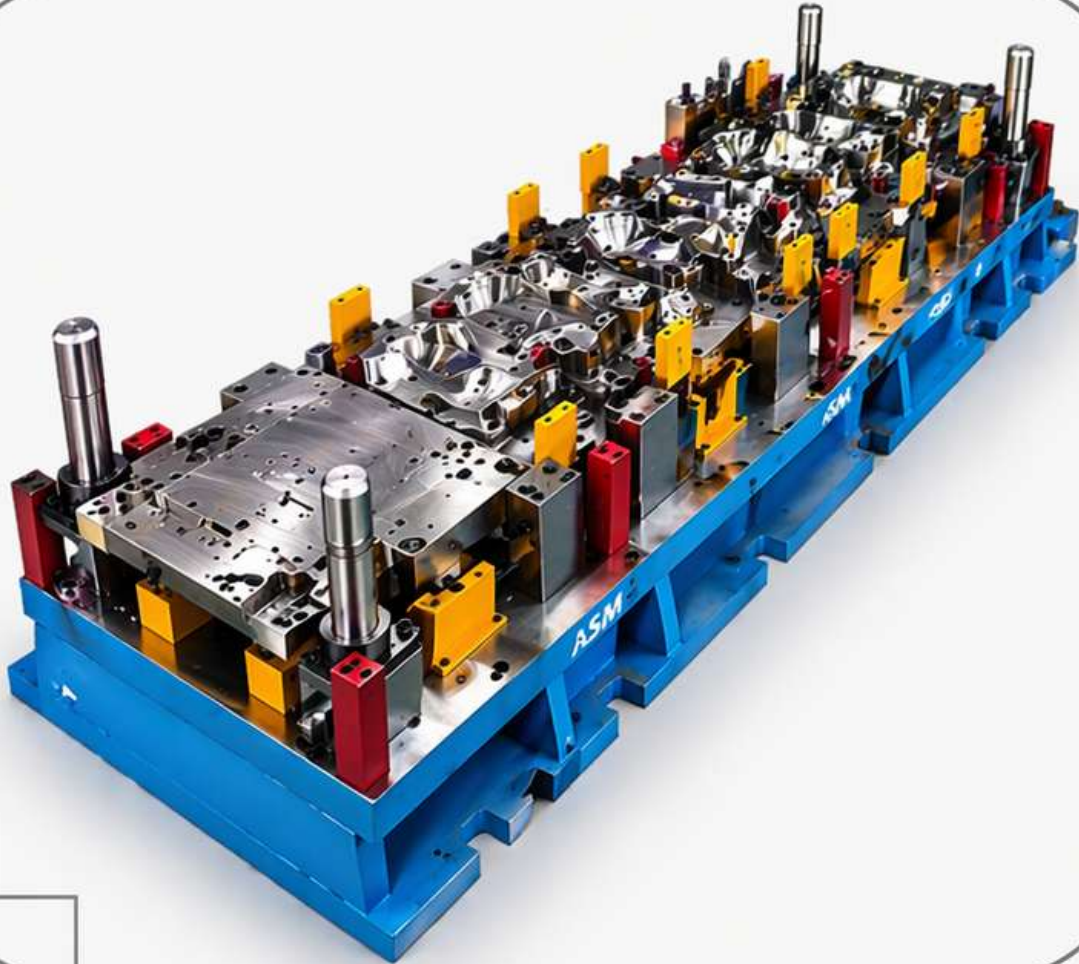
PROGRESSIVE DIE DISPLAY





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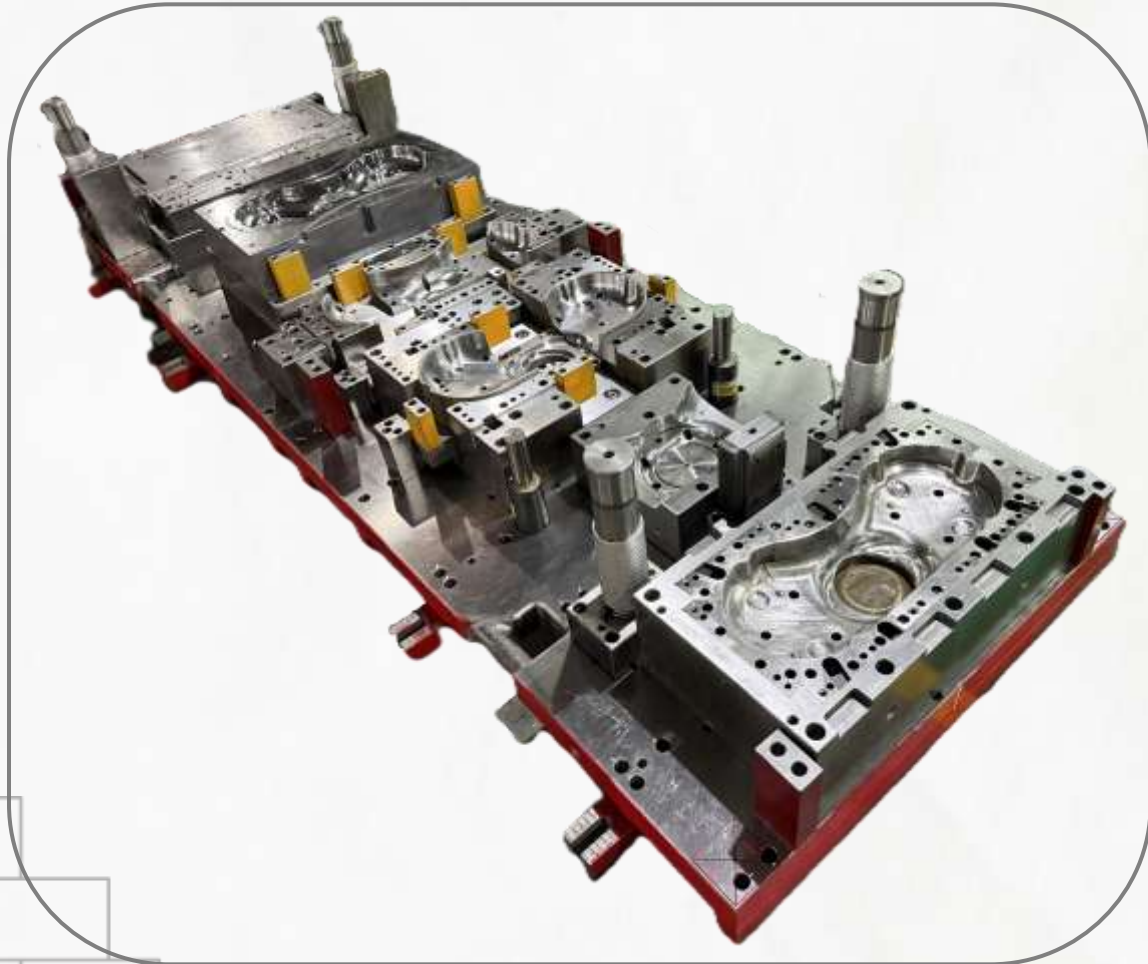
TRANSFER DIE DISPLAY





MLD

TRANSFER DIE DISPLAY





MLD

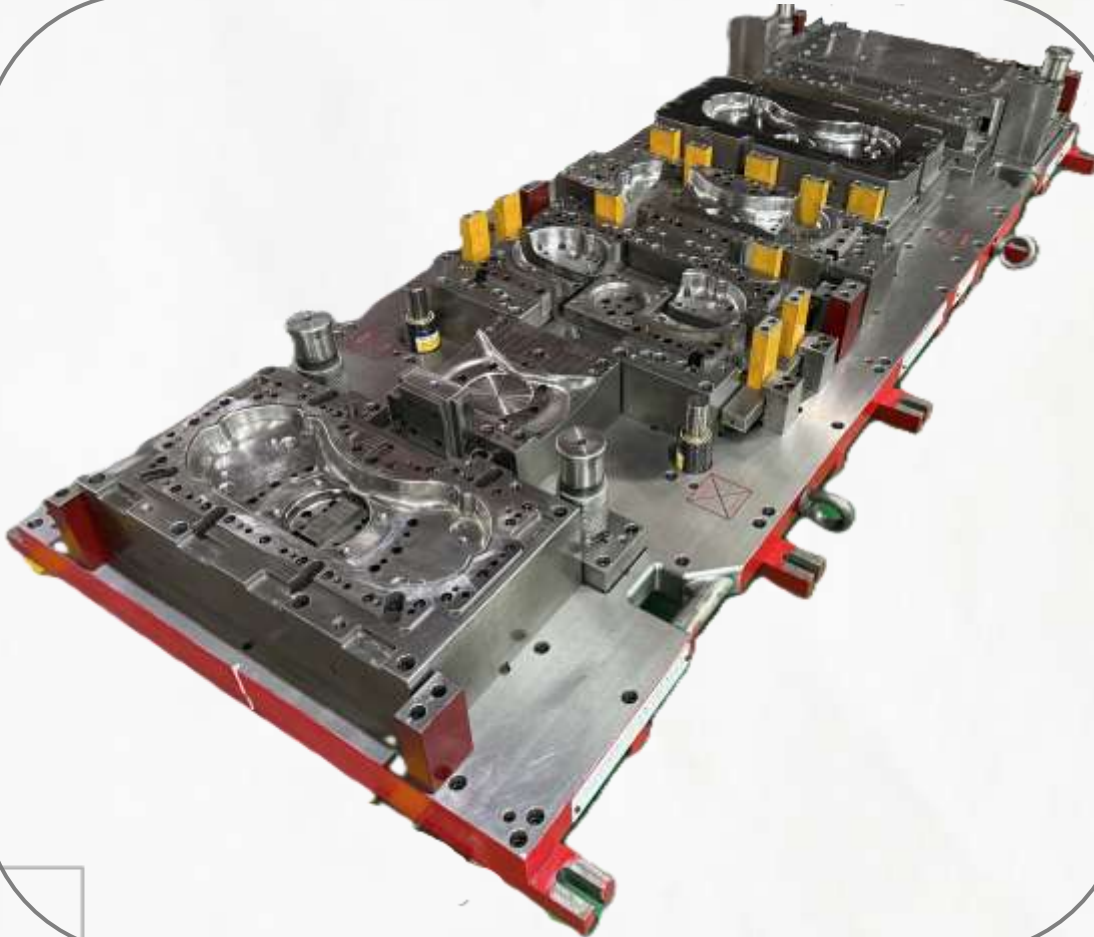
TRANSFER DIE DISPLAY





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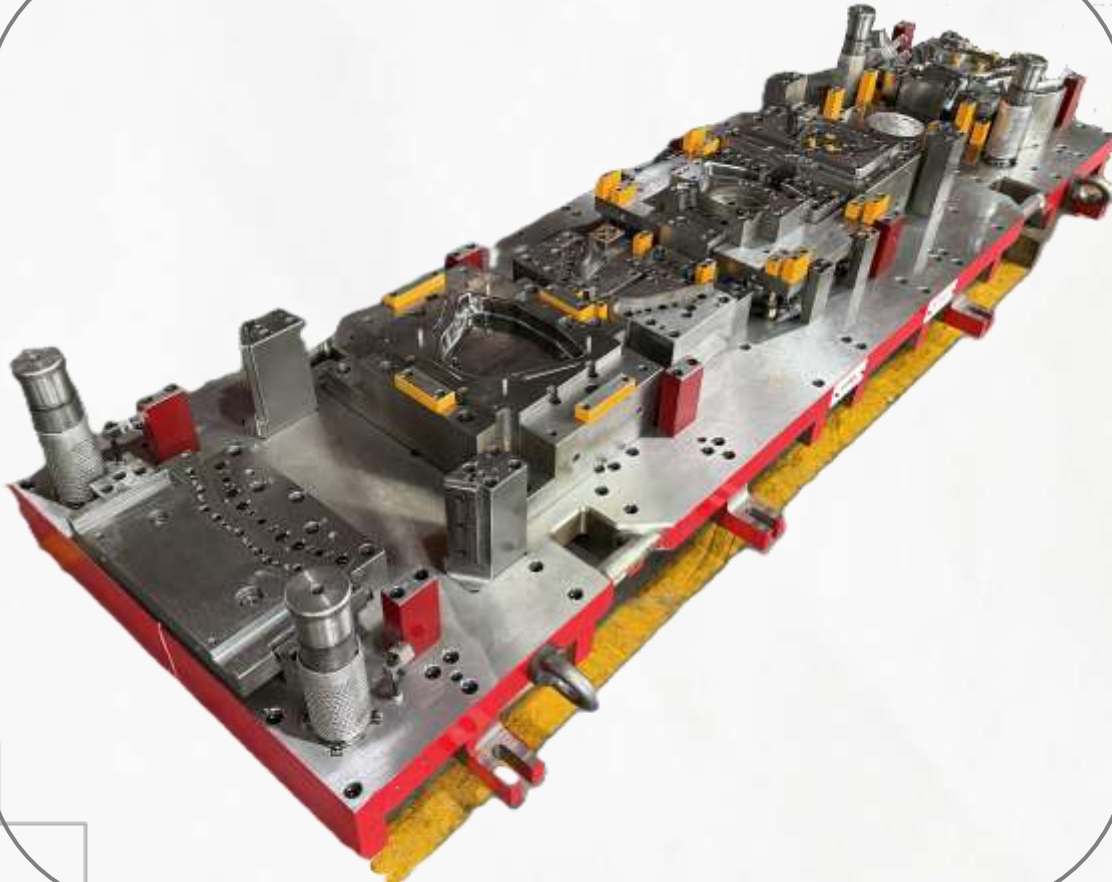
TRANSFER DIE DISPLAY





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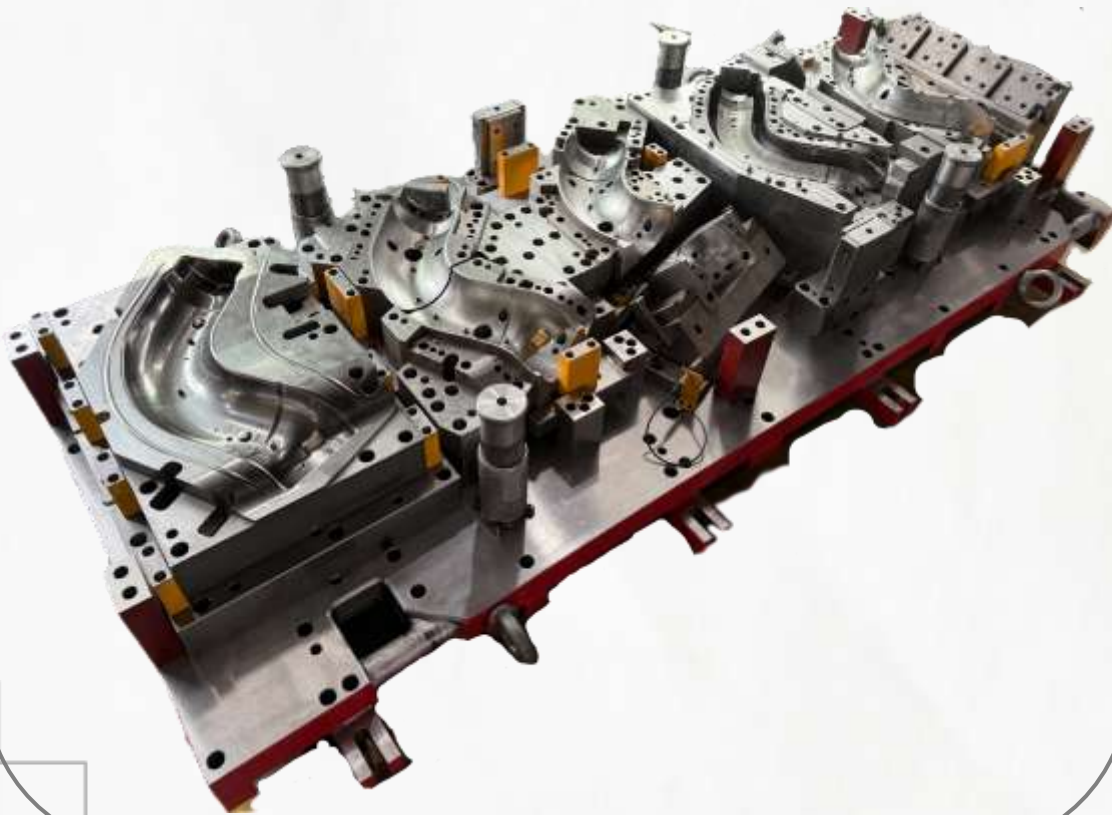
TRANSFER DIE DISPLAY





MLD

TRANSFER DIE DISPLAY





MLD

TRANSFER DIE DISPLAY





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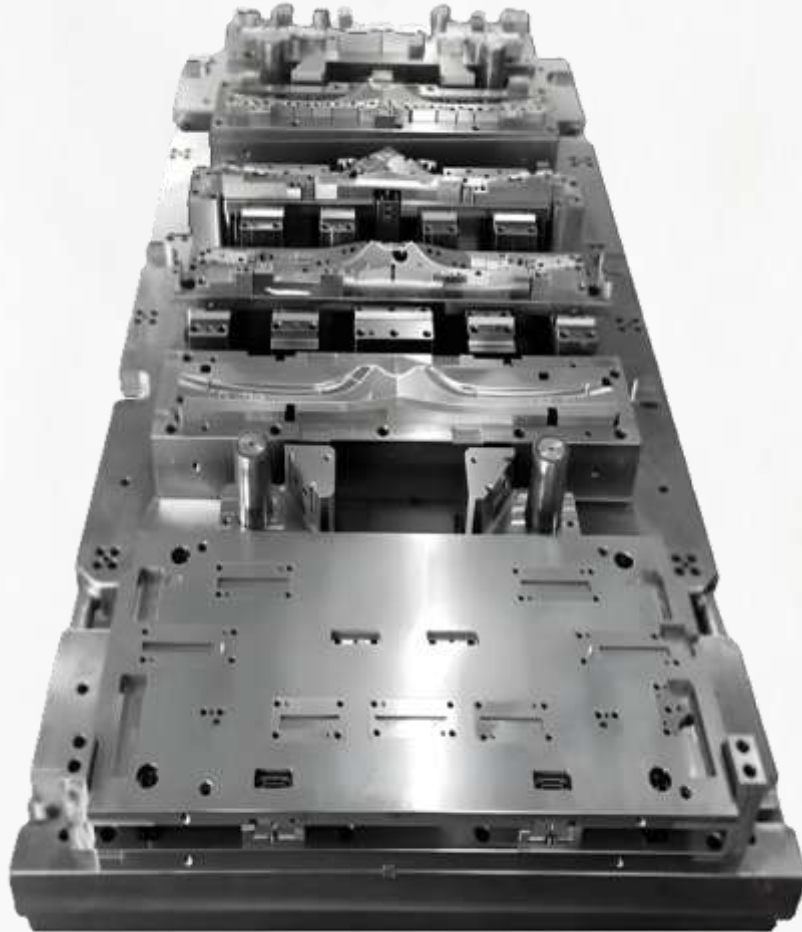
TRANSFER DIE DISPLAY





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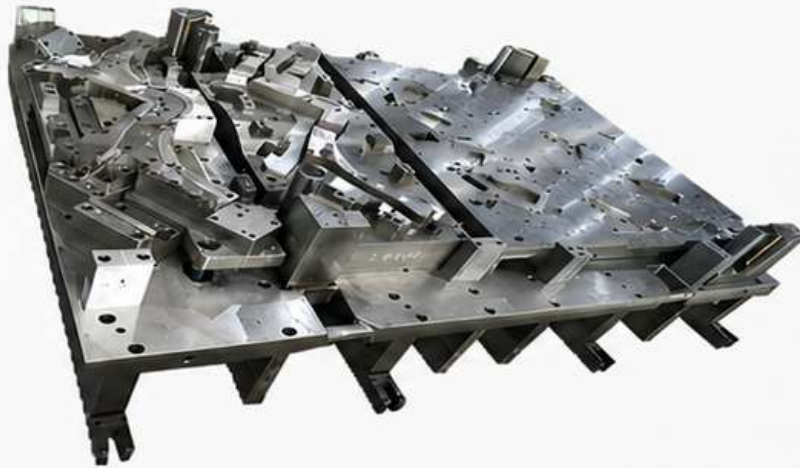
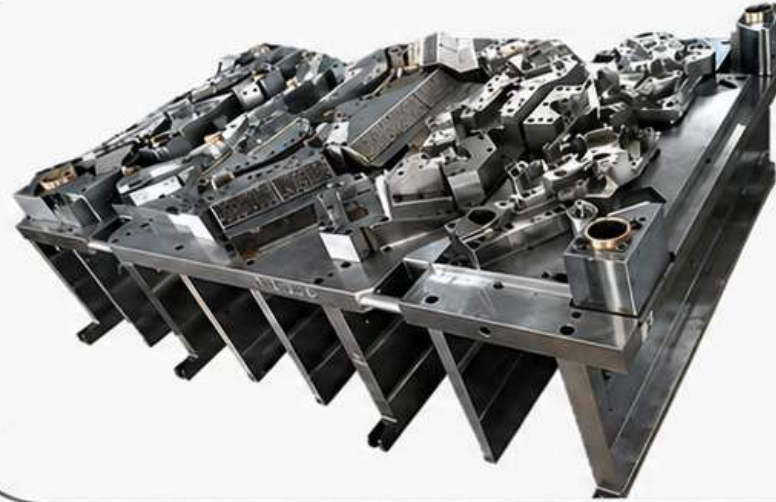
ROBOTIC TRANSFER DIE DISPLAY





MLD

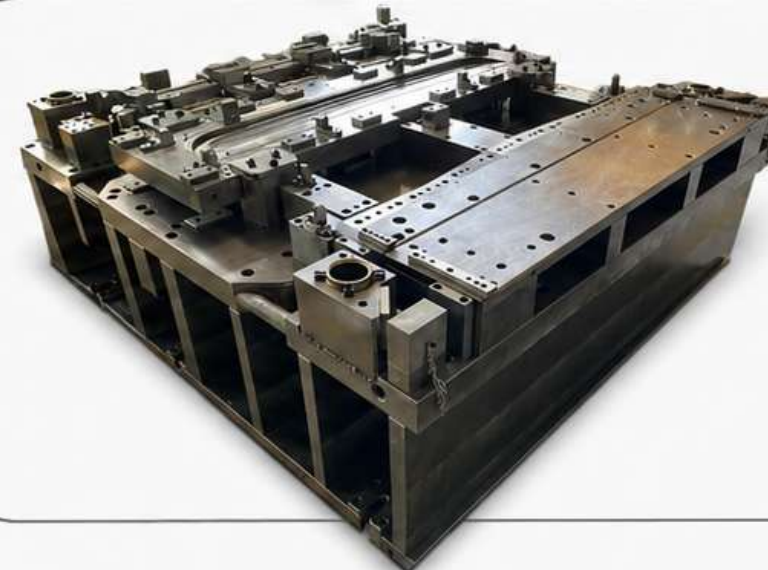
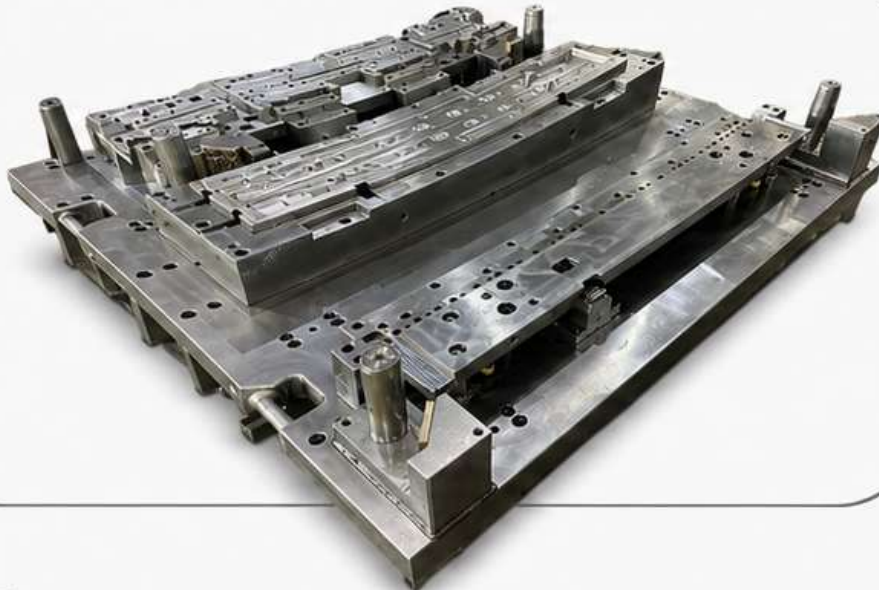
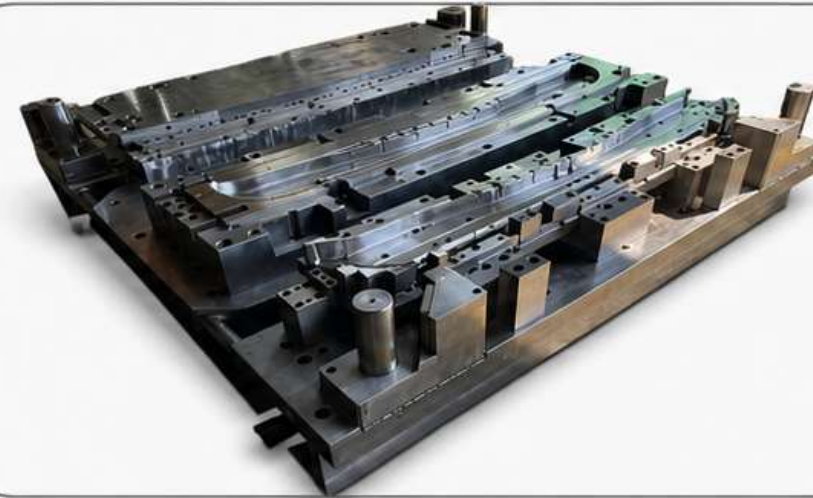
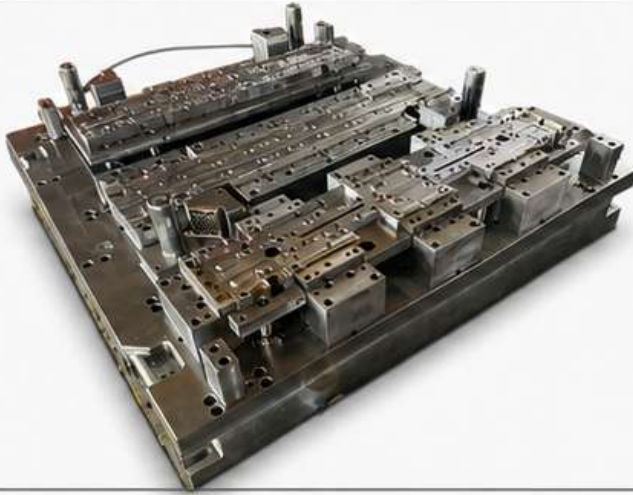
ROBOTIC TRANSFER DIE DISPLAY





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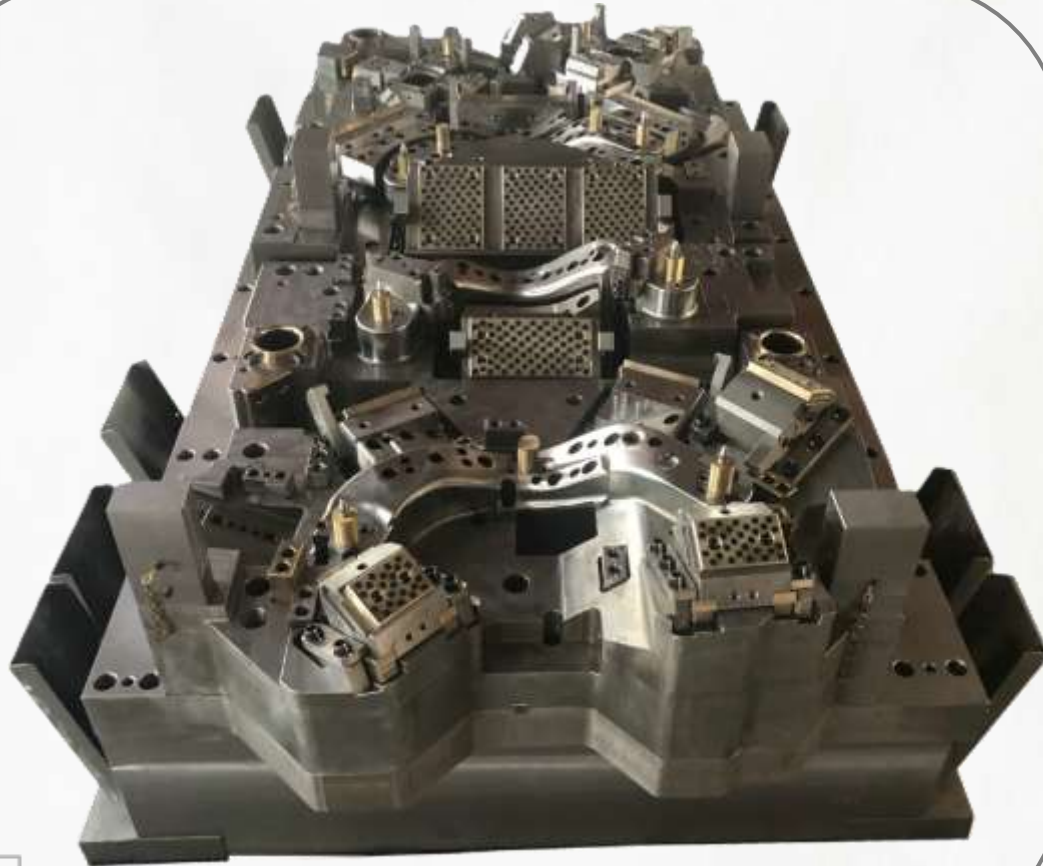
ROBOTIC TRANSFER DIE DISPLAY





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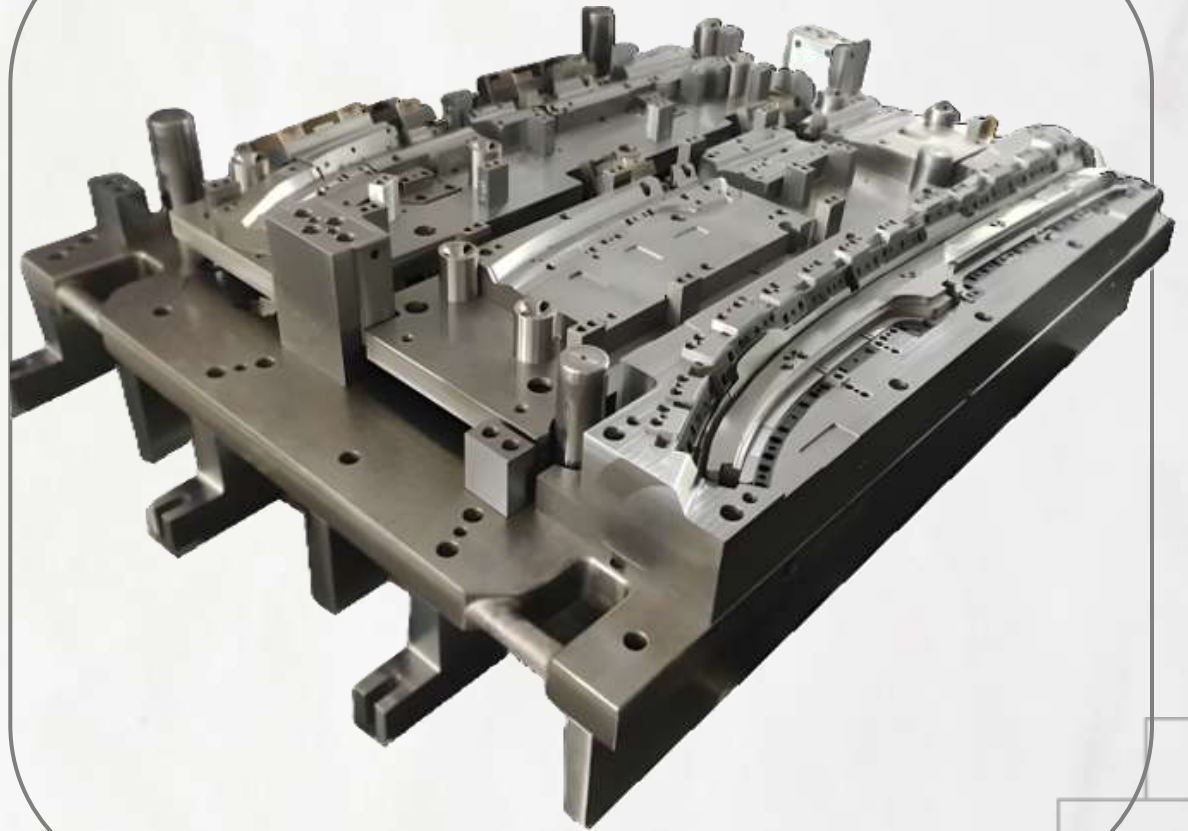
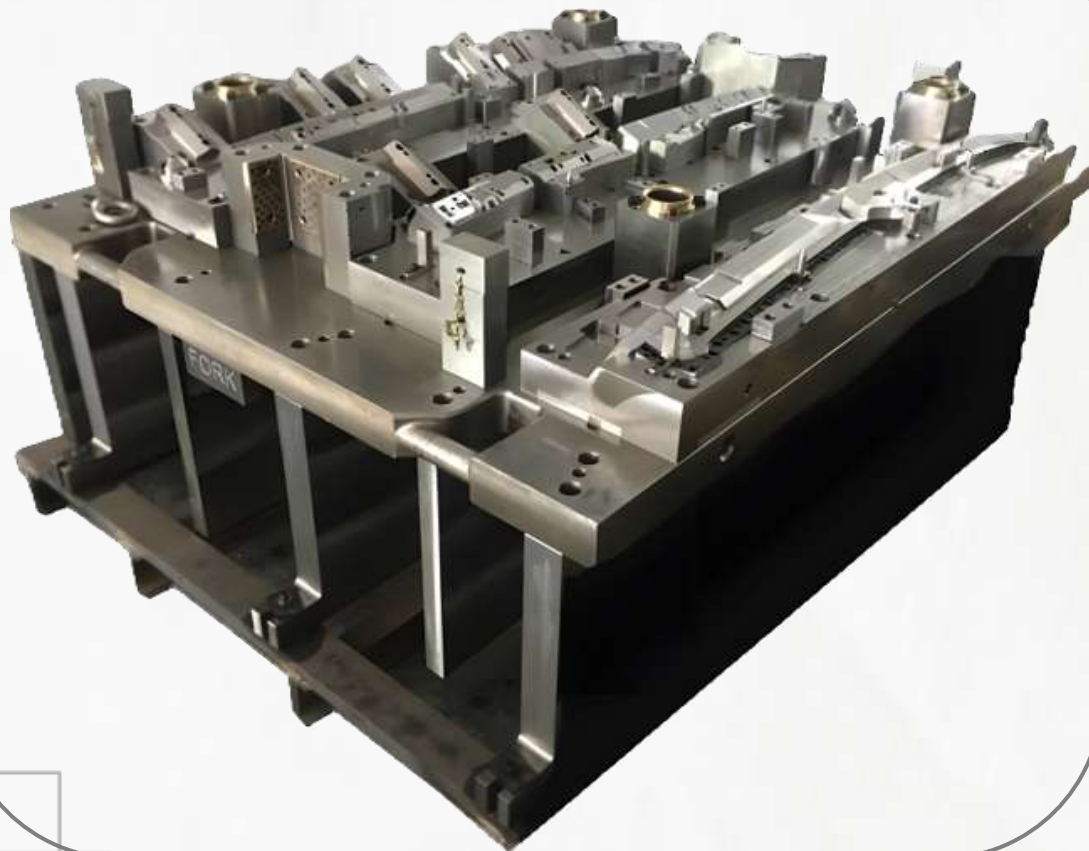
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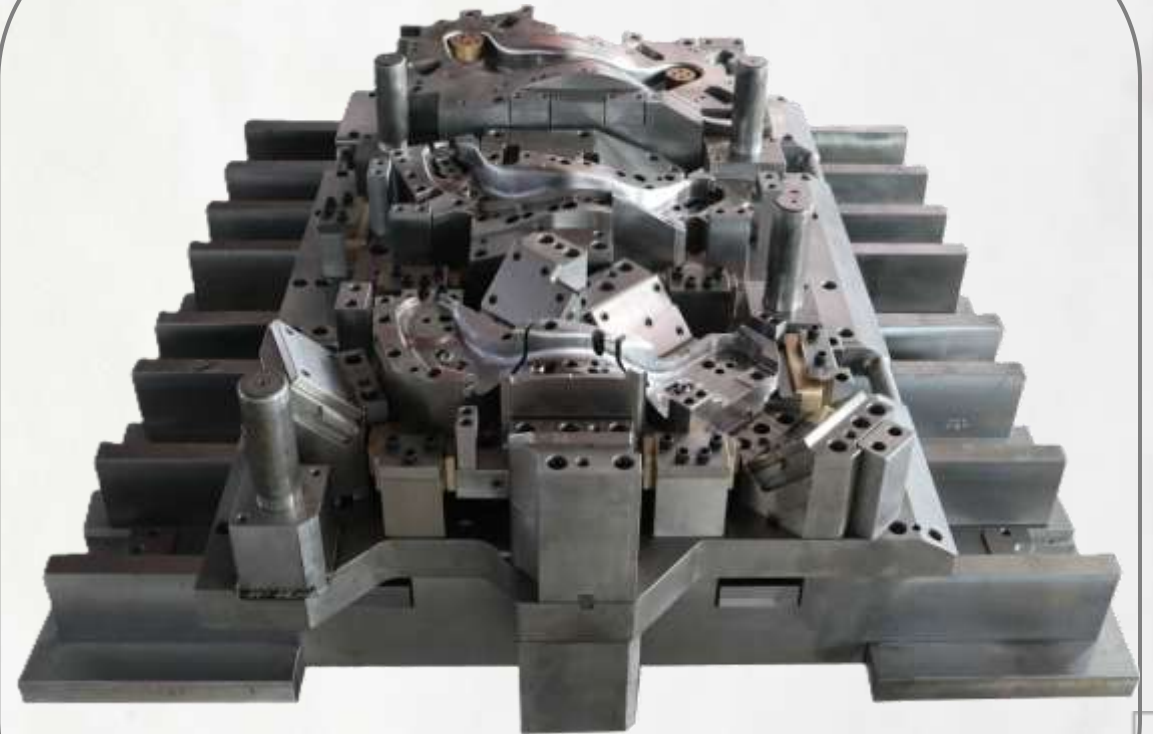
ROBOTIC TRANSFER DIE DISPLAY





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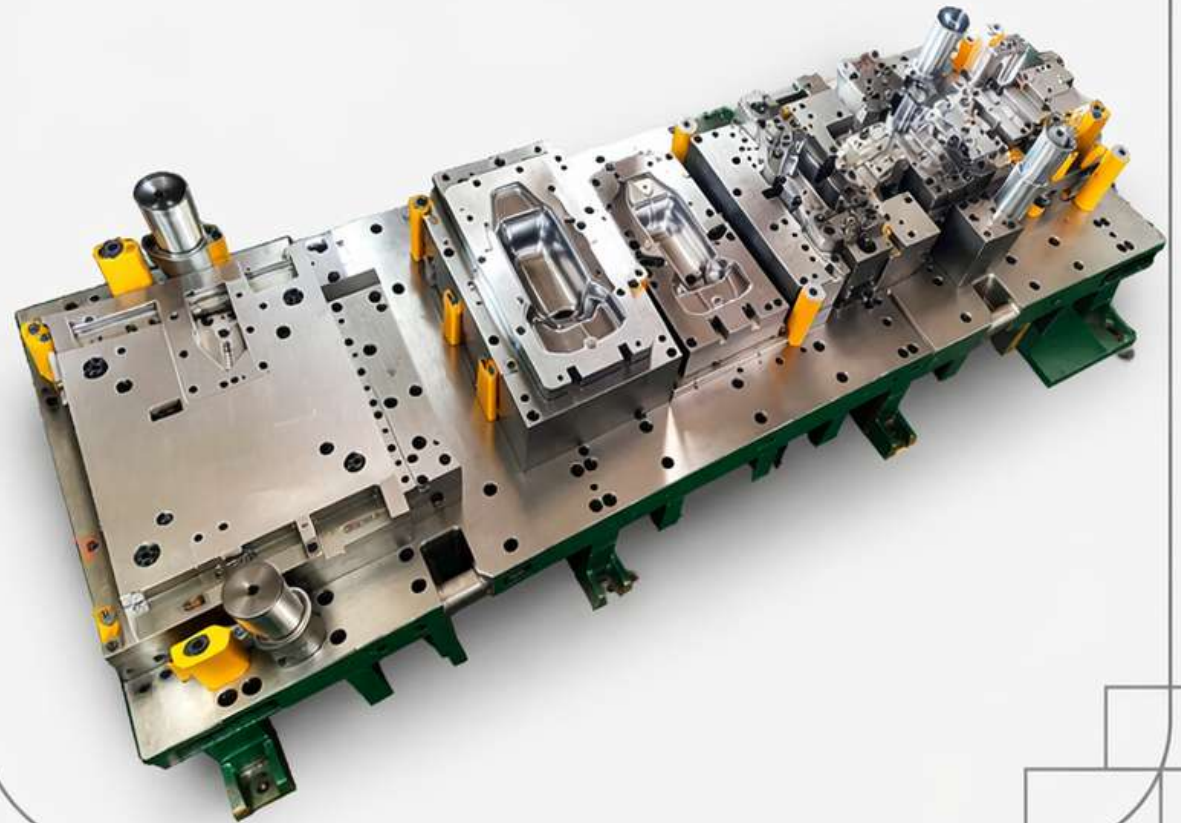
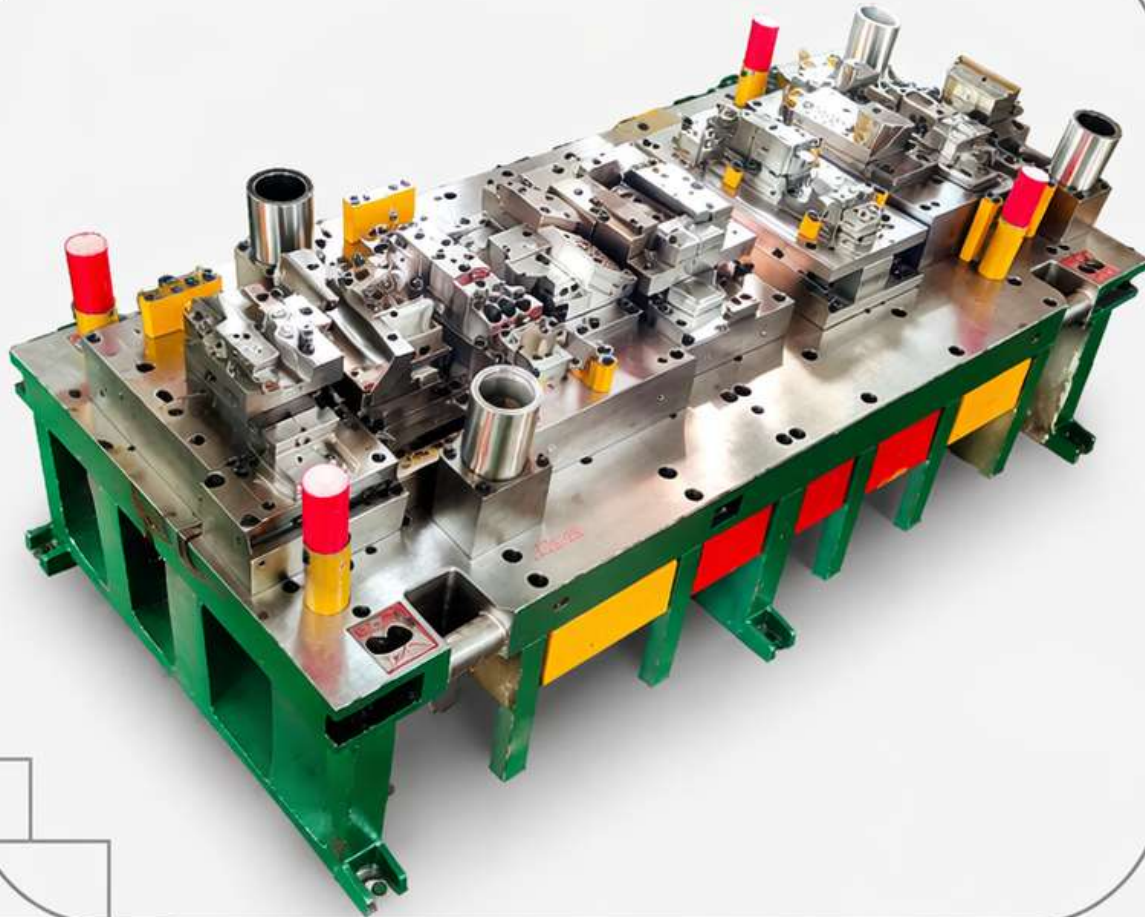
ROBOTIC TRANSFER DIE DISPLAY





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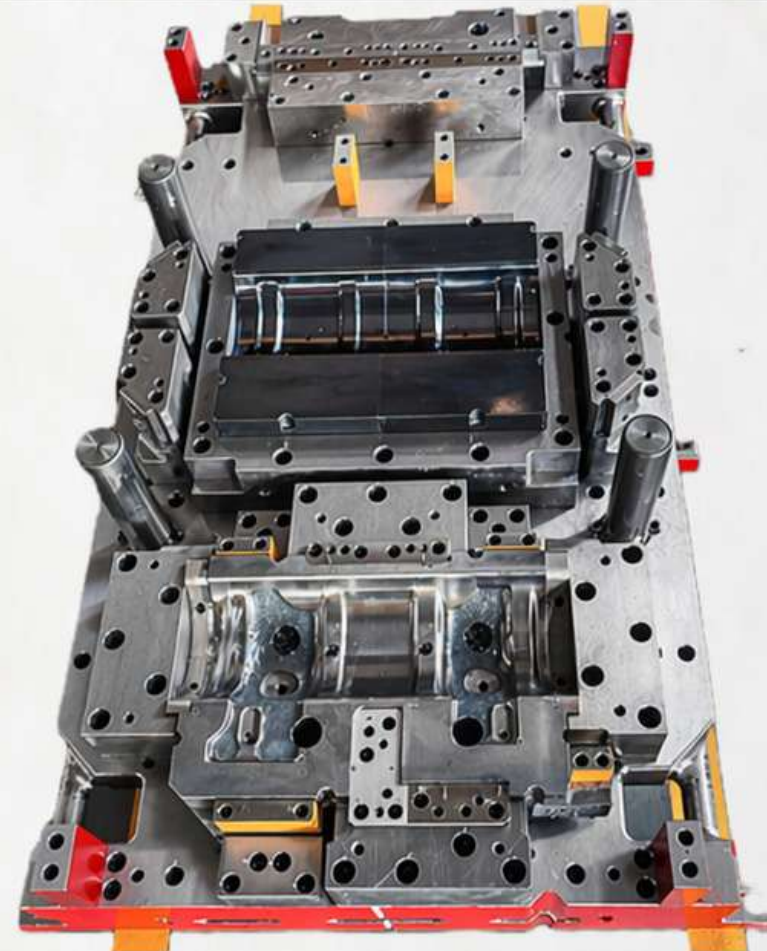
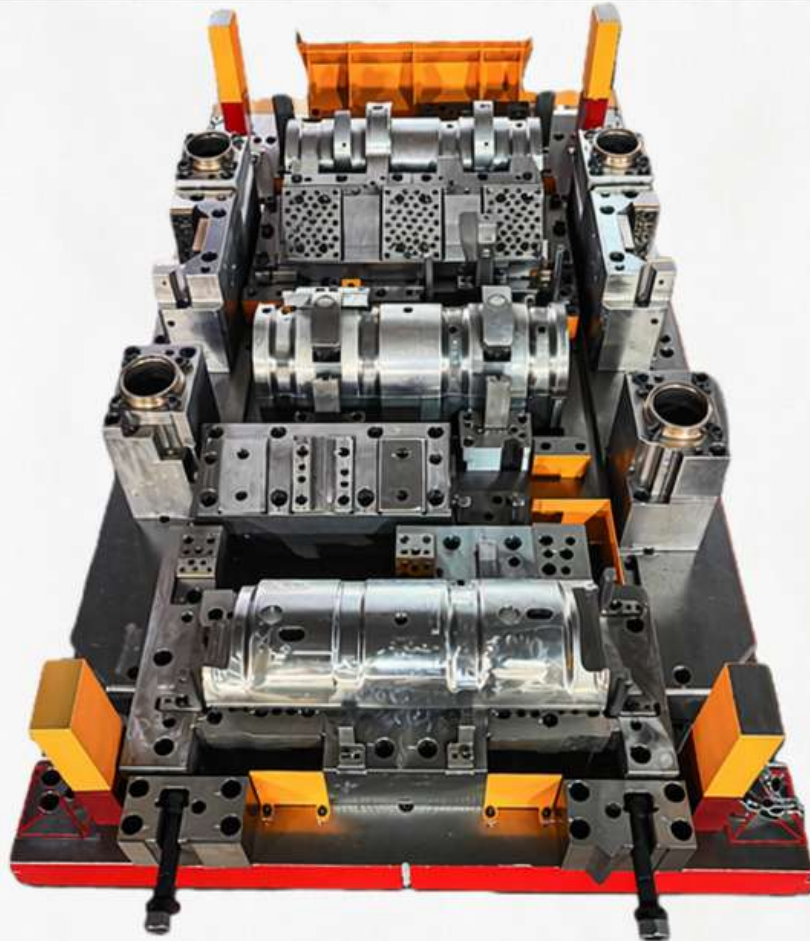
MANUAL DIE DISPLAY





MLD

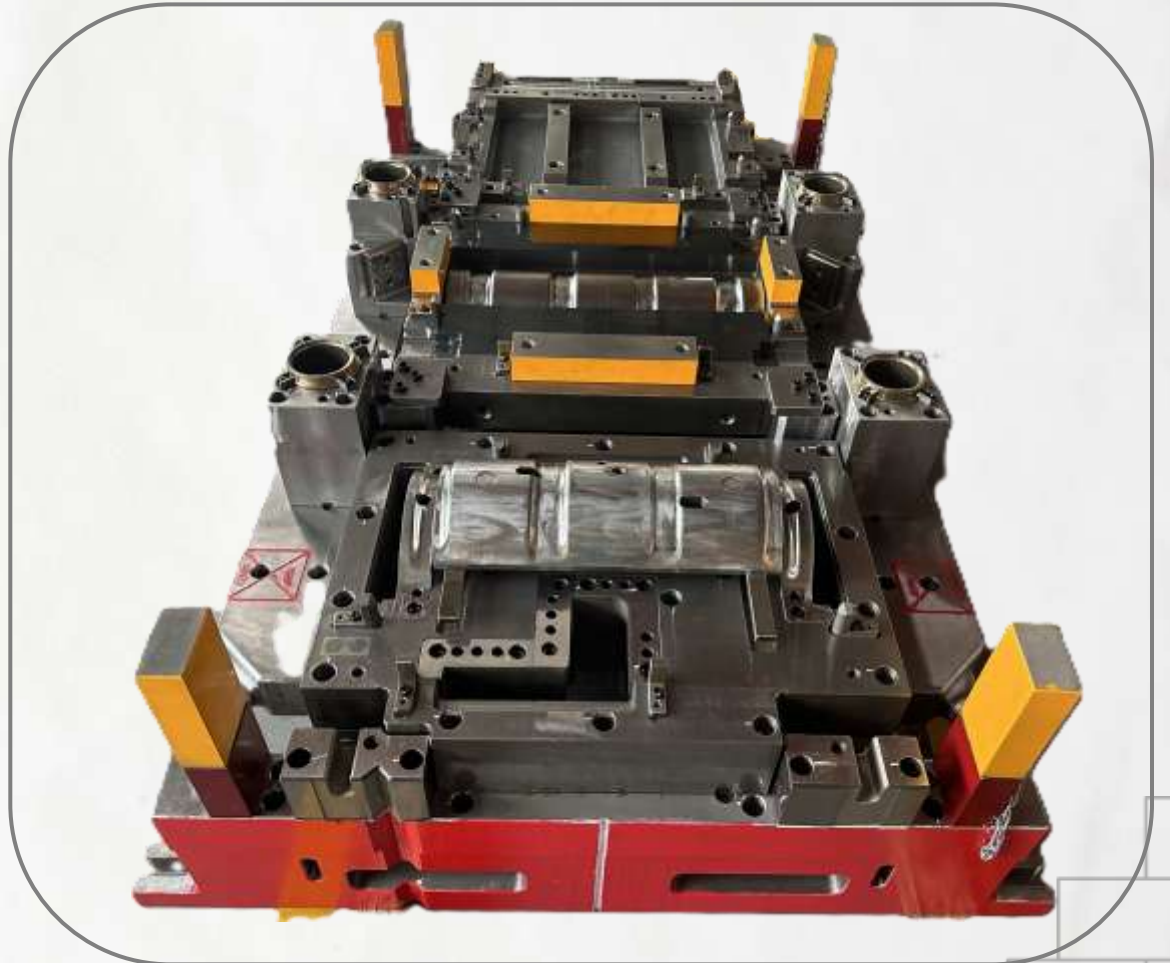
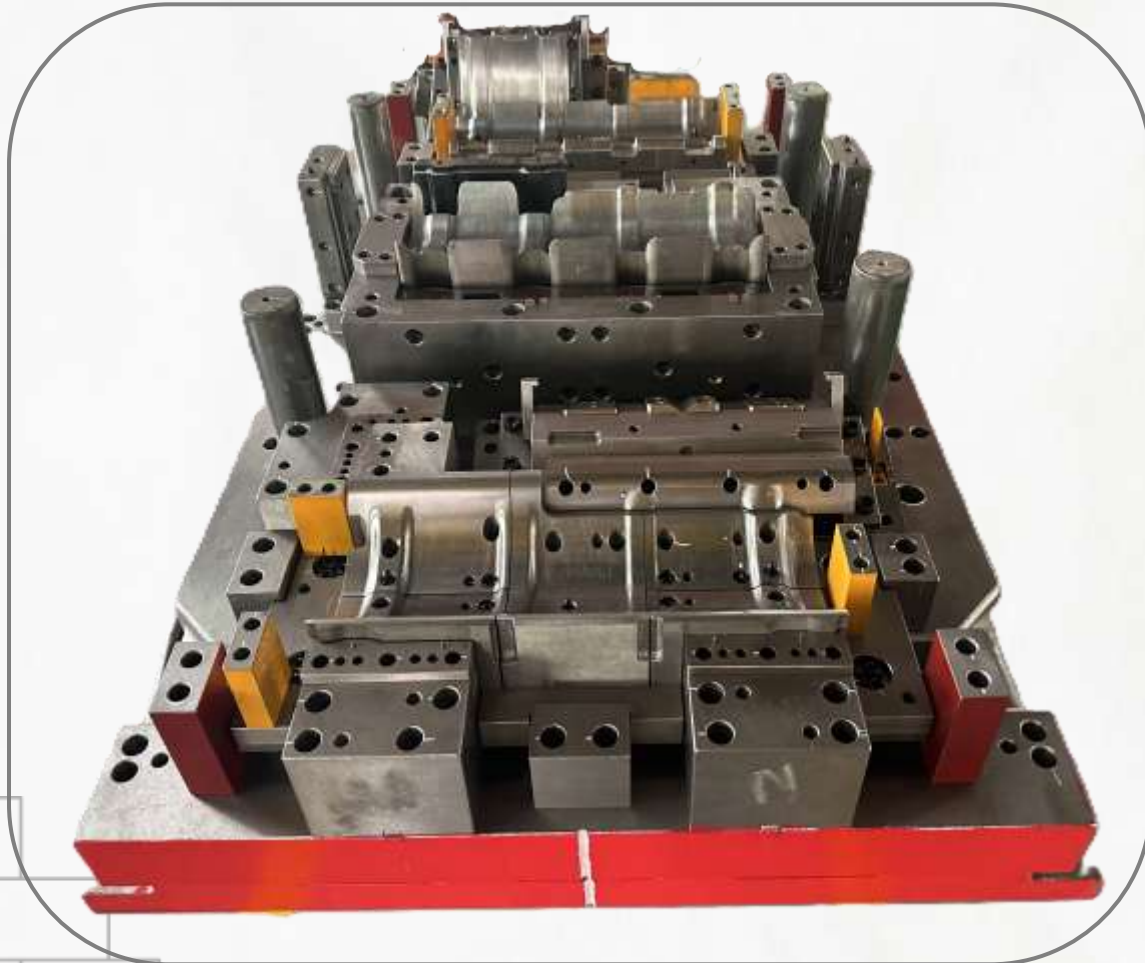
MANUAL DIE DISPLAY





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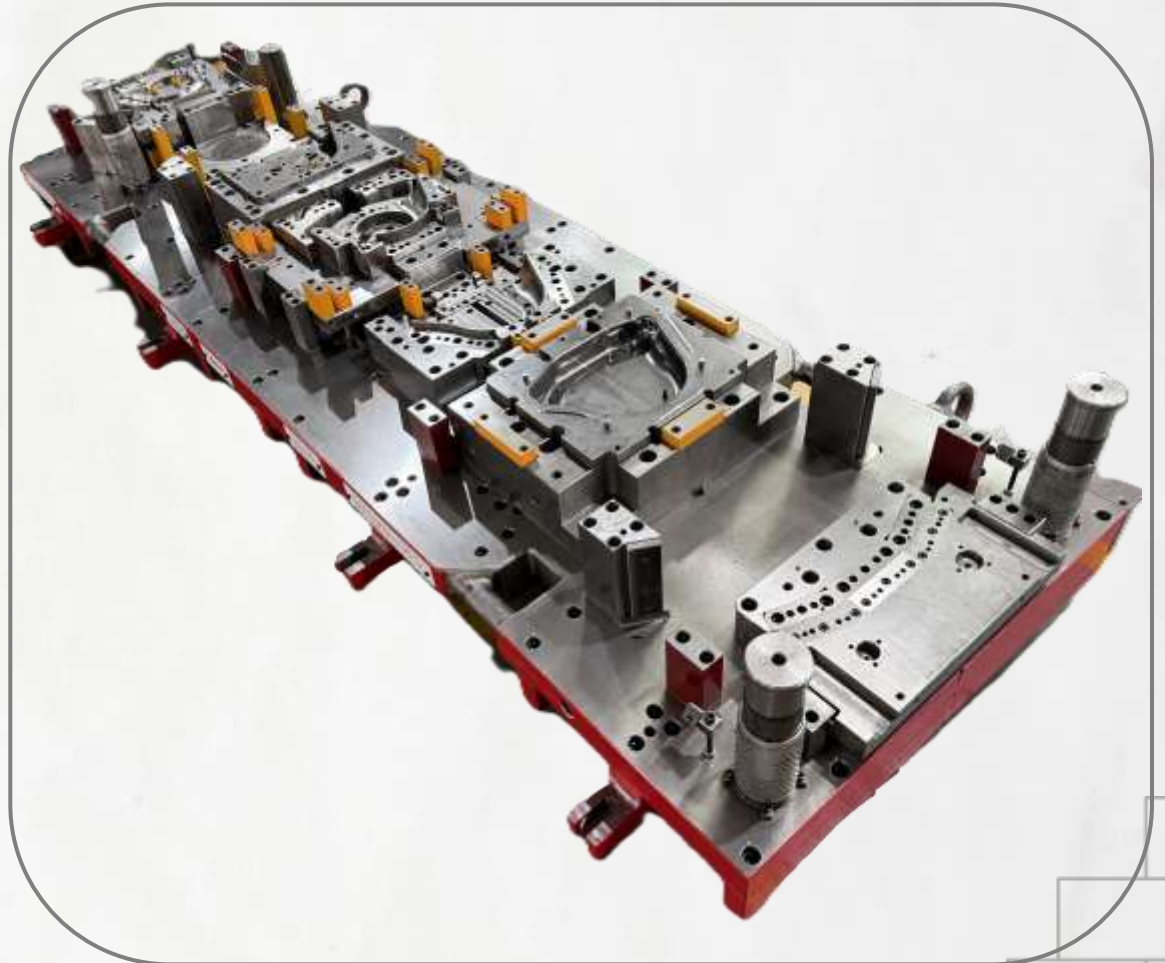
MANUAL DIE DISPLAY





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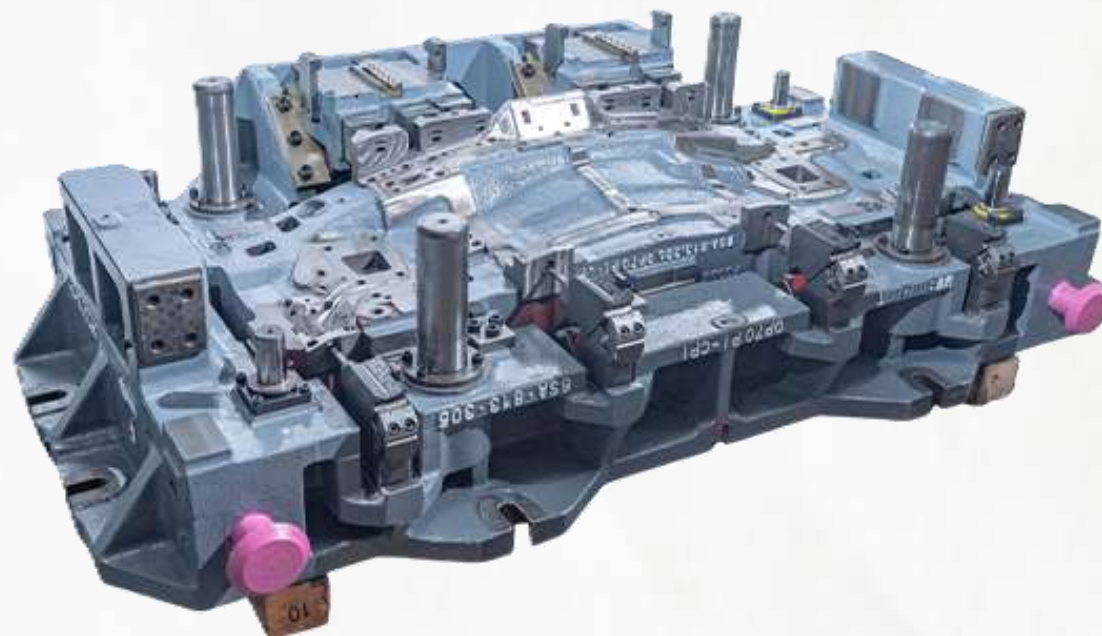
MANUAL DIE DISPLAY





MLD

TANDEM DIE DISPLAY





MLD

TANDEM DIE DISPLAY





MLD

TANDEM DIE DISPLAY





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TANDEM DIE DISPLAY





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TANDEM DIE DISPLAY





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TANDEM DIE DISPLAY





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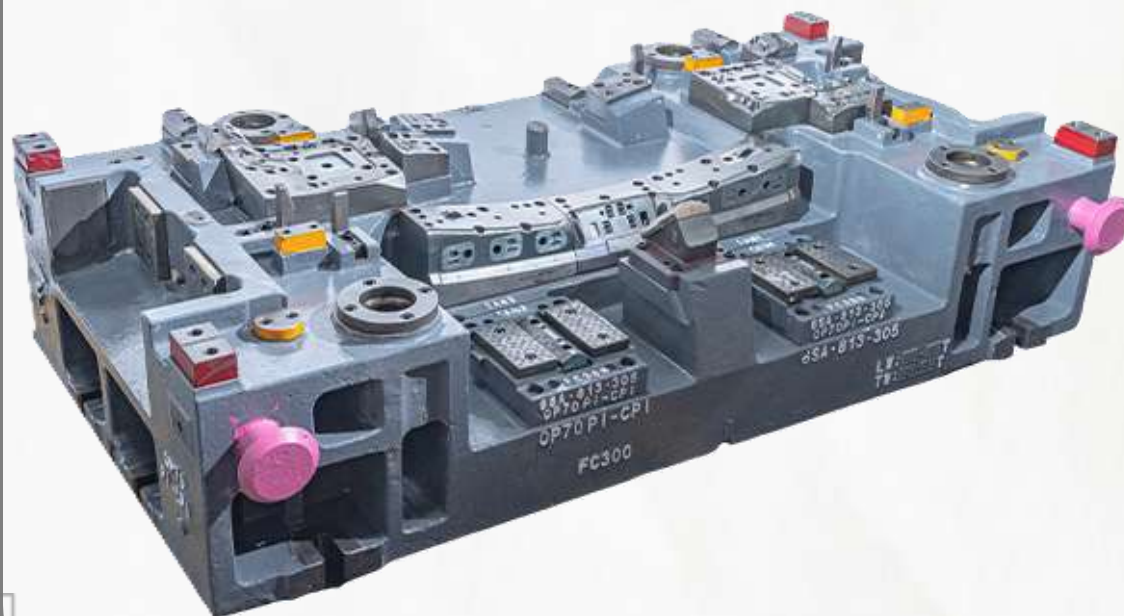
TANDE DIE DISPLAY





MLD

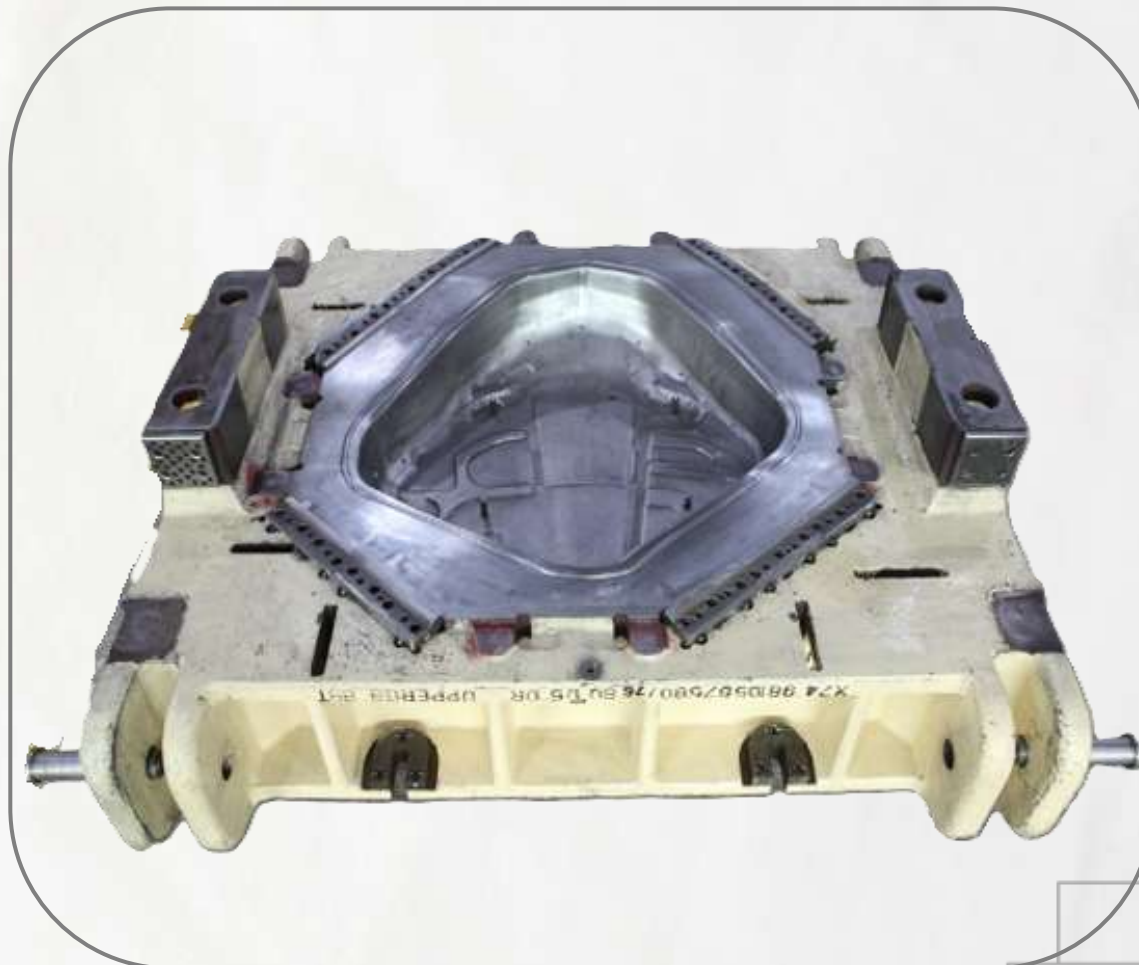
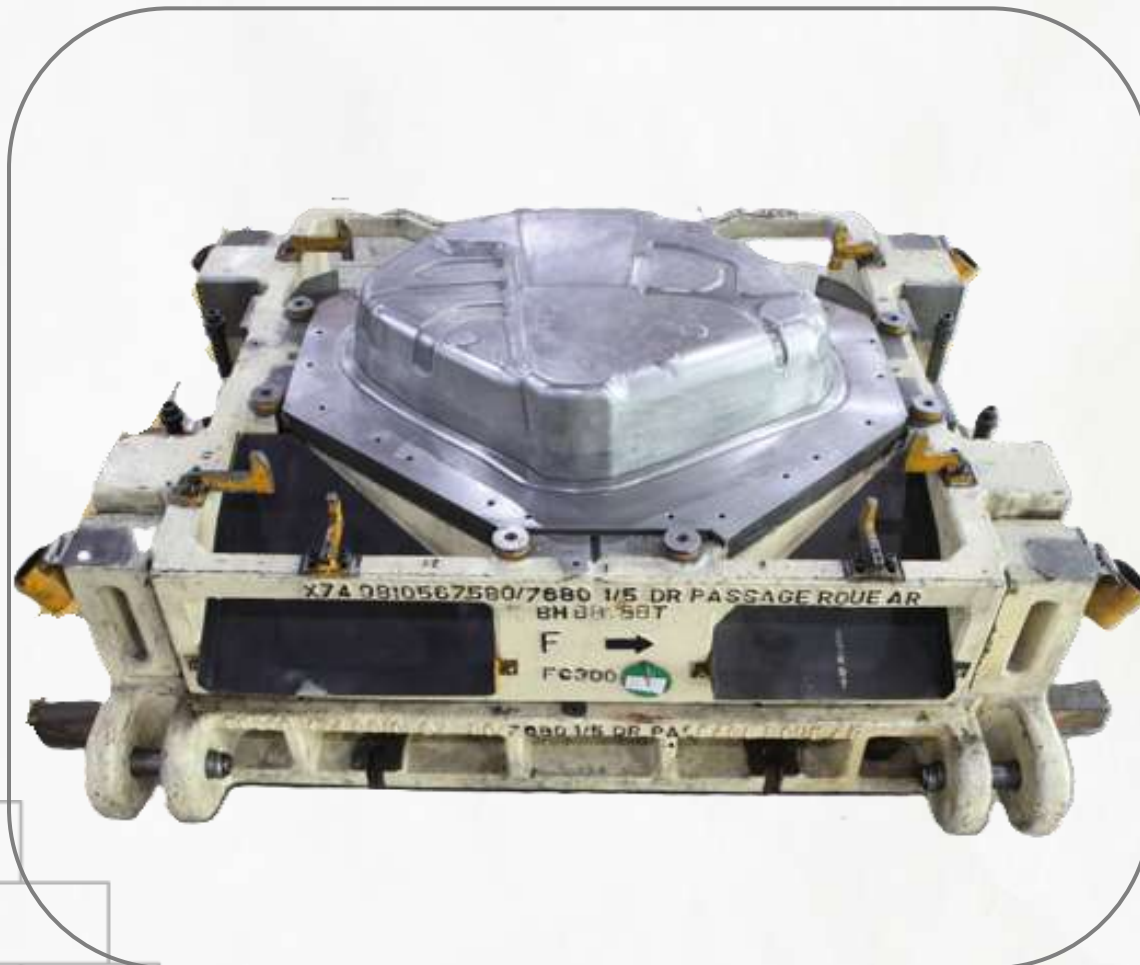
TANDEM DIE DISPLAY





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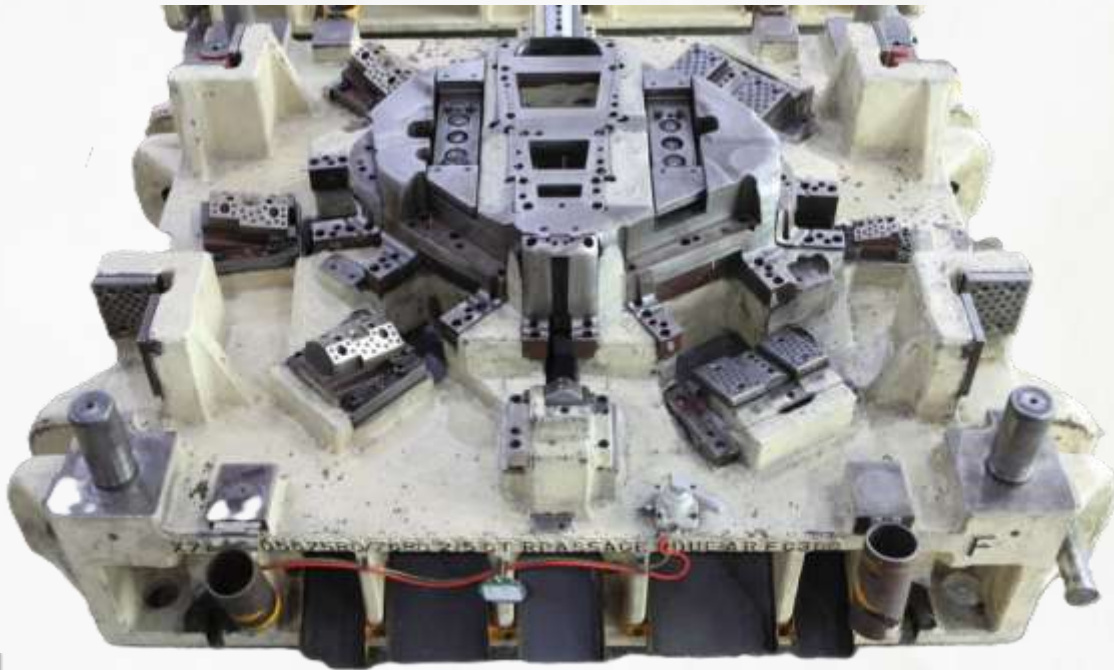
TANDEM DIE DISPLAY





MLD

TANDEM DIE DISPLAY





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TANDEM DIE DISPLAY





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TANDEM DIE DISPLAY





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TANDEM DIE DISPLAY





MLD

TANDEM DIE DISPLAY





MLD

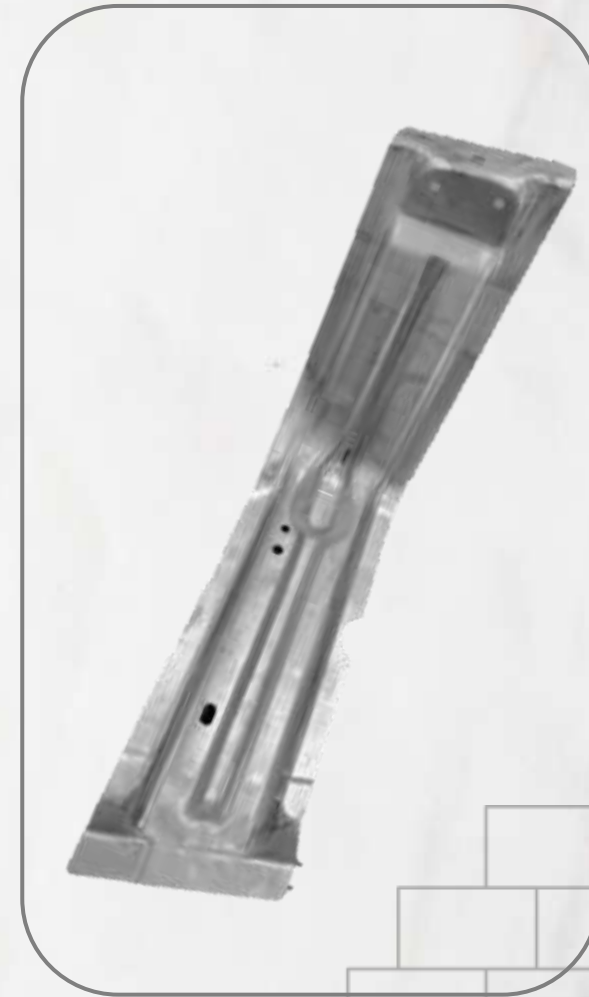
SPOTTING





MLD

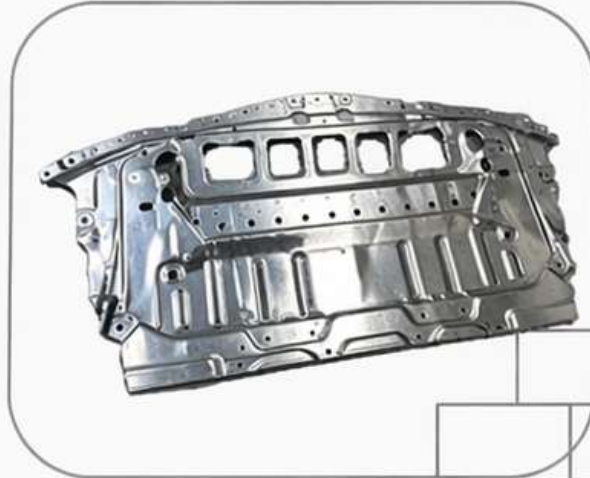
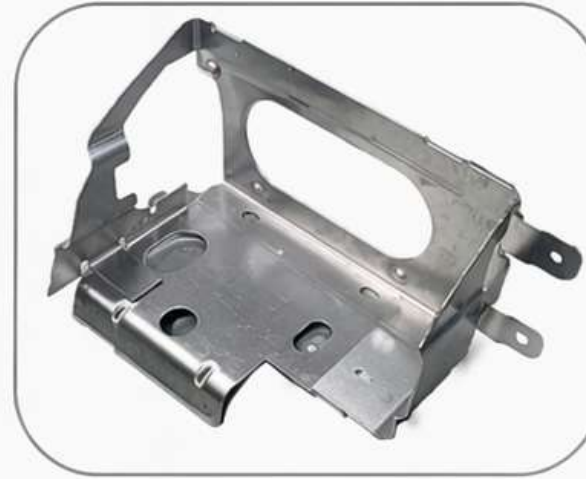
HIGH STRENGTH PART DISPLAY





MLD

VARIOUS PARTS DISPLAY





MLD

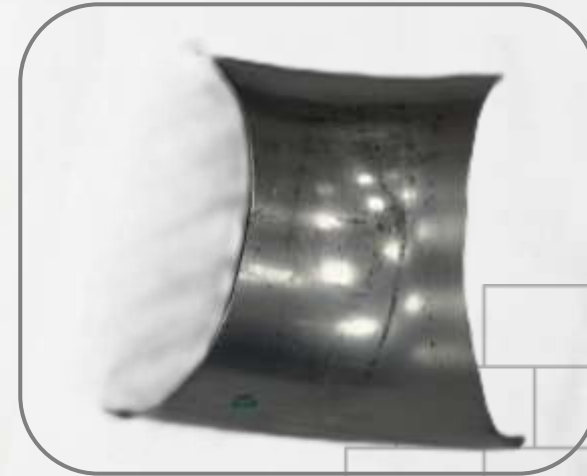
VARIOUS PARTS DISPLAY





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VARIOUS PARTS DISPLAY





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VARIOUS PARTS DISPLAY





MLD

EXHAUST PART DISPLAY





MLD

EXHAUST PART DISPLAY





MLD

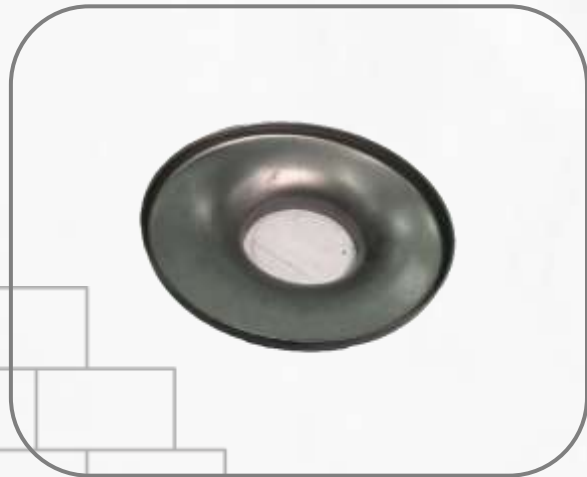
EXHAUST PART DISPLAY





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EXHAUST PART DISPLAY





MLD

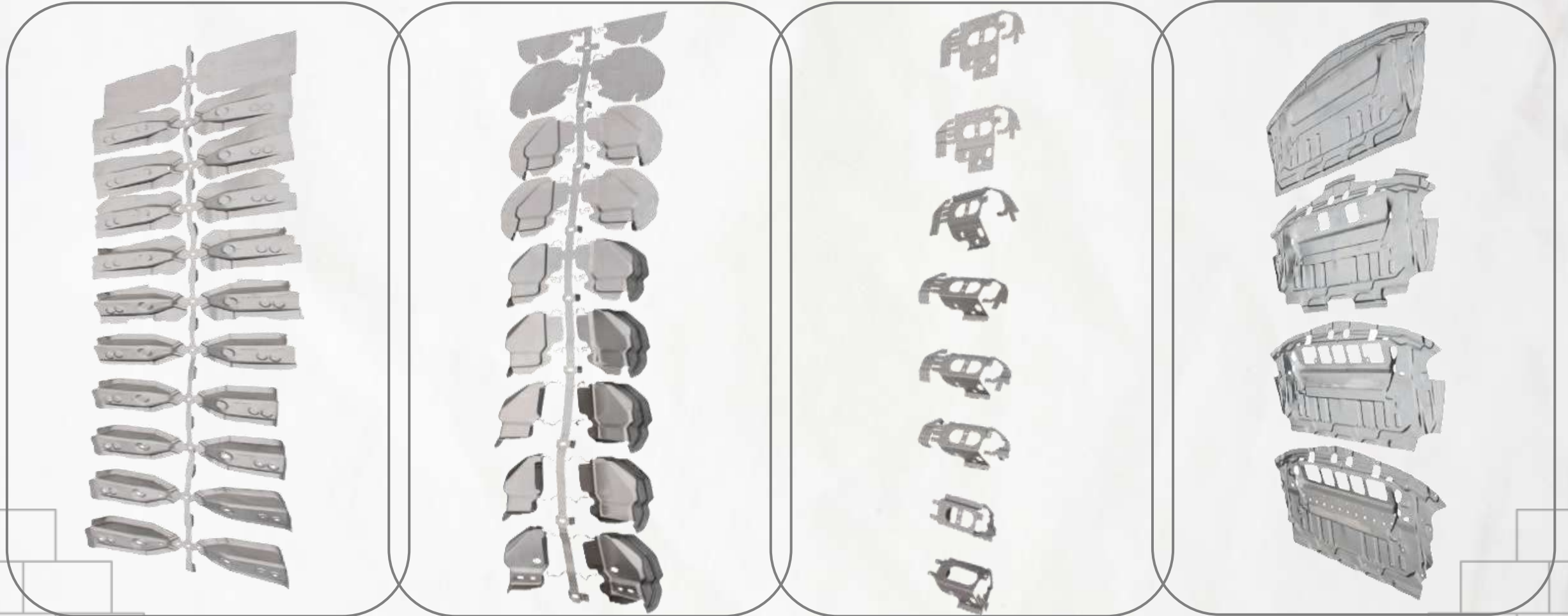
STRIP LAYOUT DISPLAY





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STRIP LAYOUT DISPLAY





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STRIP LAYOUT DISPLAY





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STRIP LAYOUT DISPLAY





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STRIP LAYOUT DISPLAY





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05

***CONTACT
US***



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CONTACT & NEXT STEPS

READY TO DISCUSS YOUR TOOLING REQUIREMENTS?
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